GRAND CANYON NATIONAL PARK

WORLD HERITAGE SITE

REHABILITATE NORTH RIM INFRASTRUCTURE

VOLUME 10 – POWER DISTRIBUTION – NORTH RIM TO ROARING SPRINGS PUMPHOUSE (RSPH)

VOLUME #	DESCRIPTION
VOLUME 0	GENERAL
VOLUME 1	PIPELINES - NORTH RIM
VOLUME 2	PIPELINES - NORTH RIM TO ROARING SPRINGS PUMPHOUSE (RSPH)
VOLUME 3	PIPELINES - NORTH RIM TO SUPAI TUNNEL
VOLUME 4	PIPELINES - ROARING SPRINGS CAVE TO ROARING SPRINGS PUMPHOUSE (RSPH)
VOLUME 5	ROARING SPRINGS PUMPHOUSE (RSPH)
VOLUME 6	NORTH RIM WATER TREATMENT PLANT
VOLUME 7	NOT USED
VOLUME 8	NORTH RIM WASTEWATER COLLECTION SYSTEM
VOLUME 9	POWER DISTRIBUTION - NORTH RIM
VOLUME 10	POWER DISTRIBUTION - NORTH RIM TO ROARING SPRINGS PUMPHOUSE (RSPH)
VOLUME 11	POWER DISTRIBUTION - ROARING SPRINGS PUMPHOUSE (RSPH) TO MANZANITA
VOLUME 12	POWER SYSTEM - COTTONWOOD CAMPGROUND

SHEET	SUB SHEET	SUB AREA	TITLE OF SHEET
1	100G001	GENERAL	SHEET INDEX
2	101X101	POWER DISTRIBUTION	OVERALL SITE DEMOLITION PLAN
3	101X102	POWER DISTRIBUTION	DEMOLITION PLAN NORTH RIM
4	101X103	POWER DISTRIBUTION	DEMOLITION PLAN UPPER BENCH
5	101X104	POWER DISTRIBUTION	DEMOLITION PLAN LOWER BENCH
6	101X105	POWER DISTRIBUTION	DEMOLITION PLAN ROARING SPRINGS
7	101E101	POWER DISTRIBUTION	SITE PLAN OVERVIEW
8	101E102	POWER DISTRIBUTION	SITE PLAN NORTH RIM
9	101E103	POWER DISTRIBUTION	SITE PLAN UPPER BENCH
10	101E104	POWER DISTRIBUTION	SITE PLAN LOWER BENCH
11	101E105	POWER DISTRIBUTION	SITE PLAN ROARING SPRINGS
12	101E106	POWER DISTRIBUTION	POWER LINE PROFILE
13	101E107	POWER DISTRIBUTION	POWER LINE CABLE ATTACHMENTS
14	101E108	POWER DISTRIBUTION	OVERHEAD POWER AND FIBER OPTIC ONE LINE DIAGRAMS
15	101E109	POWER DISTRIBUTION	CONDUCTOR SAGGING TABLES
16	101E110	POWER DISTRIBUTION	ATTACHMENTS
17	101E501	POWER DISTRIBUTION	STRUCTURE DETAIL NORTH RIM
18	101E502	POWER DISTRIBUTION	STRUCTURE DETAIL UPPER BENCH
19	101E503	POWER DISTRIBUTION	STRUCTURE DETAIL UPPER BENCH
20	101E504	POWER DISTRIBUTION	STRUCTURE DETAIL LOWER BENCH
21	101E505	POWER DISTRIBUTION	STRUCTURE DETAIL ROARING SPRINGS
22	101E506	POWER DISTRIBUTION	STRUCTURE DETAILS ASSEMBLY ATTACHMENTS
23	101E507	POWER DISTRIBUTION	STRUCTURE DETAILS ASSEMBLY ATTACHMENTS
24	101E508	POWER DISTRIBUTION	STRUCTURE DETAIL GROUNDING
25	101E509	POWER DISTRIBUTION	STRUCTURAL & ASSEMBLY LOAD AND DESIGN NORTH RIM
26	101E510	POWER DISTRIBUTION	STRUCTURAL & ASSEMBLY LOAD AND DESIGN NORTH RIM
27	101E511	POWER DISTRIBUTION	STRUCTURAL & ASSEMBLY LOAD AND DESIGN NORTH RIM
28	101E512	POWER DISTRIBUTION	STRUCTURAL & ASSEMBLY LOAD AND DESIGN UPPER BENCH
29	101E513	POWER DISTRIBUTION	STRUCTURAL & ASSEMBLY LOAD AND DESIGN UPPER BENCH
30	101E514	POWER DISTRIBUTION	STRUCTURAL & ASSEMBLY LOAD AND DESIGN UPPER BENCH
31	101E515	POWER DISTRIBUTION	STRUCTURAL & ASSEMBLY LOAD AND DESIGN UPPER BENCH
32	101E516	POWER DISTRIBUTION	STRUCTURAL & ASSEMBLY LOAD AND DESIGN UPPER BENCH
33	101E517	POWER DISTRIBUTION	STRUCTURAL & ASSEMBLY LOAD AND DESIGN LOWER BENCH
34	101E518	POWER DISTRIBUTION	STRUCTURAL & ASSEMBLY LOAD AND DESIGN LOWER BENCH
35	101E519	POWER DISTRIBUTION	STRUCTURAL & ASSEMBLY LOAD AND DESIGN LOWER BENCH
36	101E520	POWER DISTRIBUTION	STRUCTURAL & ASSEMBLY LOAD AND DESIGN ROARING SPRINGS
37	101E521	POWER DISTRIBUTION	STRUCTURAL & ASSEMBLY LOAD AND DESIGN ROARING SPRINGS
38	101E522	POWER DISTRIBUTION	STRUCTURAL & ASSEMBLY LOAD AND DESIGN ROARING SPRINGS
39	101E523	POWER DISTRIBUTION	STRUCTURAL & ASSEMBLY LOAD AND DESIGN DETAILS
40	101E524	POWER DISTRIBUTION	STRUCTURAL & ASSEMBLY LOAD AND DESIGN DETAILS
41	101E525	POWER DISTRIBUTION	STRUCTURAL & ASSEMBLY LOAD AND DESIGN DETAILS
42	101E526	POWER DISTRIBUTION	STRUCTURAL & ASSEMBLY LOAD AND DESIGN DETAILS

SHEET	SUB SHEET	SUB AREA	TITLE OF SHEET
43	101E527	POWER DISTRIBUTION	MICROPILE DETAILS
44	101E528	POWER DISTRIBUTION	MICROPILE FOUNDATION DETAILS NORTH RIM
45	101E529	POWER DISTRIBUTION	MICROPILE FOUNDATION DETAILS UPPER BENCH
46	101E530	POWER DISTRIBUTION	MICROPILE FOUNDATION DETAILS, LOWER BENCH & ROARING SPRINGS
47	101E531	POWER DISTRIBUTION	MICROPILE FOUNDATION LOADS
48	101E532	POWER DISTRIBUTION	DETAILS
49	101E533	POWER DISTRIBUTION	OPGW DETAILS
50	101E534	POWER DISTRIBUTION	STRUCTURE DETAILS MATERIAL ITEMS

REGISTERED PROFESSIONAL ENGINEER (ELECTRICAL) CERTIFICATE NO. 75014 TOU VANG Date Signed

REGISTERED PROFESSIONAL ENGINEER (CIVIL) CERTIFICATE NO. 41809 RYAN RHOADES Date Signed ARIZONA, U.S.A.

A/E FIRM	
PRIME:	HDR ENGINEERING, INC.
	DENVER, CO 80202
DESIGNED:	B. PLANT
CADD	N. LYNCH
TECH. REVIEW:	M. GOUGH
DATE:	AUG 2026
SUB SHEET NO.	100 G001
TITLE OF SHEET	GENERAL SHEET INDEX
	NORTH RIM - GRAND CANYON NATIONAL PARK
DRAWING NO.	113 182752
PMIS/PKG NO.	318719
SHEET	1 OF 50

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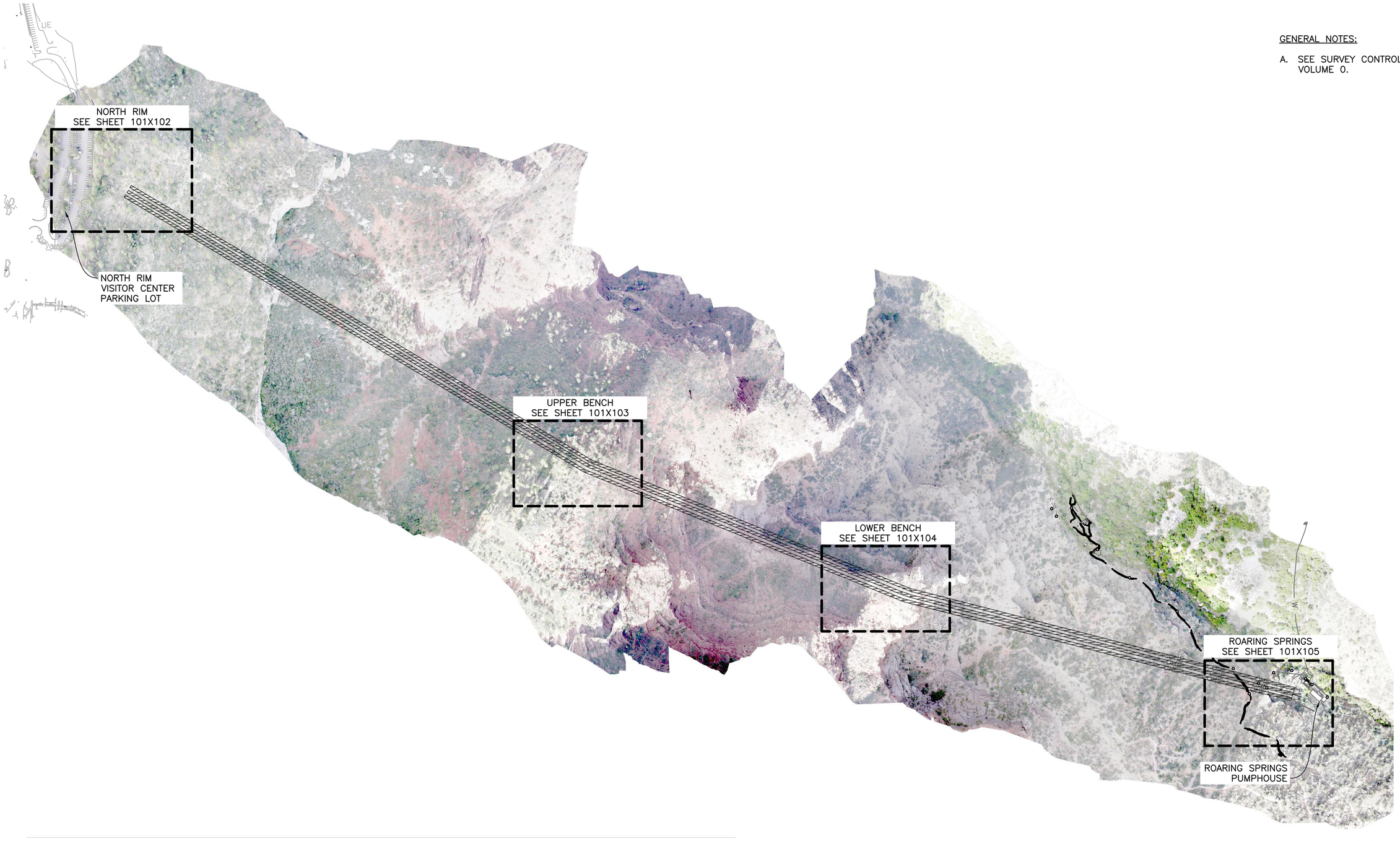

GENERAL NOTES:

A. SEE SURVEY CONTROL PLAN IN VOLUME 0.

200 0 200 400 1" = 200'
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER COLORADO
CERTIFICATE NO
75014
TOU
VANG

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: B. PLANT CADD C. WALCHUK TECH. REVIEW: J. DOHRENDORF DATE: MAY 2026	SUB SHEET NO. 101 X101	TITLE OF SHEET POWER DISTRIBUTION OVERALL SITE DEMOLITION PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 2 OF 50

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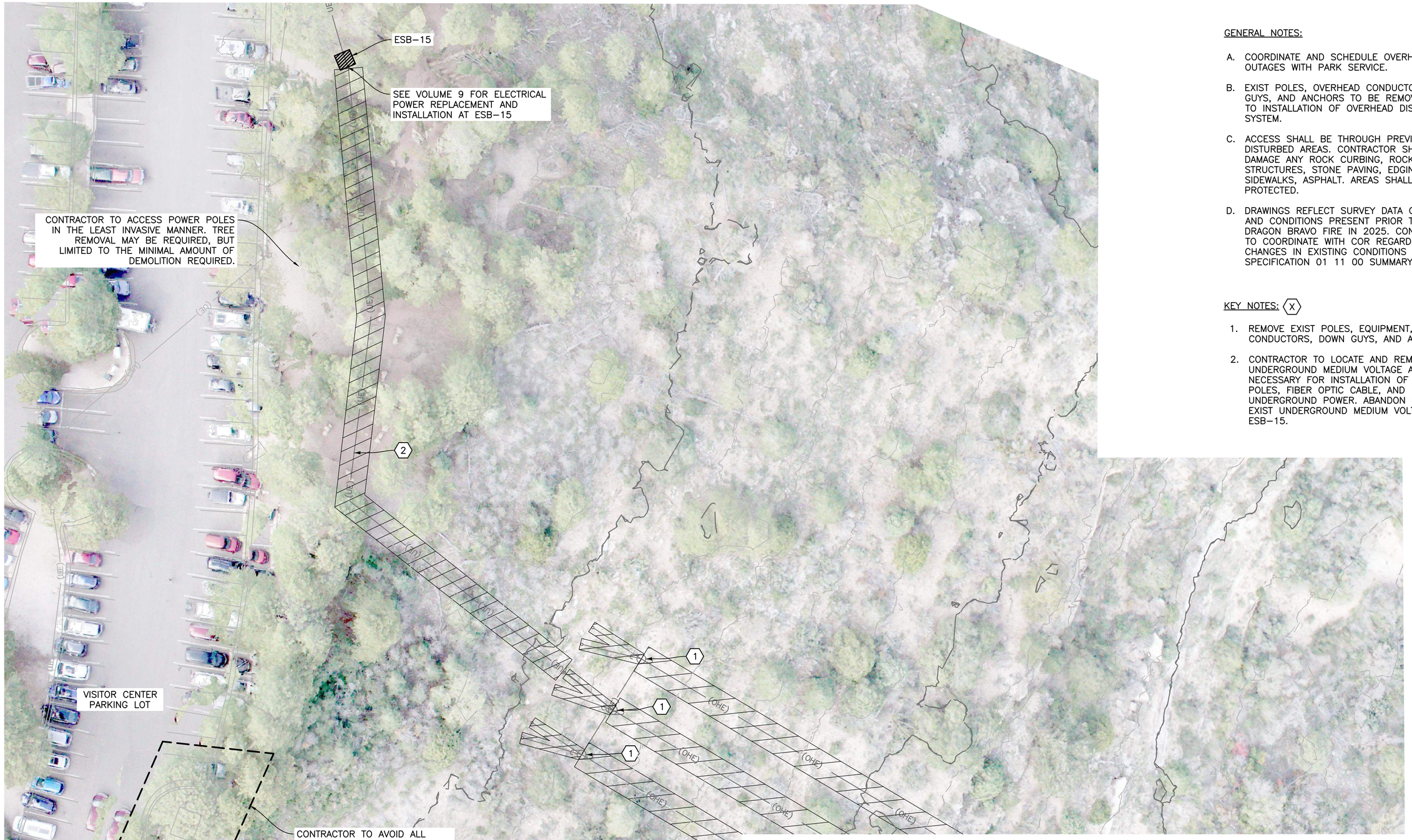

GENERAL NOTES:

A. COORDINATE AND SCHEDULE OVERHEAD POWER OUTAGES WITH PARK SERVICE.

B. EXIST POLES, OVERHEAD CONDUCTORS, DOWN GUYS, AND ANCHORS TO BE REMOVED PRIOR TO INSTALLATION OF OVERHEAD DISTRIBUTION SYSTEM.

C. ACCESS SHALL BE THROUGH PREVIOUSLY DISTURBED AREAS. CONTRACTOR SHALL NOT DAMAGE ANY ROCK CURBING, ROCK WALLS, STRUCTURES, STONE PAVING, EDGING, SIDEWALKS, ASPHALT. AREAS SHALL BE PROTECTED.

D. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

KEY NOTES: (X)

1. REMOVE EXIST POLES, EQUIPMENT, OVERHEAD CONDUCTORS, DOWN GUYS, AND ANCHORS.
2. CONTRACTOR TO LOCATE AND REMOVE EXIST UNDERGROUND MEDIUM VOLTAGE AS NECESSARY FOR INSTALLATION OF POWER POLES, FIBER OPTIC CABLE, AND UNDERGROUND POWER. ABANDON REMAINING EXIST UNDERGROUND MEDIUM VOLTAGE FROM ESB–15.

20 0 20 40
SCALE OF FEET
1" = 20'

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 75014 TOU VANG STATE OF CALIFORNIA

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: B. PLANT CADD C. WALCHUK TECH. REVIEW: J. DOHRENDORF DATE: MAY 2026	SUB SHEET NO. 101 X102	TITLE OF SHEET POWER DISTRIBUTION DEMOLTION PLAN NORTH RIM NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 3 OF 50

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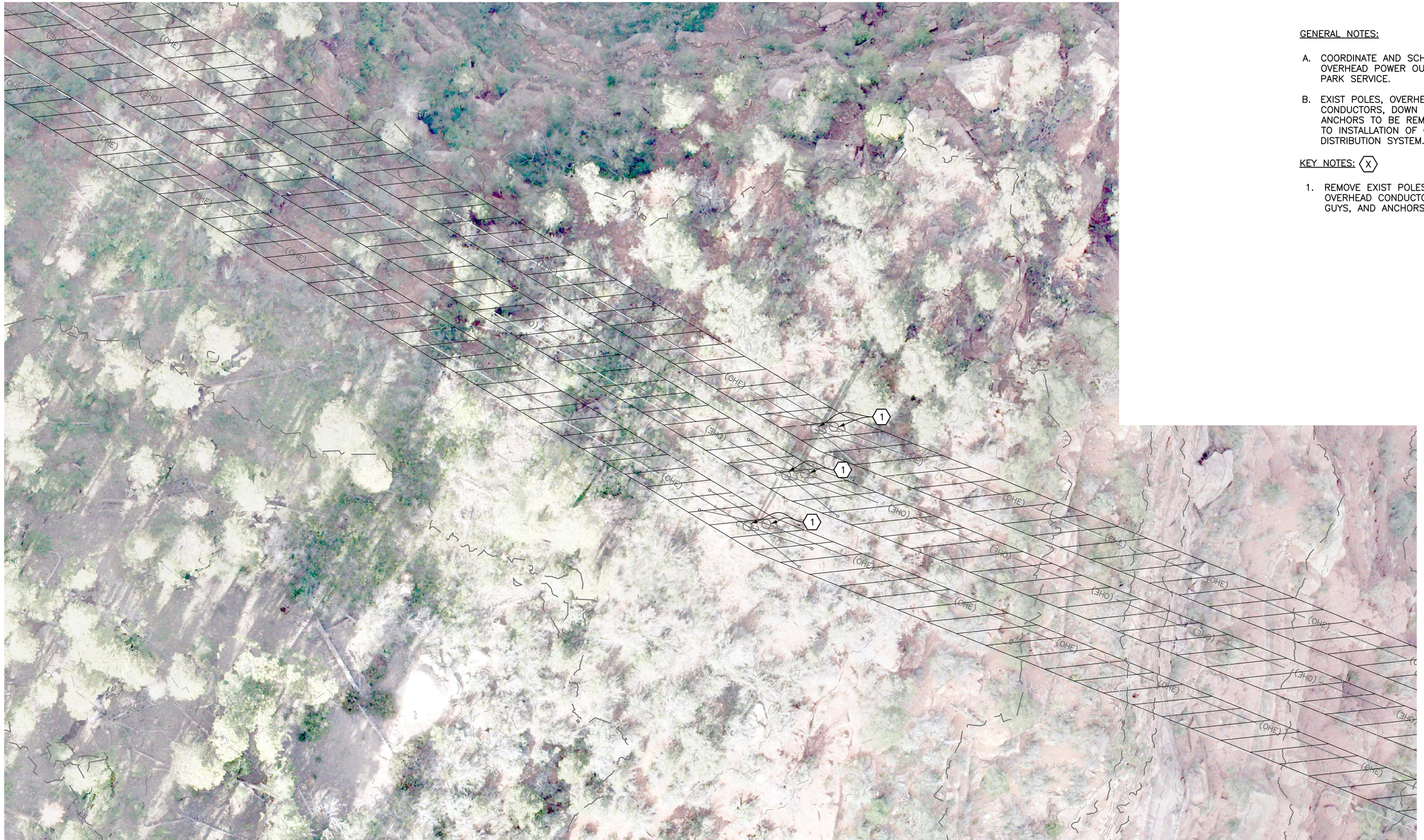

GENERAL NOTES:

A. COORDINATE AND SCHEDULE OVERHEAD POWER OUTAGES WITH PARK SERVICE.

B. EXIST POLES, OVERHEAD CONDUCTORS, DOWN GUYS AND ANCHORS TO BE REMOVED PRIOR TO INSTALLATION OF OVERHEAD DISTRIBUTION SYSTEM.

KEY NOTES: (X)

1. REMOVE EXIST POLES, EQUIPMENT, OVERHEAD CONDUCTORS, DOWN GUYS, AND ANCHORS.

20 0 20 40 1" = 20'
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 75014 TOU VANG STATE OF COLORADO

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: B. PLANT CADD C. WALCHUK TECH. REVIEW: J. DOHRENDORF DATE: MAY 2026	SUB SHEET NO. 101 X103	TITLE OF SHEET POWER DISTRIBUTION DEMOLITION PLAN UPPER BENCH NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 4 OF 50

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GENERAL NOTES:

A. COORDINATE AND SCHEDULE OVERHEAD POWER OUTAGES WITH PARK SERVICE.

B. EXIST POLES, OVERHEAD CONDUCTORS, DOWN GUYS AND ANCHORS TO BE REMOVED PRIOR TO INSTALLATION OF OVERHEAD DISTRIBUTION SYSTEM.

KEY NOTES: (X)

1. REMOVE EXIST POLES, EQUIPMENT, OVERHEAD CONDUCTORS, DOWN GUYS, AND ANCHORS.

20 0 20 40 1" = 20'
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER COLORADO
CERTIFICATE NO. 75014
TOU VANG

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: B. PLANT DRAWN: C. WALCHUK TECH. REVIEW: J. DOHRENDORF DATE: MAY 2026	SUB SHEET NO. 101 X104	TITLE OF SHEET POWER DISTRIBUTION DEMOLITION PLAN LOWER BENCH NORTH RIM - GRAND CANYON NATIONAL PARK

DRAWING NO. 113 182752
PMIS/PKG NO. 318719
SHEET 5 OF 50

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GENERAL NOTES:

A. COORDINATE AND SCHEDULE OVERHEAD POWER OUTAGES WITH PARK SERVICE.

B. EXIST POLES, OVERHEAD CONDUCTORS, DOWN GUYS AND ANCHORS TO BE REMOVED PRIOR TO INSTALLATION OF OVERHEAD DISTRIBUTION SYSTEM.

KEY NOTES: (X)

1. REMOVE EXIST POLES, EQUIPMENT, OVERHEAD CONDUCTORS, DOWN GUYS, AND ANCHORS.

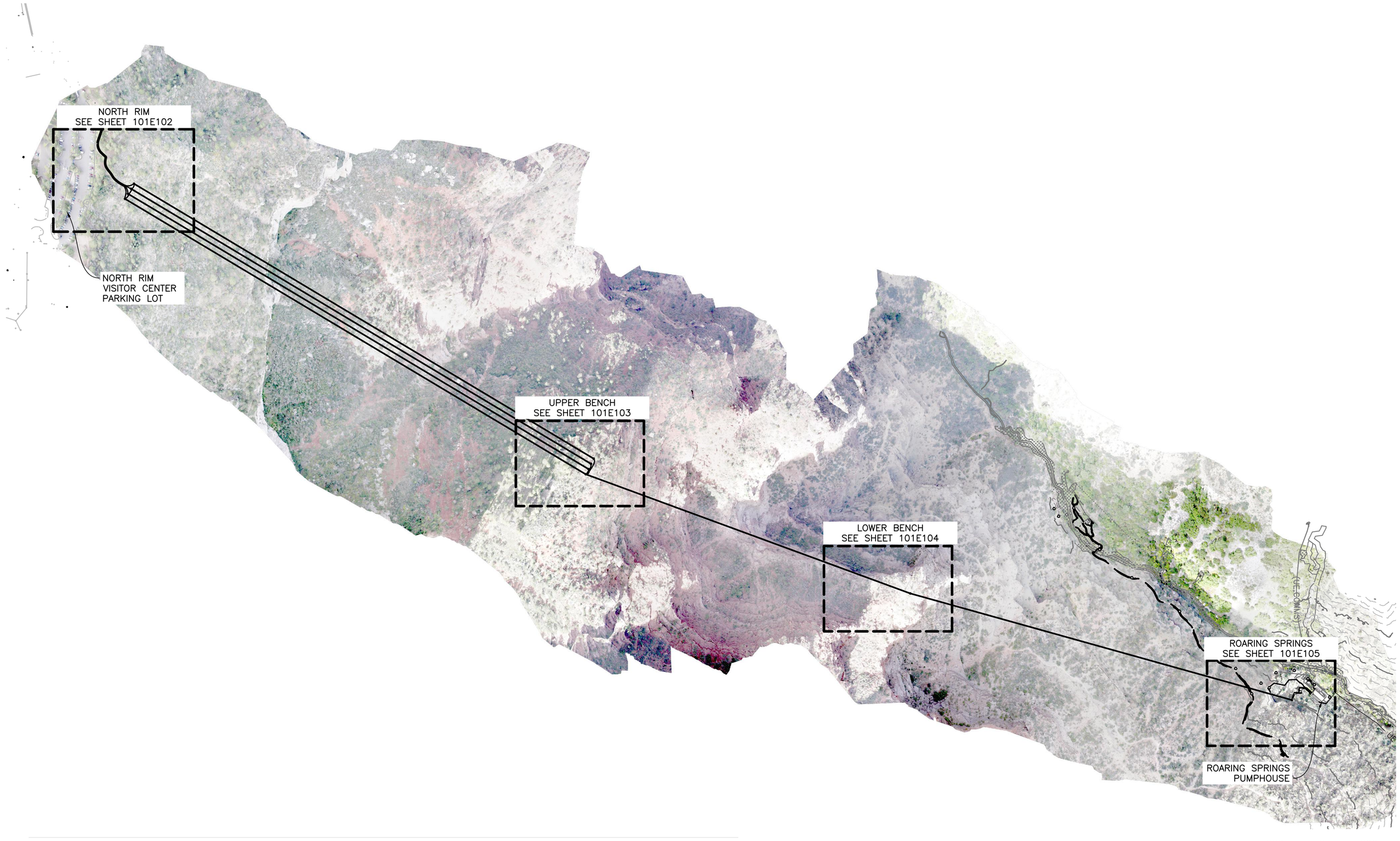
2. REMOVE EXIST UNDERGROUND MEDIUM VOLTAGE POWER CABLE AND CONDUIT FROM MIDDLE POLE TO EXIST JUNCTION ENCLOSURES.

20 0 20 40
SCALE OF FEET
1" = 20'

REGISTERED PROFESSIONAL ENGINEER ELECTRICAL
CERTIFICATE NO. 75014
TOU VANG
STATE OF COLORADO

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: B. PLANT CADD C. WALCHUK TECH. REVIEW: J. DOHRENDORF DATE: MAY 2026	SUB SHEET NO. 101 X105	TITLE OF SHEET POWER DISTRIBUTION DEMOLITION PLAN ROARING SPRINGS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 6 OF 50

4/17/2026 Vang, Tou C:\Users\TVANG\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\ELEC\101-X105.dwg



200 0 200 400
SCALE OF FEET
1" = 200'

REGISTERED PROFESSIONAL ENGINEER COLORADO
CERTIFICATE NO
75014
TOU
VANG

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: B. PLANT CADD C. WALCHUK TECH. REVIEW: J. DOHRENDORF DATE: MAY 2026	SUB SHEET NO. 101 E101	TITLE OF SHEET POWER DISTRIBUTION SITE PLAN OVERVIEW NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 7 OF 50

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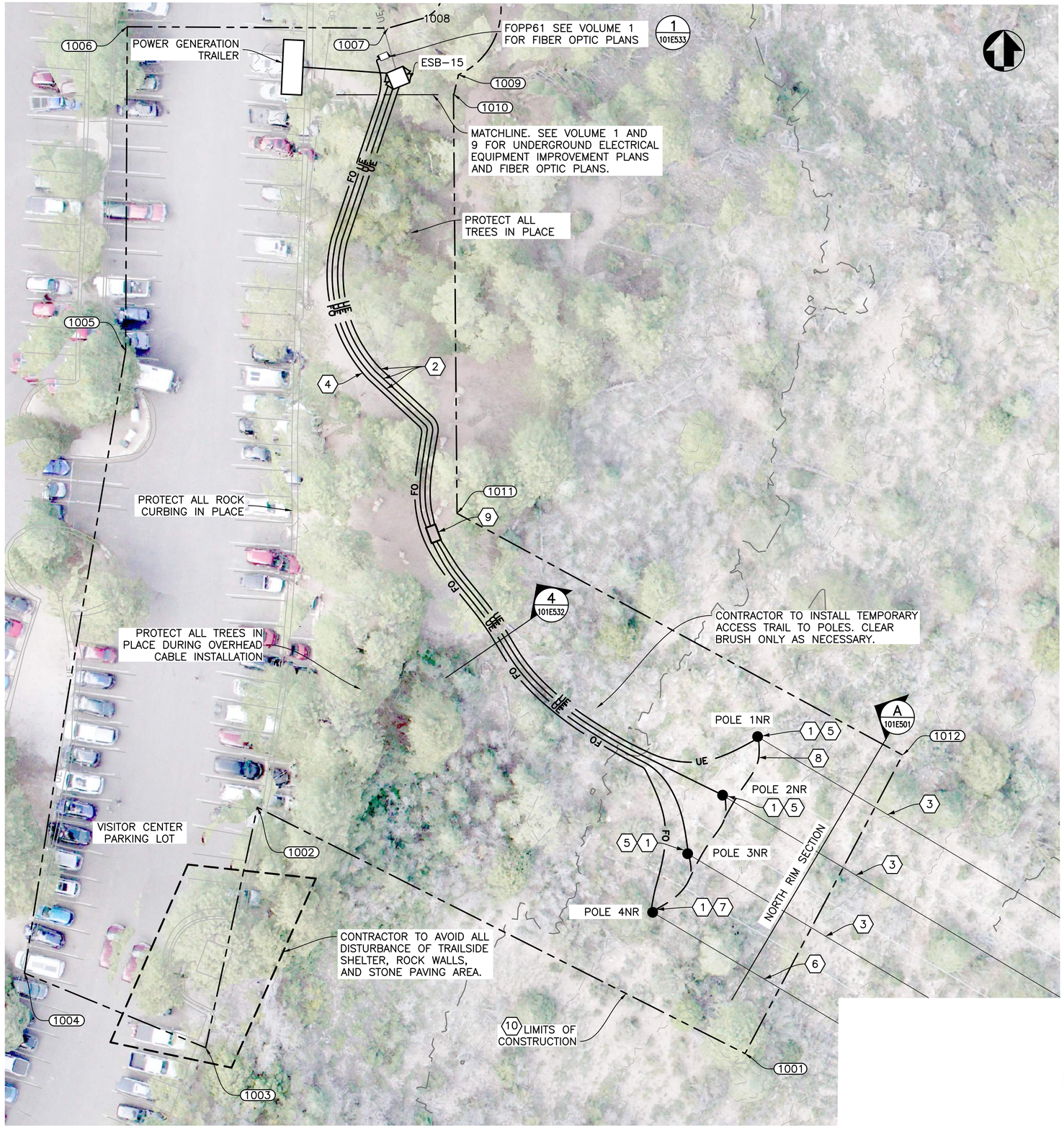

GENERAL NOTES:

A. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

B. SEE SHEET 101E501 FOR STRUCTURE SECTION LAYOUT AT NORTH RIM.

C. SEE SHEET 101E509 THROUGH 101E511 FOR STRUCTURAL ASSEMBLY LOAD AND DESIGN DETAILS AT NORTH RIM.

D. OVERHEAD POWER CABLE SPLICES TO BE APPROVED BY THE CONTRACTING OFFICER REPRESENTATIVE BEFORE CONTRACTOR PROCURES SPLICE MATERIAL AND BEGINS WORK.

E. ACCESS SHALL BE THROUGH PREVIOUSLY DISTURBED AREAS. CONTRACTOR SHALL NOT DAMAGE ANY ROCK CURBING, ROCK WALLS, STRUCTURES, STONE PAVING, EDGING, SIDEWALKS, OR ASPHALT. AREAS SHALL BE PROTECTED.

F. DRAWINGS REFLECT SURVEY DATA OF FEATURES AND CONDITIONS PRESENT PRIOR TO THE DRAGON BRAVO FIRE IN 2025. CONTRACTOR TO COORDINATE WITH COR REGARDING CHANGES IN EXISTING CONDITIONS PER SPECIFICATION 01 11 00 SUMMARY OF WORK.

KEY NOTES: (X)

1. INSTALL 55' SELF SUPPORTING STEEL POLE ON MICROPILE FOUNDATION APPROXIMATELY 12.5' FROM EXIST POLE LOCATIONS, SEE SHEET 101E528 FOR POLE COORDINATES. FINAL POLE AND FOUNDATION SIZE TO BE DETERMINED BY CONTRACTOR'S MANUFACTURER, BASED ON DESIGN REQUIREMENTS AND APPROVED BY CONTRACTING OFFICER REPRESENTATIVE. SEE SHEET 101E501 FOR NORTH RIM OVERHEAD STRUCTURE LAYOUT DETAILS.
2. TRENCH 15KV (3) 1/C 4/0 AWG URD CABLE WITH CONCENTRIC NEUTRAL IN (3) 3" CONDUIT (ONE CABLE PER CONDUIT) FROM POLES 1NR, 2NR, AND 3NR TO 4–WAY SWITCH ESB–15. CONTRACTOR TO FOLLOW EXIST UNDERGROUND ELECTRICAL CORRIDOR TO THE OVERHEAD POWER POLES AND BE WITHIN PREVIOUSLY DISTURBED AREAS. TERMINATE UNDERGROUND POWER CABLE IN 4–WAY LOADBREAK INTERRUPTING PADMOUNTED SWITCH WITH 200 AMP LOAD BREAK ELBOWS. CONTRACTOR SHALL ADJUST ALIGNMENT AS NEEDED TO AVOID TREES. PROTECT TREES IN PLACE.
3. INSTALL 183 KCMIL ALUMINUM CONDUCTOR, COMPOSITE SUPPORTED CORE (ACCR) WITH CABLE MARKER BALLS. SEE SHEET 101E107 FOR CABLE MARKER BALL DETAILS. SEE DETAIL 6 SHEET 101E532 FOR ACCR DETAIL.
4. INSTALL 144–STRAND SM FIBER IN 2" CONDUIT ALONG THE SAME TRENCH AS THREE–PHASE UNDERGROUND CABLE FROM POLE 4NR TO FOPP61. SEE DETAIL 4 SHEET 101E532.
5. INSTALL PRIMARY SINGLE–PHASE UNDERGROUND RISER AND NEUTRAL RISER. BOND UNDERGROUND POWER CABLE CONCENTRIC NEUTRAL TO RISER NEUTRAL. SEE SHEET 101E507 DETAILS 4, 5, & 6 FOR TYPICAL DOUBLE RISER DETAIL.
6. INSTALL 144–STRAND SM OPTICAL GROUND WIRE (OPGW) WITH CABLE MARKER BALLS. SEE SHEET 101E107 FOR CABLE MARKER BALL DETAILS.
7. INSTALL OPTICAL GROUND WIRE RISER. SEE SHEET 101E507 DETAIL 7 FOR TYPICAL RISER DETAIL. SEE SHEET 101E533 FOR OPGW VAULT TERMINATION.
8. INSTALL UNDERGROUND POWER CABLE NEUTRAL AT CABLE RISER TERMINATION TO POLE GROUND AT BASE OF POLE. SEE SHEET 101E508 FOR POLE GROUNDING DETAIL.
9. INSTALL PULL BOX AND 15KV (3) 1/C 4/0 AWG URD CABLES FROM ESB–15 TO POLES 1NR, 2NR, 3NR. INSTALL ZIPPERED CABLE SUPPORT GRIPS ON CABLES LEADING TO POLES PER ROCK ANCHOR CABLE SUPPORT DETAIL ON SHEET 101E532. SEE SHEET 101E532 FOR CABLE VAULT DETAIL. DO NOT SPLICE FIBER HERE. ROUTE FIBER AROUND CABLE VAULT.
10. AVAILABLE CONSTRUCTION LIMITS ARE SHOWN IN THE LIMITS OF CONSTRUCTION TABLE.

LIMITS OF CONSTRUCTION		
POINT NO.	NORTHING	EASTING
1001	13144762.37	1330358.40
1002	13144852.59	1330179.66
1003	13144764.50	1330160.76
1004	13144791.84	1330094.32
1005	13145019.83	1330131.41
1006	13145138.63	1330132.01
1007	13145138.15	1330227.62
1008	13145141.65	1330240.15
1009	13145121.03	1330253.39
1010	13145113.95	1330251.56
1011	13144960.31	1330252.86
1012	13144871.77	1330416.12

20 0 20 40
SCALE OF FEET
1" = 20'

REGISTERED PROFESSIONAL ENGINEER AND LAND SURVEYOR
CERTIFICATE NO. 75014
TOU VANG

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	B. PLANT CADD C. WALCHUK TECH. REVIEW: J. DOHRENDORF DATE: JUNE 2026	101 E102	POWER DISTRIBUTION SITE PLAN NORTH RIM NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752 PMIS/PKG NO. 318719 SHEET 8 OF 50

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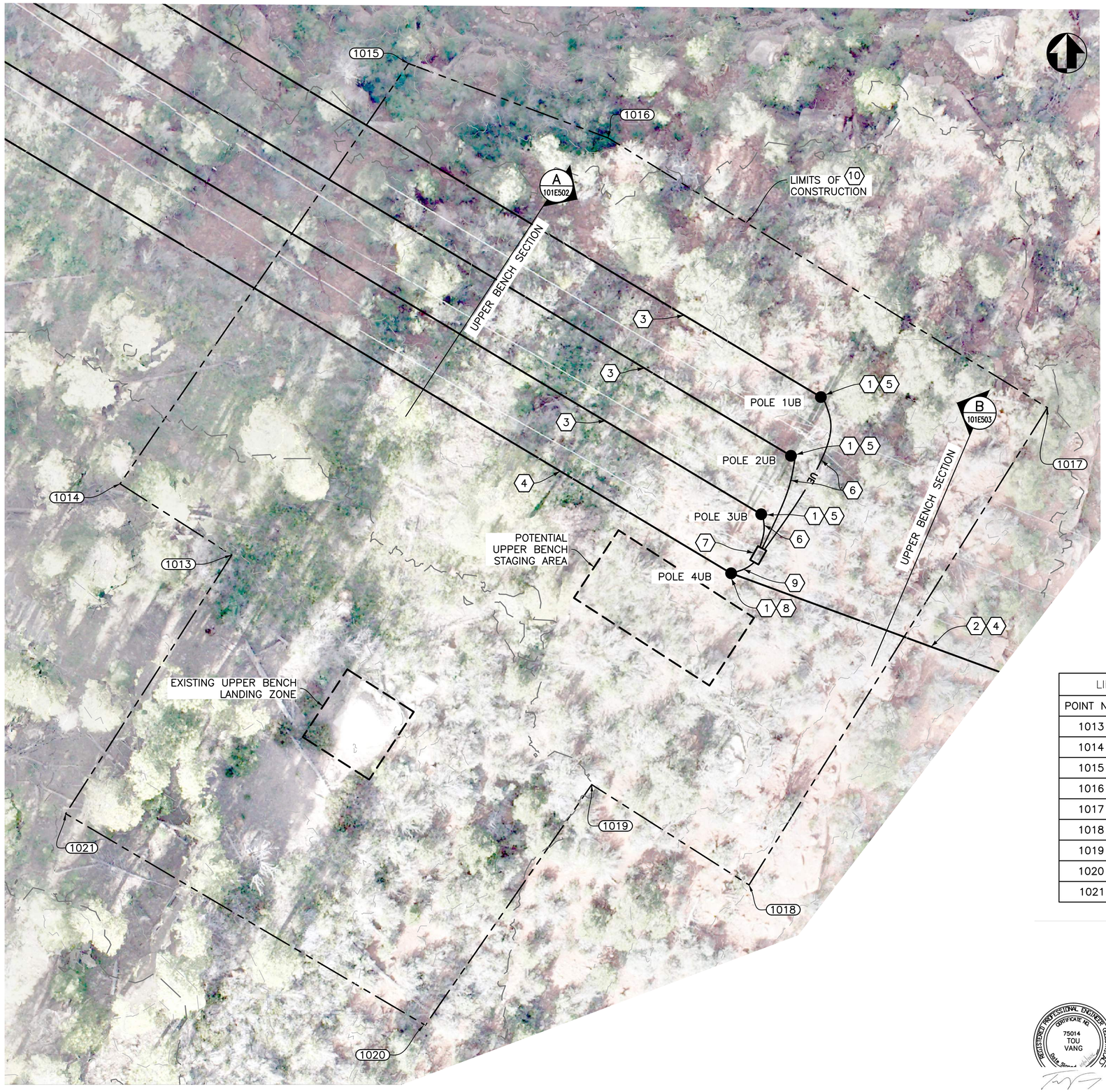


GENERAL NOTES:

A. SEE SHEET 101E502 AND 101E503 FOR STRUCTURE SECTION LAYOUT AT UPPER BENCH.

B. SEE SHEET 101E512 THROUGH 101E516 FOR STRUCTURAL ASSEMBLY LOAD AND DESIGN DETAILS AT UPPER BENCH.

C. OVERHEAD POWER CABLE SPLICES TO BE APPROVED BY THE CONTRACTING OFFICER REPRESENTATIVE BEFORE CONTRACTOR PROCURES SPLICE MATERIAL AND BEGINS WORK.

D. SEE GRCA 318719 UPPER BENCH SITE TRIP REPORT FOR HIKING ACCESS FROM PICKLE TOWN LANDING ZONE TO THE UPPER BENCH SITE, SITE CONDITIONS AT UPPER BENCH EXISTING LANDING ZONE, AND POTENTIAL STAGING AREA.

KEY NOTES: (X)

1. INSTALL (4) 90' SELF SUPPORTING STEEL POLES ON MICROPILE FOUNDATIONS APPROXIMATELY 12.5' FROM EXIST. POLE LOCATIONS, SEE SHEET 101E529 FOR POLE COORDINATES. FINAL POLE AND FOUNDATION SIZE TO BE DETERMINED BY CONTRACTOR'S MANUFACTURER, BASED ON DESIGN REQUIREMENTS AND APPROVED BY CONTRACTING OFFICER REPRESENTATIVE. SEE SHEET 101E502 AND 101E503 FOR UPPER BENCH OVERHEAD STRUCTURE LAYOUT DETAILS.
2. INSTALL (2) 37 #8 ALUMINUM CLAD STEEL MESSENGER WITH (3) 1/0 INSULATED SPACER CABLES AND SPACERS. SEE SHEET 101E107 FOR SPACER CABLE DETAILS.
3. INSTALL 183 KCMIL ALUMINUM CONDUCTOR, COMPOSITE SUPPORTED CORE (ACCR) WITH CABLE MARKER BALLS. SEE SHEET 101E107 FOR CABLE MARKER BALL DETAILS. SEE DETAIL 6 SHEET 101E532 FOR ACCR DETAIL.
4. INSTALL 144-STRAND SM OPTICAL GROUND WIRE (OPGW) WITH CABLE MARKER BALLS. SEE SHEET 101E107 FOR CABLE MARKER BALL DETAILS.
5. INSTALL SINGLE-PHASE UNDERGROUND RISER. BOND UNDERGROUND POWER CABLE CONCENTRIC NEUTRAL TO POLE GROUND LUG. SEE SHEET 101E507 DETAILS 4, 5, & 6 FOR TYPICAL RISER DETAILS.
6. INSTALL 15KV 1/C 4/0 AWG URD CABLE WITH CONCENTRIC NEUTRAL IN 3" CONDUIT. USE A SHALLOW TRENCH IF 3FT BURIAL DEPTH IS NOT POSSIBLE. SEE DETAIL 008E510 IN VOLUME 0.
7. INSTALL PULL BOX AND INTERCEPT 15KV (3) 1/C 4/0 AWG URD CABLES FROM POLE 4UB. PULL CABLES TO POLE 1UB, 2UB, AND 3UB (ONE PHASE PER POLE). SEE SHEET 101E532 FOR CABLE VAULT DETAILS.
8. INSTALL THREE-PHASE UNDERGROUND RISER. SEE SHEET 101E507 FOR RISER DETAILS.
9. TRENCH 15KV (3) 1/C 4/0 AWG URD CABLE WITH CONCENTRIC NEUTRAL IN (1) 4" CONDUIT FROM POLE 4UB TO CABLE VAULT. SEE SHEET 101E532 FOR TRENCHING DETAILS.
10. AVAILABLE CONSTRUCTION LIMITS ARE SHOWN IN THE LIMITS OF CONSTRUCTION TABLE.

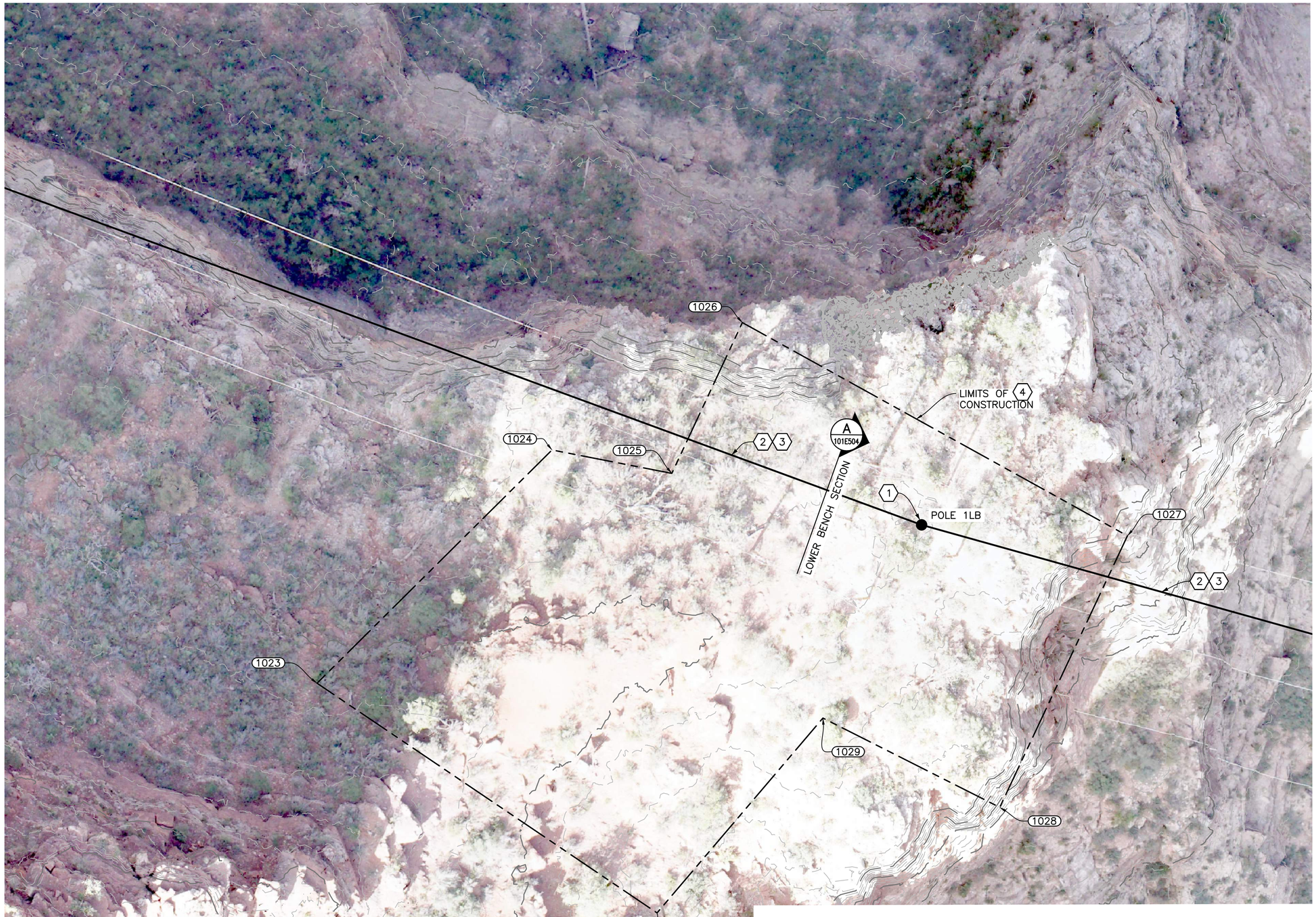
LIMITS OF CONSTRUCTION		
POINT NO.	NORTHING	EASTING
1013	13143647.09	1332099.25
1014	13143673.99	1332057.07
1015	13143832.66	1332166.00
1016	13143804.59	1332241.34
1017	13143702.72	1332407.24
1018	13143522.55	1332294.55
1019	13143560.42	1332235.18
1020	13143469.50	1332171.67
1021	13143549.45	1332036.56

20 0 20 40
SCALE OF FEET
1" = 20'

REGISTERED PROFESSIONAL ENGINEER ELECTRICITY
CERTIFICATE NO. 75014
TOU VANG
State of Colorado

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	B. PLANT CADD C. WALCHUK TECH. REVIEW: J. DOHRENDORF DATE: MAY 2026	101 E103	POWER DISTRIBUTION SITE PLAN UPPER BENCH NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752 PMIS/PKG NO. 318719 SHEET 9 OF 50

4/17/2026 Walchuk, Chris C:\Users\cwalchuk\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\ELEC\101-E103.dwg



GENERAL NOTES:

A. SEE SHEET 101E504 FOR STRUCTURE SECTION LAYOUT AT LOWER BENCH.

B. SEE SHEET 101E517 THROUGH 101E519 FOR STRUCTURAL ASSEMBLY LOAD AND DESIGN DETAILS AT LOWER BENCH.

C. OVERHEAD POWER CABLE SPLICES TO BE APPROVED BY THE CONTRACTING OFFICER REPRESENTATIVE BEFORE CONTRACTOR PROCURES SPLICE MATERIAL AND BEGINS WORK.

KEY NOTES: (X)

1. INSTALL (1) 100' SELF SUPPORTING STEEL POLE ON MICROPILE FOUNDATION APPROXIMATELY 10' FROM EXIST. OVERHEAD CONDUCTOR ALIGNMENT, SEE SHEET 101E530 FOR POLE COORDINATES. FINAL POLE AND FOUNDATION SIZE TO BE DETERMINED BY CONTRACTOR'S MANUFACTURER, BASED ON DESIGN REQUIREMENTS AND APPROVED BY CONTRACTING OFFICER REPRESENTATIVE. SEE SHEET 101E504 FOR LOWER BENCH OVERHEAD STRUCTURE LAYOUT DETAILS.

2. INSTALL (2) 37 #8 ALUMINUM CLAD STEEL MESSENGER WITH (3) 1/0 INSULATED SPACER CABLES AND SPACERS.

3. INSTALL 144-STRAND SM OPTICAL GROUND WIRE (OPGW) WITH CABLE MARKER BALLS. SEE SHEET 101E107 FOR CABLE MARKER BALL DETAILS.

4. AVAILABLE CONSTRUCTION LIMITS ARE SHOWN IN THE LIMITS OF CONSTRUCTION TABLE.

LIMITS OF CONSTRUCTION		
POINT NO.	NORTHING	EASTING
1022	13142980.77	1333567.87
1023	13143070.36	1333435.23
1024	13143161.33	1333526.90
1025	13143152.20	1333573.64
1026	13143211.01	1333601.16
1027	13143127.84	1333751.39
1028	13143021.94	1333701.59
1029	13143056.76	1333632.60

20 0 20 40 1" = 20'
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 75014 TOU VANG STATE OF COLORADO

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	B. PLANT CADD C. WALCHUK TECH. REVIEW: J. DOHRENDORF DATE: MAY 2026	101 E104	POWER DISTRIBUTION SITE PLAN LOWER BENCH NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752 PMIS/PKG NO. 318719 SHEET 10 OF 50

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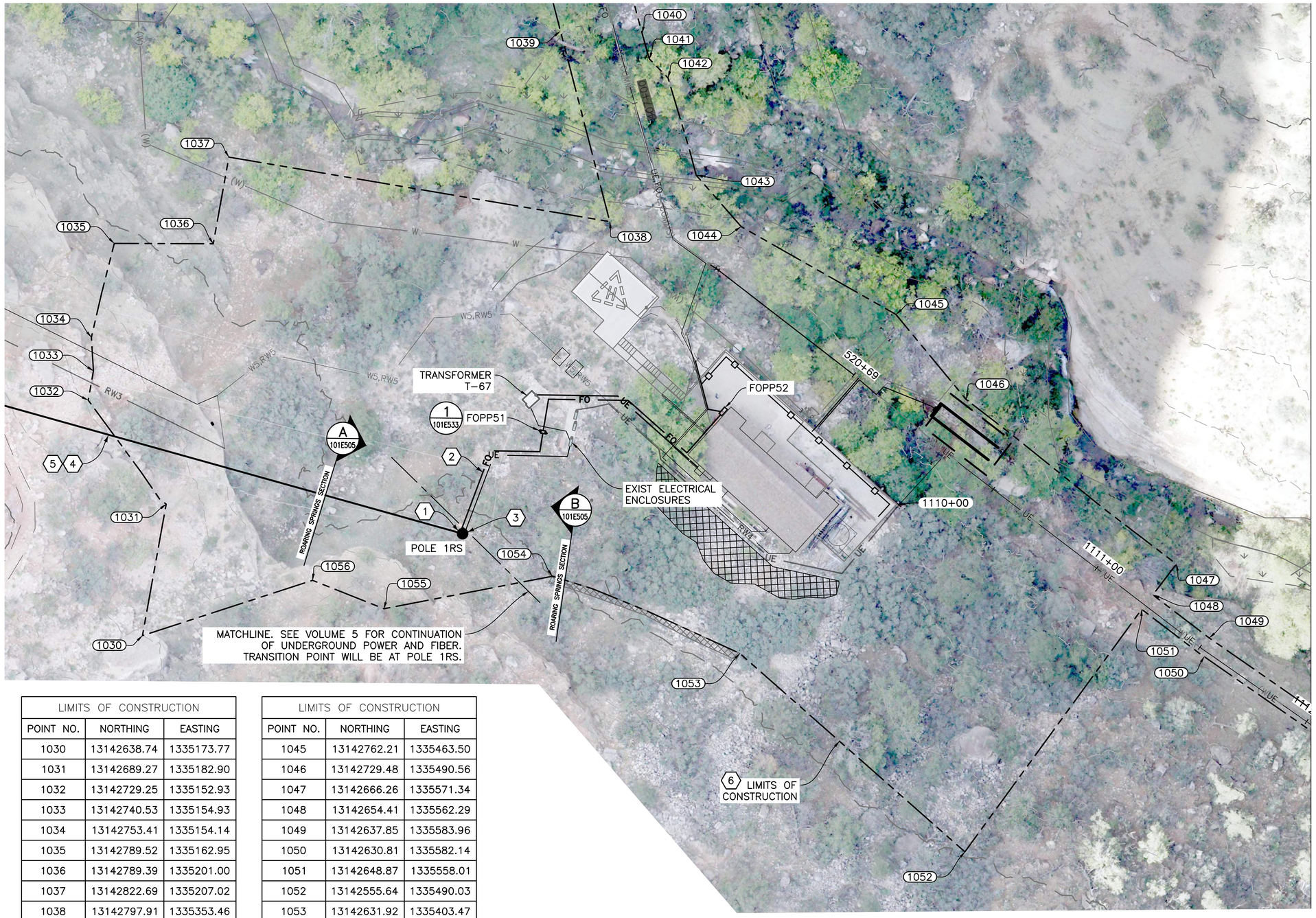

LIMITS OF CONSTRUCTION		
POINT NO.	NORTHING	EASTING
1030	13142638.74	1335173.77
1031	13142689.27	1335182.90
1032	13142729.25	1335152.93
1033	13142740.53	1335154.93
1034	13142753.41	1335154.14
1035	13142789.52	1335162.95
1036	13142789.39	1335201.00
1037	13142822.69	1335207.02
1038	13142797.91	1335353.46
1039	13142871.02	1335334.39
1040	13142870.30	1335365.91
1041	13142860.42	1335368.69
1042	13142853.16	1335375.87
1043	13142815.74	1335386.71
1044	13142795.96	1335404.03

LIMITS OF CONSTRUCTION		
POINT NO.	NORTHING	EASTING
1045	13142762.21	1335463.50
1046	13142729.48	1335490.56
1047	13142666.26	1335571.34
1048	13142654.41	1335562.29
1049	13142637.85	1335583.96
1050	13142630.81	1335582.14
1051	13142648.87	1335558.01
1052	13142555.64	1335490.03
1053	13142631.92	1335403.47
1054	13142661.57	1335331.40
1055	13142649.08	1335266.64
1056	13142660.00	1335239.17

GENERAL NOTES:

A. SEE SHEET 101E505 FOR STRUCTURE SECTION LAYOUT AT ROARING SPRINGS.

B. SEE SHEET 101E520 THROUGH 101E522 FOR STRUCTURAL ASSEMBLY LOAD AND DESIGN DETAILS AT ROARING SPRINGS.

C. OVERHEAD POWER CABLE SPLICES TO BE APPROVED BY THE CONTRACTING OFFICER REPRESENTATIVE BEFORE CONTRACTOR PROCURES SPLICE MATERIAL AND BEGINS WORK.

KEY NOTES: (X)

1. INSTALL (1) 60' SUPPORTING STEEL POLE ON MICROPILE FOUNDATION, SEE SHEET 101E530 FOR POLE COORDINATES. FINAL POLE AND FOUNDATION SIZE TO BE DETERMINED BY CONTRACTOR'S MANUFACTURER, BASED ON DESIGN REQUIREMENTS AND APPROVED BY CONTRACTING OFFICER REPRESENTATIVE. SEE SHEET 101E505 FOR ROARING SPRINGS OVERHEAD STRUCTURE LAYOUT DETAILS.
2. REFER TO VOLUME 5 FOR INSTALLATION OF UNDERGROUND MEDIUM VOLTAGE POWER CABLE AND FIBER OPTIC CABLE FROM POLE 1RS TO EXISTING ENCLOSURES.
3. INSTALL THREE-PHASE UNDERGROUND RISER AND OPTICAL GROUND WIRE RISER. SEE SHEET 101E507 FOR TYPICAL RISER DETAILS. SEE SHEET 101E533 FOR OPGW VAULT TERMINATION DETAILS.
4. INSTALL 144-STRAND SM OPTICAL GROUND WIRE (OPGW) WITH CABLE MARKER BALLS. SEE SHEET 101E107 FOR CABLE MARKER BALL DETAILS.
5. INSTALL (2) 37 #8 ALUMINUM CLAD STEEL MESSENGER WITH (3) 1/0 INSULATED SPACER CABLES AND SPACERS.
6. AVAILABLE CONSTRUCTION LIMITS ARE SHOWN IN THE LIMITS OF CONSTRUCTION TABLE.

20 0 20 40 1" = 20'
SCALE OF FEET

REGISTERED PROFESSIONAL ENGINEER OF COLORADO
CERTIFICATE NO. 75014
TOU VANG

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	B. PLANT	101 E105	POWER DISTRIBUTION SITE PLAN ROARING SPRINGS	113 182752
	CADD C. WALCHUK			PMIS/PKG NO. 318719
	TECH. REVIEW: J. DOHRENDORF		NORTH RIM - GRAND CANYON NATIONAL PARK	SHEET 11 OF 50
	DATE: MAY 2026			

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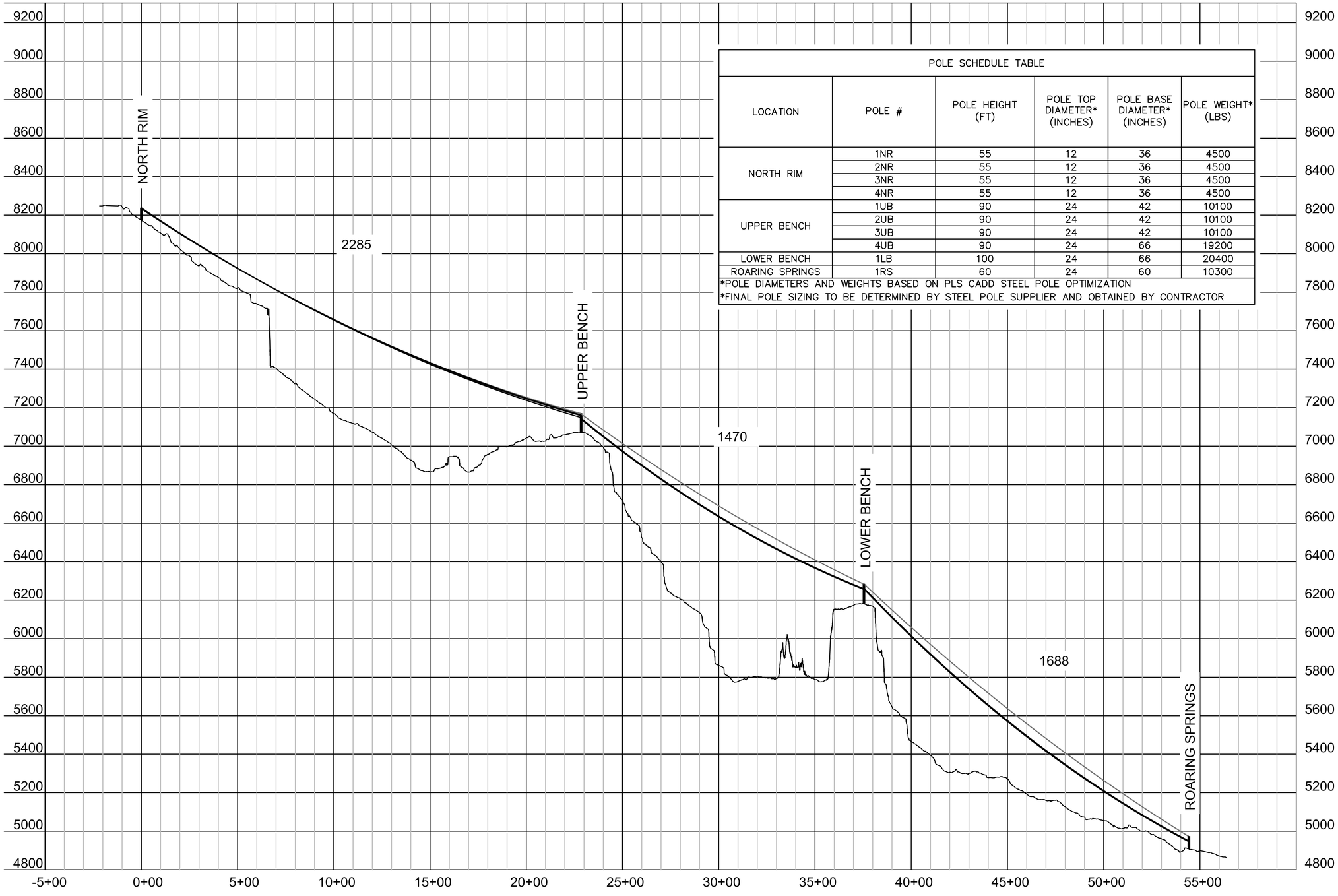

POLE SCHEDULE TABLE					
LOCATION	POLE #	POLE HEIGHT (FT)	POLE TOP DIAMETER* (INCHES)	POLE BASE DIAMETER* (INCHES)	POLE WEIGHT* (LBS)
NORTH RIM	1NR	55	12	36	4500
	2NR	55	12	36	4500
	3NR	55	12	36	4500
	4NR	55	12	36	4500
UPPER BENCH	1UB	90	24	42	10100
	2UB	90	24	42	10100
	3UB	90	24	42	10100
	4UB	90	24	66	19200
LOWER BENCH	1LB	100	24	66	20400
ROARING SPRINGS	1RS	60	24	60	10300
*POLE DIAMETERS AND WEIGHTS BASED ON PLS CADD STEEL POLE OPTIMIZATION					
*FINAL POLE SIZING TO BE DETERMINED BY STEEL POLE SUPPLIER AND OBTAINED BY CONTRACTOR					

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 75014 TOU VANG STATE OF COLORADO

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: B. PLANT CADD C. WALCHUK TECH. REVIEW: J. DOHRENDORF DATE: MAY 2026	SUB SHEET NO. 101 E106	TITLE OF SHEET POWER DISTRIBUTION POWER LINE PROFILE NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 12 OF 50

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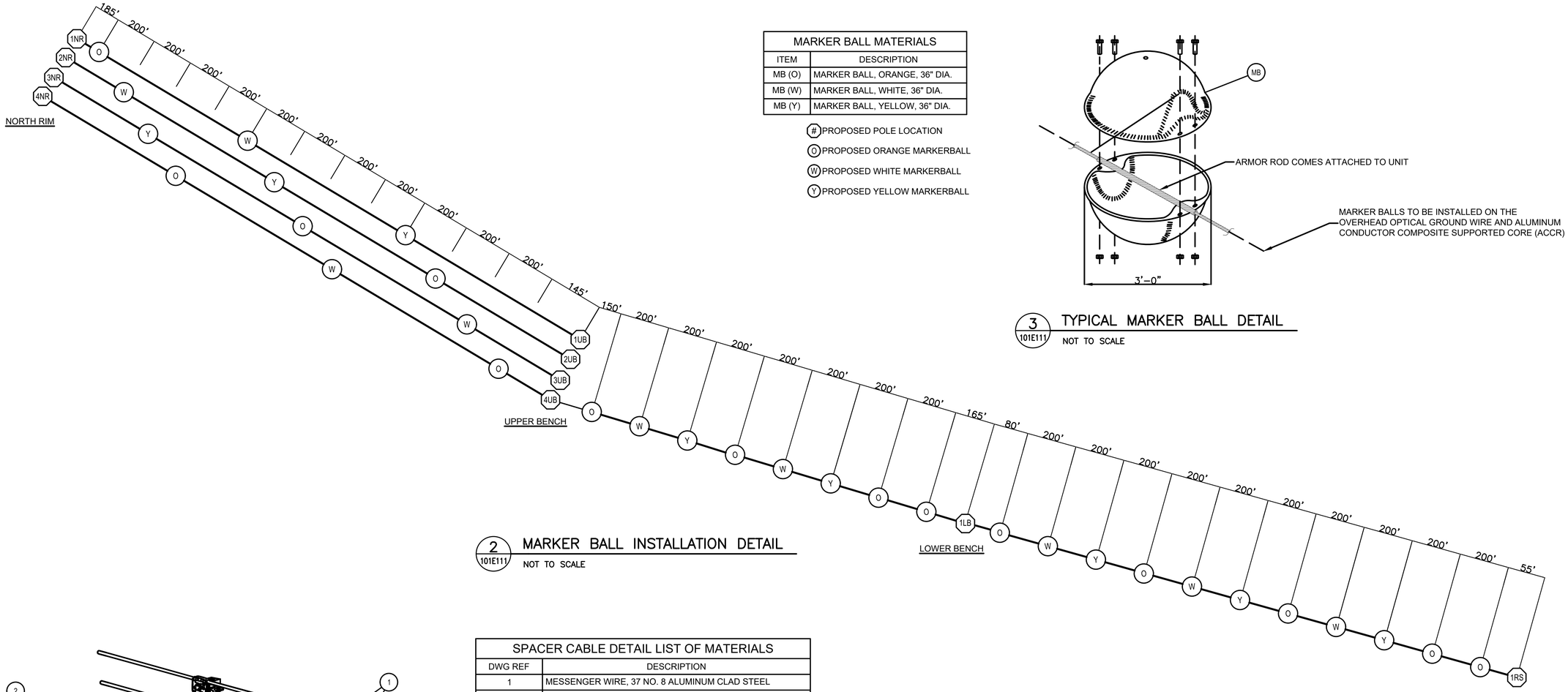


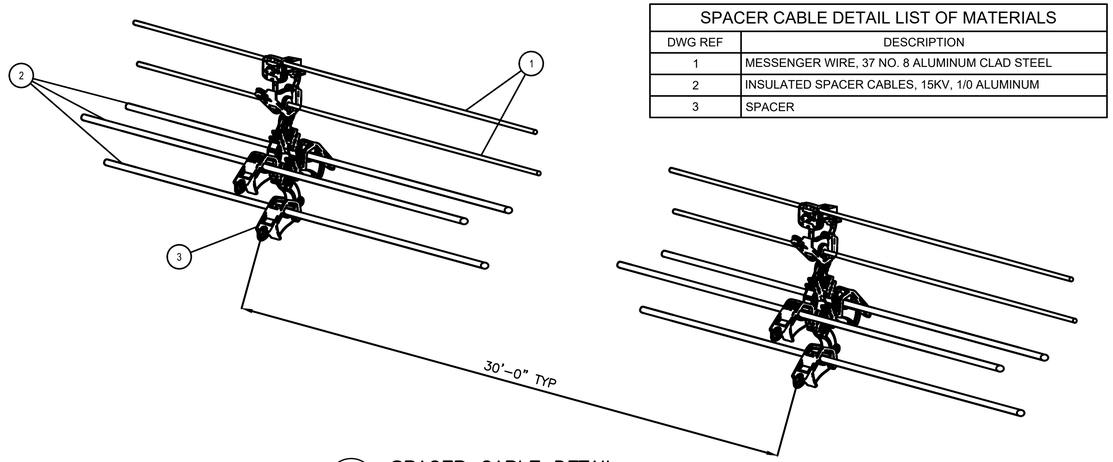
MARKER BALL MATERIALS	
ITEM	DESCRIPTION
MB (O)	MARKER BALL, ORANGE, 36" DIA.
MB (W)	MARKER BALL, WHITE, 36" DIA.
MB (Y)	MARKER BALL, YELLOW, 36" DIA.

(#) PROPOSED POLE LOCATION
(O) PROPOSED ORANGE MARKERBALL
(W) PROPOSED WHITE MARKERBALL
(Y) PROPOSED YELLOW MARKERBALL

3 / 101E111 TYPICAL MARKER BALL DETAIL
NOT TO SCALE

2 / 101E111 MARKER BALL INSTALLATION DETAIL
NOT TO SCALE

SPACER CABLE DETAIL LIST OF MATERIALS	
DWG REF	DESCRIPTION
1	MESSENGER WIRE, 37 NO. 8 ALUMINUM CLAD STEEL
2	INSULATED SPACER CABLES, 15KV, 1/0 ALUMINUM
3	SPACER



1 / 101E111 SPACER CABLE DETAIL
NOT TO SCALE

NOTE: CONTRACTOR TO CONFIRM 30 FOOT SPACING BETWEEN SPACERS WITH MANUFACTURER

REGISTERED PROFESSIONAL ENGINEER COLORADO
CERTIFICATE NO. 75014
TOU VANG
DATE SIGNED

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	B. PLANT	101 E107	POWER DISTRIBUTION POWER LINE CABLE ATTACHMENTS	113 182752
	CADD C. WALCHUK			PMIS/PKG NO. 318719
	TECH. REVIEW: J. DOHRENDORF			SHEET 13 OF 50
	DATE: MAY 2026		NORTH RIM - GRAND CANYON NATIONAL PARK	

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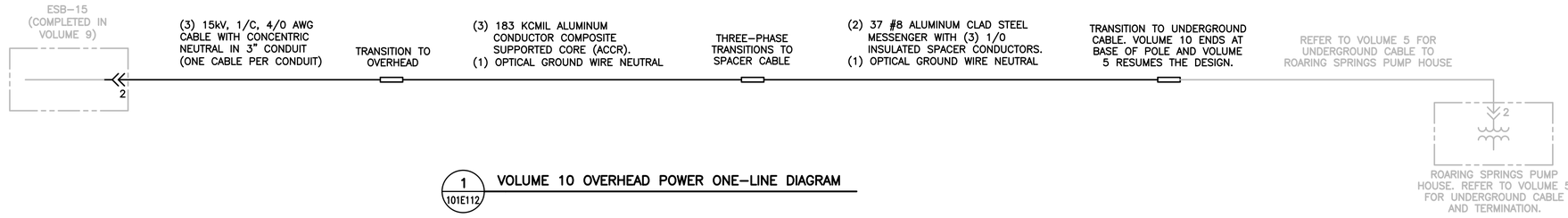

1 / 101E112 **VOLUME 10 OVERHEAD POWER ONE–LINE DIAGRAM**

2 / 101E112 **VOLUME 10 FIBER OPTIC ONE–LINE DIAGRAM**

REGISTERED PROFESSIONAL ENGINEER DEPARTMENT OF STATE — CERTIFICATE NO. 75014 TOU VANG

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: B. PLANT CADD C. WALCHUK TECH. REVIEW: J. DOHRENDORF DATE: MAY 2026	SUB SHEET NO. 101 E108	TITLE OF SHEET POWER DISTRIBUTION OVERHEAD POWER AND FIBER OPTIC ONE LINE DIAGRAMS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 14 OF 50

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ACCR_183-T114 v4.wir			183 KCMIL T114 12000# @ NESC HEAVY INITIAL														NORTH RIM to UPPER BENCH												
			SAG FT														3RD WAVE TIME SEC.												
STR #	RULING SPAN FT	AHEAD SPAN FT	-20 F	-10 F	0 F	10 F	20 F	30 F	40 F	50 F	60 F	70 F	80 F	90 F	100 F	110 F	120 F	-20 F	-10 F	0 F	10 F	20 F	30 F	40 F	50 F	60 F	70 F	80 F	90 F	100 F	110 F	120 F
NORTH RIM	2064	2282	95.59	96.22	96.87	97.49	98.11	98.73	99.35	99.98	100.55	101.20	101.82	102.43	103.05	103.66	104.27	29.24	29.34	29.44	29.53	29.62	29.72	29.81	29.90	29.99	30.09	30.18	30.27	30.36	30.45	30.54
SUB-PHASE TENS. LBS			3135	3115	3092	3072	3053	3034	3015	2997	2972	2955	2937	2920	2902	2885	2868															

DNO-13549.wir			DNO-13549 13800# @ NESC HEAVY INITIAL														NORTH RIM to UPPER BENCH												
			SAG FT														3RD WAVE TIME SEC.												
STR #	RULING SPAN FT	AHEAD SPAN FT	-20 F	-10 F	0 F	10 F	20 F	30 F	40 F	50 F	60 F	70 F	80 F	90 F	100 F	110 F	120 F	-20 F	-10 F	0 F	10 F	20 F	30 F	40 F	50 F	60 F	70 F	80 F	90 F	100 F	110 F	120 F
NORTH RIM	2071	2285	92.43	93.45	94.48	95.49	96.50	97.51	98.51	99.51	100.51	101.50	102.49	103.46	104.43	105.40	106.36	28.75	28.91	29.07	29.23	29.38	29.53	29.68	29.83	29.98	30.13	30.28	30.42	30.56	30.70	30.84
SUB-PHASE TENS. LBS			5499	5438	5379	5322	5266	5212	5159	5107	5058	5001	4953	4906	4860	4815	4772															

DNO-13549_UB.wir			DNO-13549 12700# @ NESC HEAVY INITIAL														UPPER BENCH to LOWER BENCH												
			SAG FT														3RD WAVE TIME SEC.												
STR #	RULING SPAN FT	AHEAD SPAN FT	-20 F	-10 F	0 F	10 F	20 F	30 F	40 F	50 F	60 F	70 F	80 F	90 F	100 F	110 F	120 F	-20 F	-10 F	0 F	10 F	20 F	30 F	40 F	50 F	60 F	70 F	80 F	90 F	100 F	110 F	120 F
UPPER BENCH	1261	1471	37.84	38.72	39.60	40.50	41.40	42.30	43.20	44.11	45.05	45.92	46.82	47.73	48.63	49.53	50.42	18.40	18.61	18.82	19.03	19.24	19.45	19.66	19.86	20.07	20.27	20.46	20.66	20.86	21.05	21.24
SUB-PHASE TENS. LBS			5884	5751	5622	5498	5379	5265	5154	5049	4953	4842	4748	4658	4571	4488	4408															

DNO-13549_LB.wir			DNO-13549 12900# @ NESC HEAVY INITIAL														LOWER BENCH to ROARING SPRINGS												
			SAG FT														3RD WAVE TIME SEC.												
STR #	RULING SPAN FT	AHEAD SPAN FT	-20 F	-10 F	0 F	10 F	20 F	30 F	40 F	50 F	60 F	70 F	80 F	90 F	100 F	110 F	120 F	-20 F	-10 F	0 F	10 F	20 F	30 F	40 F	50 F	60 F	70 F	80 F	90 F	100 F	110 F	120 F
LOWER BENCH	1331	1686	65.95	67.11	68.28	69.44	70.59	71.75	72.90	74.04	75.22	76.31	77.44	78.56	79.67	80.78	81.88	24.29	24.50	24.71	24.92	25.13	25.33	25.54	25.73	25.94	26.13	26.32	26.51	26.69	26.88	27.06
SUB-PHASE TENS. LBS			4810	4726	4646	4568	4493	4422	4352	4285	4222	4150	4089	4031	3974	3919	3866															

37#8_alumoweld.wir			37 No. 8 AW 17025# @ NESC HEAVY INITIAL														UPPER BENCH to LOWER BENCH												
			SAG FT														3RD WAVE TIME SEC.												
STR #	RULING SPAN FT	AHEAD SPAN FT	-20 F	-10 F	0 F	10 F	20 F	30 F	40 F	50 F	60 F	70 F	80 F	90 F	100 F	110 F	120 F	-20 F	-10 F	0 F	10 F	20 F	30 F	40 F	50 F	60 F	70 F	80 F	90 F	100 F	110 F	120 F
UPPER BENCH	1259	1470	52.38	53.19	54.00	54.81	55.60	56.40	57.18	57.96	58.74	59.51	60.28	61.11	61.87	62.62	63.36	21.64	21.81	21.98	22.14	22.30	22.46	22.62	22.77	22.92	23.07	23.22	23.38	23.52	23.67	23.81
SUB-PHASE TENS. LBS			8416	8287	8163	8043	7927	7815	7707	7603	7502	7405	7311	7211	7122	7037	6954															

37#8_alumoweld.wir			37 No. 8 AW 16900# @ NESC HEAVY INITIAL														LOWER BENCH to ROARING SPRINGS												
			SAG FT														3RD WAVE TIME SEC.												
STR #	RULING SPAN FT	AHEAD SPAN FT	-20 F	-10 F	0 F	10 F	20 F	30 F	40 F	50 F	60 F	70 F	80 F	90 F	100 F	110 F	120 F	-20 F	-10 F	0 F	10 F	20 F	30 F	40 F	50 F	60 F	70 F	80 F	90 F	100 F	110 F	120 F
LOWER BENCH	1330	1684	89.16	90.11	91.06	92.00	92.93	93.85	94.77	95.69	96.60	97.50	98.39	99.40	100.28	101.16	102.03	28.24	28.39	28.54	28.69	28.83	28.97	29.12	29.26	29.39	29.53	29.67	29.82	29.95	30.08	30.21
SUB-PHASE TENS. LBS			7047	6972	6899	6828	6759	6692	6627	6563	6501	6441	6382	6317	6261	6206	6152															

37no8AW_1_0AAC_15KV_ 3PH_RTL-15DM_Spaced bundled			37 No. 8 AW WITH 3PH 1/0 AAC SPACER CABLE 35475# @ NESC HEAVY INITIAL														UPPER BENCH to LOWER BENCH												
			SAG FT														3RD WAVE TIME SEC.												
STR #	RULING SPAN FT	AHEAD SPAN FT	-20 F	-10 F	0 F	10 F	20 F	30 F	40 F	50 F	60 F	70 F	80 F	90 F	100 F	110 F	120 F	-20 F	-10 F	0 F	10 F	20 F	30 F	40 F	50 F	60 F	70 F	80 F	90 F	100 F	110 F	120 F
UPPER BENCH	1259	1470	57.47	58.19	58.90	59.61	60.32	61.02	61.73	62.43	63.12	63.81	64.50	65.26	65.94	66.62	67.30	22.67	22.81	22.95	23.09	23.23	23.36	23.50	23.63	23.76	23.89	24.02	24.16	24.29	24.41	24.53
SUB-PHASE TENS. LBS			11673	11529	11389	11252	11120	10991	10865	10743	10625	10509	10397	10275	10168	10064	9963															

37no8AW_1_0AAC_15KV_ 3PH_RTL-15DM_Spaced bundled			37 No. 8 AW WITH 3PH 1/0 AAC SPACER CABLE 37825# @ NESC HEAVY INITIAL														LOWER BENCH to ROARING SPRINGS												
			SAG FT														3RD WAVE TIME SEC.												
STR #	RULING SPAN FT	AHEAD SPAN FT	-20 F	-10 F	0 F	10 F	20 F	30 F	40 F	50 F	60 F	70 F	80 F	90 F	100 F	110 F	120 F	-20 F	-10 F	0 F	10 F	20 F	30 F	40 F	50 F	60 F	70 F	80 F	90 F	100 F	110 F	120 F
LOWER BENCH	1329	1684	94.76	95.63	96.49	97.34	98.19	99.04	99.88	100.72	101.55	102.38	103.21	104.14	104.96	105.78	106.59	29.11	29.25	29.38	29.51	29.64	29.76	29.89	30.01	30.14	30.26	30.38	30.52	30.64	30.76	30.88
SUB-PHASE TENS. LBS			10089	9997	9907	9819	9734	9650	9568	9488	9409	9333	9257	9174	9102	9031	8962															

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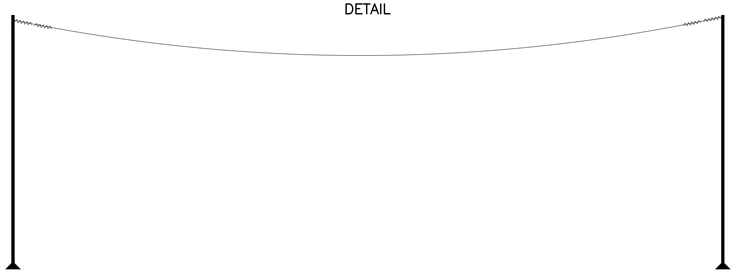
REGISTERED PROFESSIONAL ENGINEER DEPARTMENT OF DEFENSE CERTIFICATE NO. 75014 TOU VANG

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	B. PLANT C. WALCHUK TECH. REVIEW: J. DOHRENDORF DATE: MAY 2026	101 E109	POWER DISTRIBUTION CONDUCTOR SAGGING TABLES NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752 PMIS/PKG NO. 318719 SHEET 15 OF 50

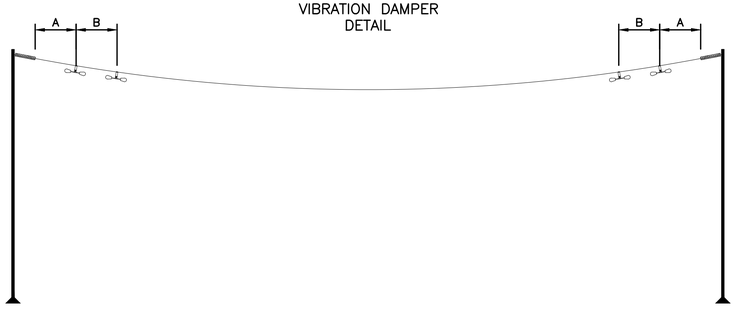
OPGW & MESSENGER

VIBRATION DAMPER
DETAIL



COMPOSITE CORE

VIBRATION DAMPER
DETAIL



GENERAL NOTES:

A. QUANTITY AND INSTALLATION DISTANCES (A&B), TO BE DETERMINED BY MANUFACTURER.

B. CONTRACTOR TO INSTALL VIBRATION DAMPERS ON ALL COMPOSITE CORE SPANS PER CONDUCTOR AND DAMPER MANUFACTURER RECOMMENDATIONS.

C. CONTRACTOR TO INSTALL VIBRATION DAMPERS ON ALL OPGW SPANS PER FIBER MANUFACTURER AND DAMPER MANUFACTURER RECOMMENDATIONS.

D. CONTRACTOR TO INSTALL VIBRATION DAMPERS ON ALL MESSENGER SPANS PER SPACER CABLE MANUFACTURER AND DAMPER MANUFACTURER RECOMMENDATIONS.

4/17/2026 Walchuk, Chris C:\Users\Cwalchuk\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\ELEC\101-E110.dwg

REGISTERED PROFESSIONAL ENGINEER ELECTRICAL
CERTIFICATE NO.
75014
TOU
VANG

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: B. PLANT CADD C. WALCHUK TECH. REVIEW: J. DOHRENDORF DATE: MAY 2026	SUB SHEET NO. 101 E110	TITLE OF SHEET POWER DISTRIBUTION ATTACHMENTS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 16 OF 50

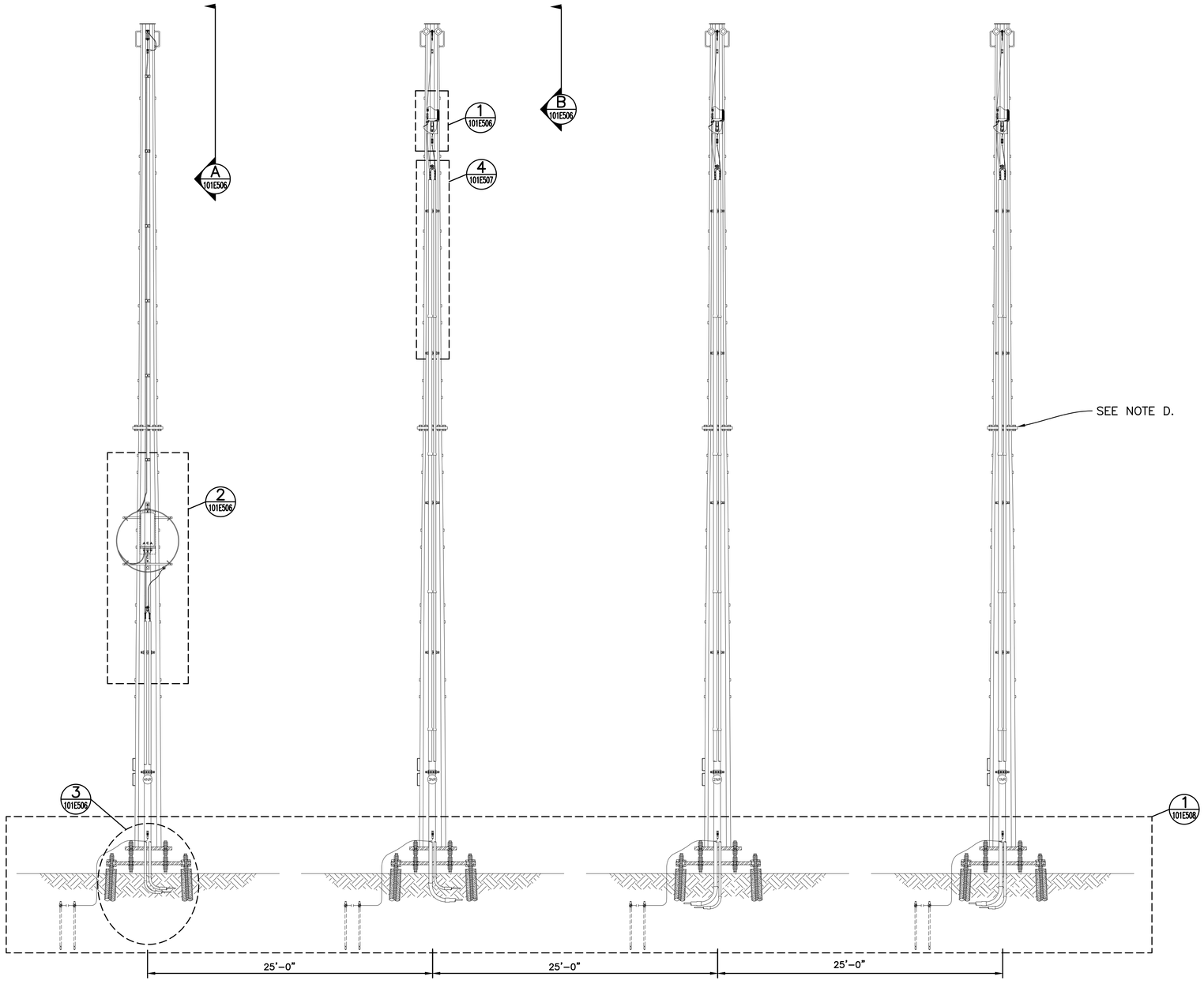
GENERAL NOTES:

A. SEE SHEET 101E506, 101E507, AND 101E508 FOR STRUCTURE DETAILS AND SECTION VIEWS.

B. SEE SHEETS 101E509 THROUGH 101E511 FOR STRUCTURAL ASSEMBLY LOAD AND DESIGN DETAILS AT THE NORTH RIM.

C. SEE SHEET 101E102 FOR NORTH RIM SITE PLAN.

D. ALL POLE JOINTS TO BE FLANGE JOINTS. QUANTITY & LOCATION TO BE DETERMINED BY POLE MANUFACTURER.

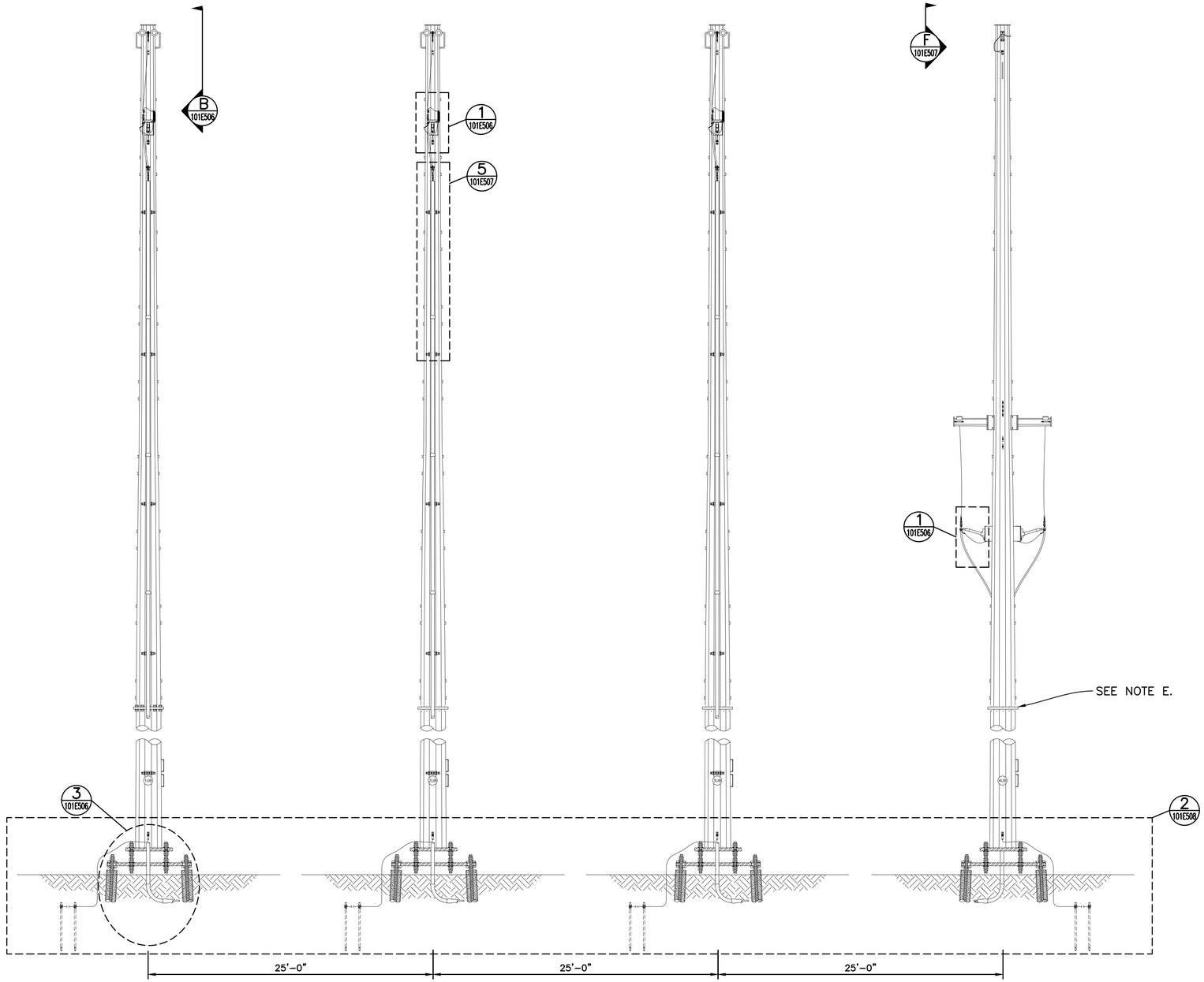


A 101E102 NORTH RIM SECTION VIEW (LOOKING NORTHWEST)
NOT TO SCALE

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 75014 TOU VANG STATE OF COLORADO

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: B. PLANT DRAWN: C. WALCHUK TECH. REVIEW: J. DOHRENDORF DATE: MAY 2026	SUB SHEET NO. 101 E501	TITLE OF SHEET POWER DISTRIBUTION STRUCTURE DETAIL NORTH RIM NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 17 OF 50

4/17/2026 Walchuk, Chris C:\Users\cwalchuk\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\ELEC\101-E501.dwg



A 101E103 UPPER BENCH SECTION VIEW (LOOKING SOUTHEAST)

NOT TO SCALE

GENERAL NOTES:

A. SEE SHEET 101E503 FOR SECTION VIEW OF UPPER BENCH STRUCTURE LOOKING NORTH WEST.

B. SEE SHEET 101E506, 101E507, AND 101E508 FOR STRUCTURE DETAILS AND SECTION VIEWS.

C. SEE SHEETS 101E512 THROUGH 101E515 FOR STRUCTURAL ASSEMBLY LOAD AND DESIGN DETAILS AT THE UPPER BENCH.

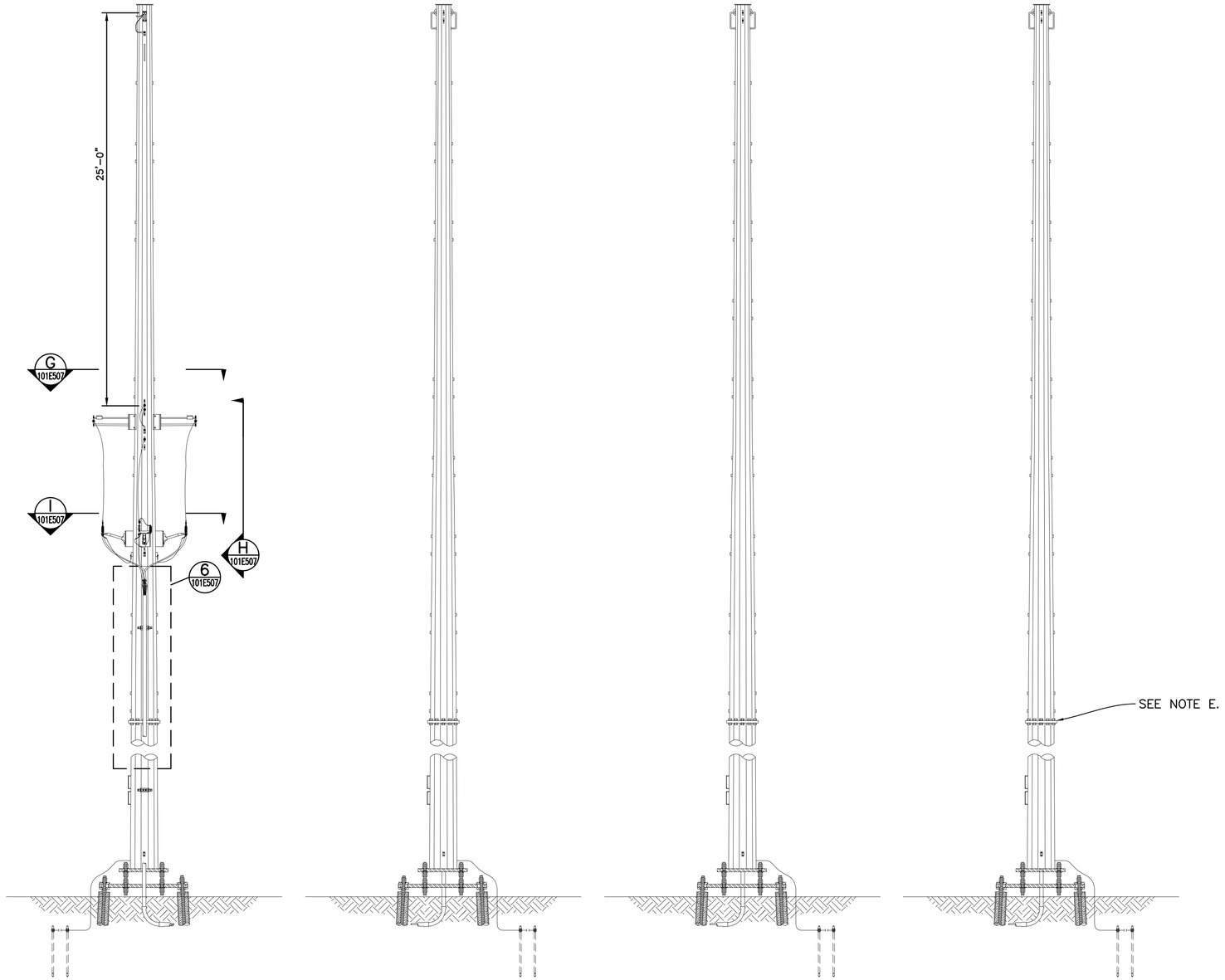
D. SEE SHEET 101E103 FOR UPPER BENCH SITE PLAN.

E. ALL POLE JOINTS TO BE FLANGE JOINTS. QUANTITY & LOCATION TO BE DETERMINED BY POLE MANUFACTURER.

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 75014 TOU VANG STATE OF OHIO

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: B. PLANT DRAWN: C. WALCHUK TECH. REVIEW: J. DOHRENDORF DATE: MAY 2026	SUB SHEET NO. 101 E502	TITLE OF SHEET POWER DISTRIBUTION STRUCTURE DETAIL UPPER BENCH NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 18 OF 50

4/17/2026 Walchuk, Chris C:\Users\Cwalchuk\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\ELEC\101-E502.dwg



B 101E103 UPPER BENCH SECTION VIEW (LOOKING NORTHWEST)
NOT TO SCALE

GENERAL NOTES:

A. SEE SHEET 101E502 FOR SECTION VIEW OF UPPER BENCH STRUCTURE LOOKING SOUTH EAST.

B. SEE SHEET 101E506 & 101E507 FOR STRUCTURE DETAILS AND SECTION VIEWS.

C. SEE SHEETS 101E512 THROUGH 101E515 FOR STRUCTURAL ASSEMBLY LOAD AND DESIGN DETAILS AT THE UPPER BENCH.

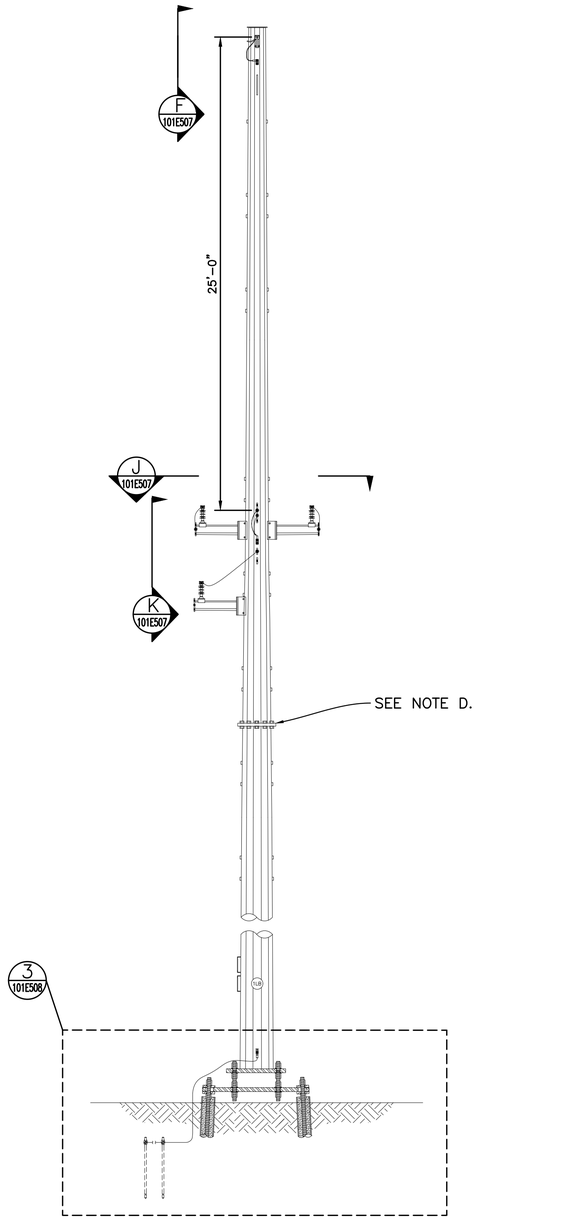
D. SEE SHEET 101E103 FOR UPPER BENCH SITE PLAN.

E. ALL POLE JOINTS TO BE FLANGE JOINTS. QUANTITY & LOCATION TO BE DETERMINED BY POLE MANUFACTURER.

REGISTERED PROFESSIONAL ENGINEER OF THE STATE OF COLORADO
CERTIFICATE NO. 75014
TOU VANG

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: B. PLANT CADD C. WALCHUK TECH. REVIEW: J. DOHRENDORF DATE: MAY 2026	SUB SHEET NO. 101 E503	TITLE OF SHEET POWER DISTRIBUTION STRUCTURE DETAIL UPPER BENCH NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 19 OF 50

4/17/2026 Walchuk, Chris C:\Users\cwalchuk\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\ELEC\101-E503.dwg



A 101E104 LOWER BENCH SECTION VIEW (LOOKING NORTHWEST)
NOT TO SCALE

GENERAL NOTES:

A. SEE SHEET 101E506, 101E507, AND 101E508 FOR STRUCTURE DETAILS AND SECTION VIEWS.

B. SEE SHEETS 101E516 AND 101E517 FOR STRUCTURAL ASSEMBLY LOAD AND DESIGN DETAILS AT THE LOWER BENCH.

C. SEE SHEET 101E104 FOR LOWER BENCH SITE PLAN.

D. ALL POLE JOINTS TO BE FLANGE JOINTS. QUANTITY & LOCATION TO BE DETERMINED BY POLE MANUFACTURER.

REGISTERED PROFESSIONAL ENGINEER DEPARTMENT OF THE INTERIOR
CERTIFICATE NO. 75014
TOU VANG
Date Signed

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: B. PLANT DRAWN: C. WALCHUK TECH. REVIEW: J. DOHRENDORF DATE: MAY 2026	SUB SHEET NO. 101 E504	TITLE OF SHEET POWER DISTRIBUTION STRUCTURE DETAIL LOWER BENCH NORTH RIM - GRAND CANYON NATIONAL PARK

DRAWING NO.	113
	182752
PMIS/PKG NO.	318719
SHEET	20 OF 50

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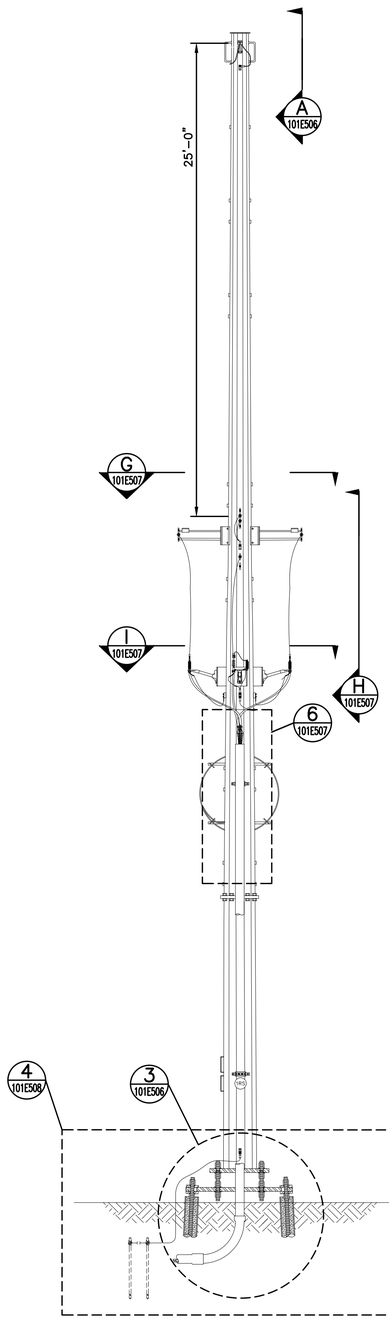
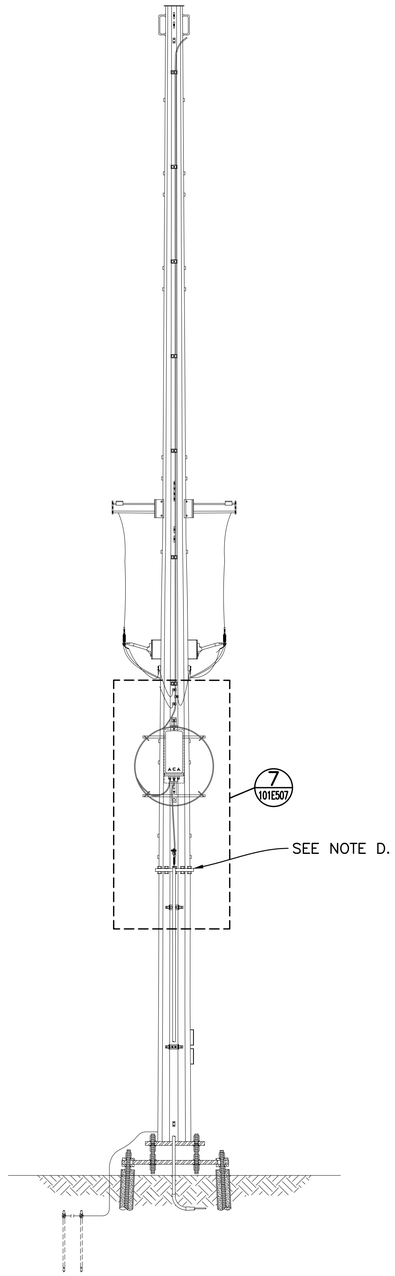


A 101E105 ROARING SPRINGS SECTION VIEW (LOOKING SOUTHEAST)
NOT TO SCALE



B 101E105 ROARING SPRINGS SECTION VIEW (LOOKING NORTHEAST)
NOT TO SCALE

GENERAL NOTES:

A. SEE SHEET 101E506, 101E507, AND 101E508 FOR STRUCTURE DETAILS AND SECTION VIEWS.

B. SEE SHEETS 101E518 THROUGH 101E520 FOR STRUCTURAL ASSEMBLY LOAD AND DESIGN DETAILS AT THE UPPER BENCH.

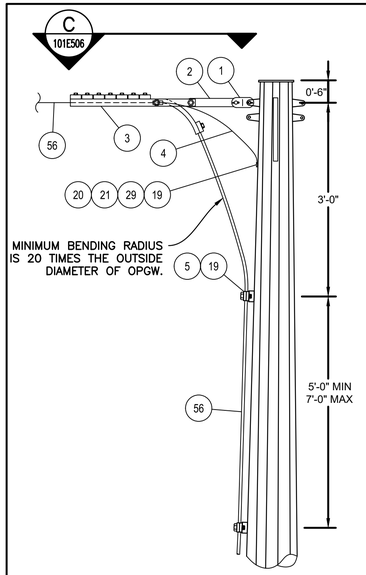
C. SEE SHEET 101E105 FOR UPPER BENCH SITE PLAN.

D. ALL POLE JOINTS TO BE FLANGE JOINTS. QUANTITY & LOCATION TO BE DETERMINED BY POLE MANUFACTURER.

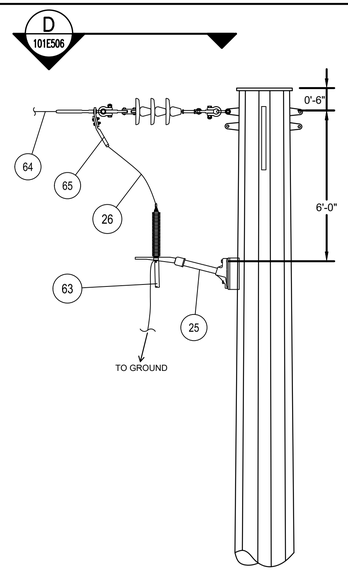
REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 75014 TOU VANG DATE Signed

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: B. PLANT DRAWN: C. WALCHUK TECH. REVIEW: J. DOHRENDORF DATE: MAY 2026	SUB SHEET NO. 101 E505	TITLE OF SHEET POWER DISTRIBUTION STRUCTURE DETAIL ROARING SPRINGS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 21 OF 50

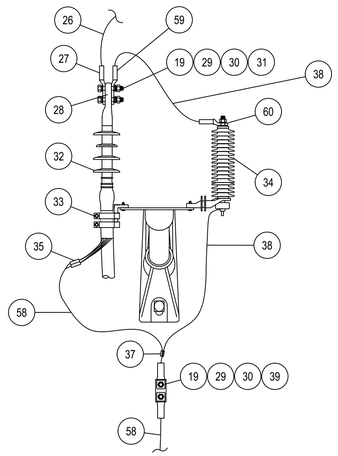
4/17/2026 Vang, Tou C:\Users\TVANG\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\ELEC\101-E505.dwg



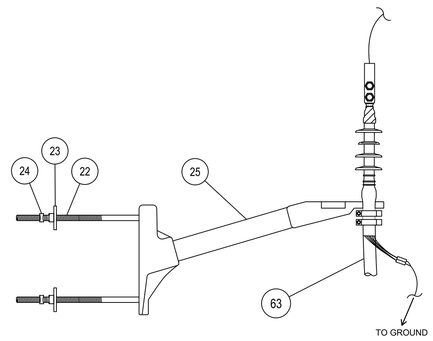
A 101E506 VIEW: OPGW NEUTRAL DEAD END ATTACHMENT DETAIL
NOT TO SCALE



B 101E506 VIEW: PRIMARY ATTACHMENT DETAIL
NOT TO SCALE



FRONT SIDE VIEW



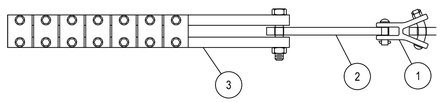
LEFT SIDE VIEW

1 101E506 DETAIL: TERMINATION AND ARRESTER (FRONT VIEW)
NOT TO SCALE

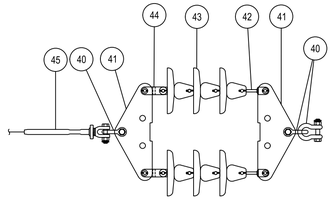
GENERAL NOTES:

A. SEE SHEET 101E501 THROUGH 101E505 FOR STRUCTURE DETAILS AT THE NORTH RIM, UPPER BENCH, LOWER BENCH, AND ROARING SPRINGS.

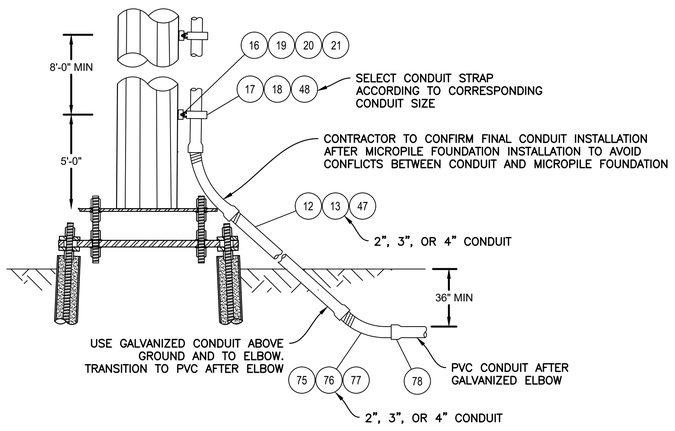
B. SEE SHEET 101E534 FOR MATERIAL UNITS.



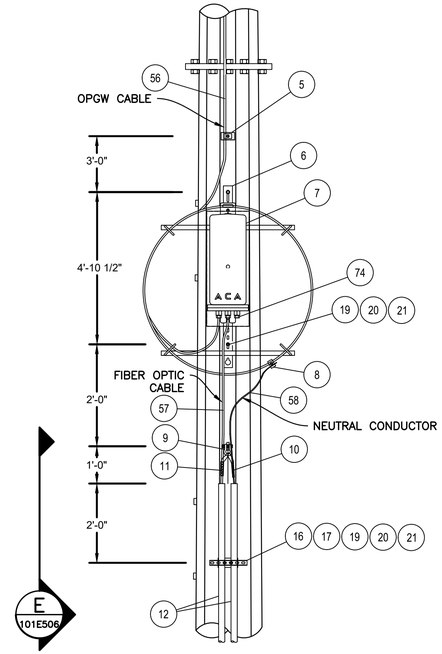
C 101E506 VIEW: OPGW DEADEND GRIP (TOP VIEW)
NOT TO SCALE



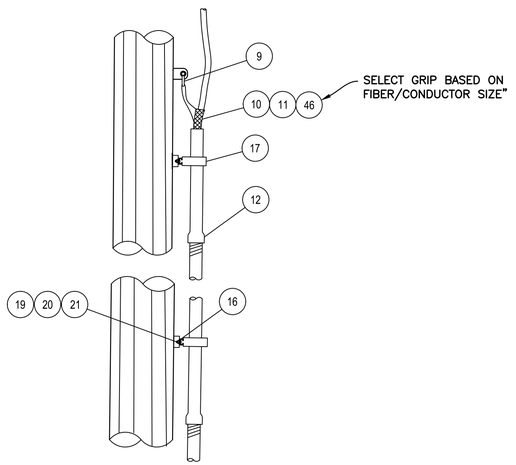
D 101E506 VIEW: PRIMARY ATTACHMENT DETAIL (TOP VIEW)
NOT TO SCALE



3 101E506 DETAIL: UNDERGROUND RISER TRANSITION
NOT TO SCALE



2 101E506 DETAIL: OPGW TRANSITION TO FIBER OPTIC CABLE
NOT TO SCALE

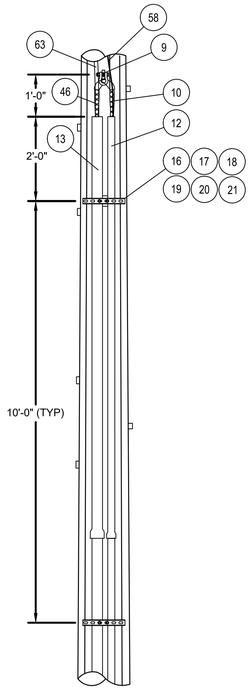


E 101E506 VIEW: RISER ATTACHMENT
NOT TO SCALE

REGISTERED PROFESSIONAL ENGINEER DEPARTMENT OF ARIZONA
CERTIFICATE NO. 75014
TOU VANG
Date Signed

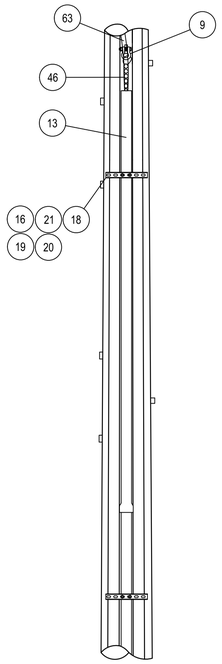
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: B. PLANT DRAWN: C. WALCHUK TECH. REVIEW: J. DOHRENDORF DATE: MAY 2026	SUB SHEET NO. 101 E506	TITLE OF SHEET POWER DISTRIBUTION STRUCTURE DETAILS ASSEMBLY ATTACHMENTS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 22 OF 50

4/17/2026 Walchuk, Chris C:\Users\cwalchuk\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\ELEC\101-E506.dwg



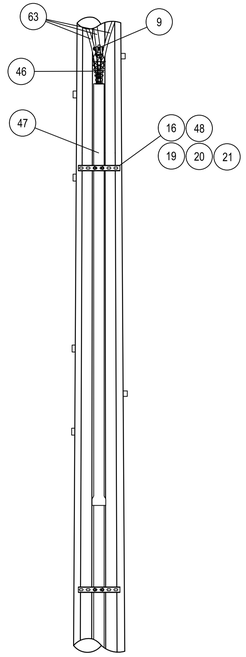
4 101E507 DETAIL: TYPICAL DOUBLE RISER

NOT TO SCALE



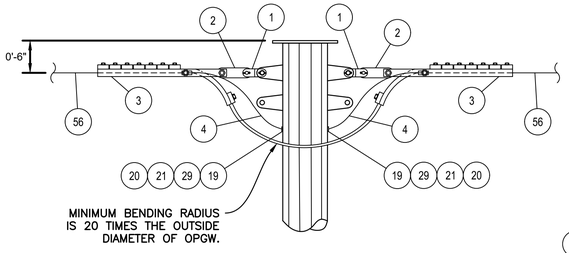
5 101E507 DETAIL: TYPICAL SINGLE RISER

NOT TO SCALE



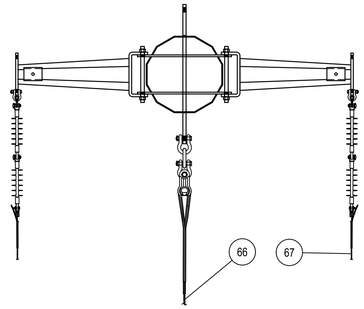
6 101E507 DETAIL: TYPICAL THREE–PHASE RISER

NOT TO SCALE



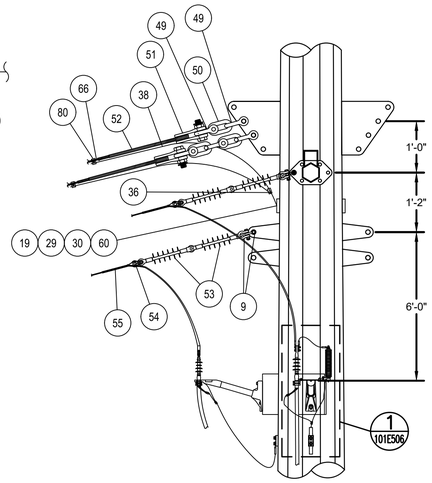
F 101E507 VIEW: OPGW NEUTRAL DOUBLE DEADEND ATTACHMENT DETAIL

NOT TO SCALE



G 101E507 VIEW: TYPICAL SPACER CABLE DEADEND ATTACHMENT DETAIL (TOP VIEW)

NOT TO SCALE



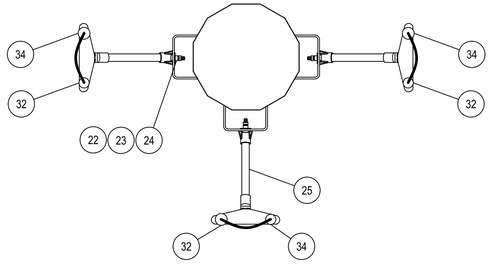
H 101E507 VIEW: TYPICAL SPACER CABLE DEADEND ATTACHMENT DETAIL (SIDE VIEW)

NOT TO SCALE

GENERAL NOTES:

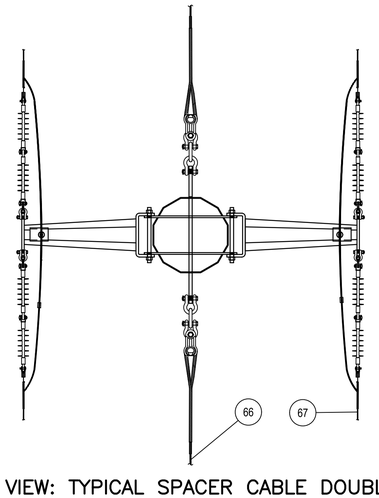
A. SEE SHEET 101E501 THROUGH 101E505 FOR STRUCTURE SECTION LAYOUTS AT THE NORTH RIM, UPPER BENCH, LOWER BENCH, AND ROARING SPRINGS.

B. SEE SHEET 101E534 FOR MATERIAL UNITS.



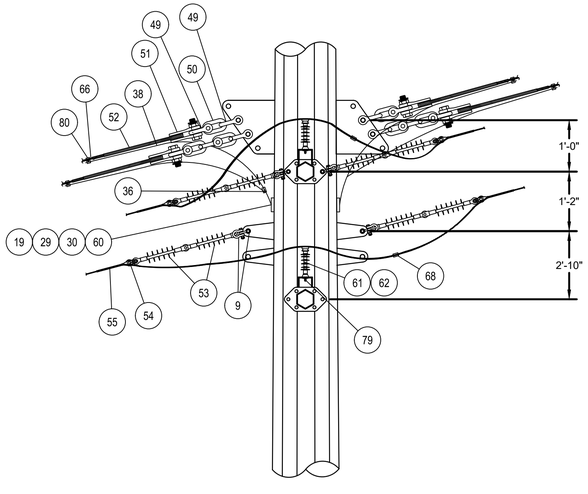
I 101E507 VIEW: TYPICAL THREE–PHASE TERMINATION AND ARRESTER MOUNTING DETAIL (TOP VIEW)

NOT TO SCALE



J 101E507 VIEW: TYPICAL SPACER CABLE DOUBLE DEADEND ATTACHMENT DETAIL (TOP VIEW)

NOT TO SCALE



K 101E507 VIEW: TYPICAL SPACER CABLE DOUBLE DEADEND ATTACHMENT DETAIL (SIDE VIEW)

NOT TO SCALE

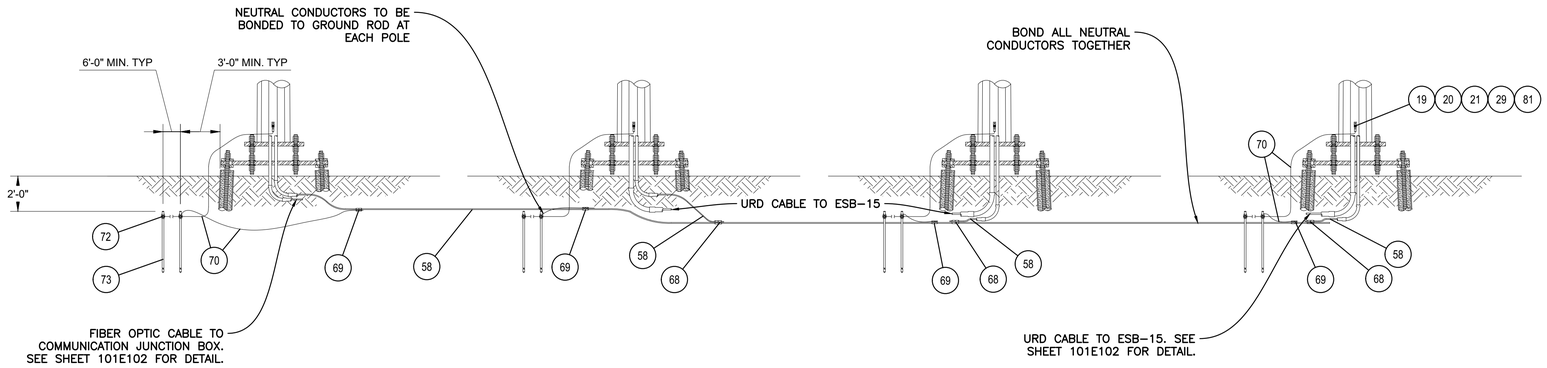
7 101E507 DETAIL: OPGW TRANSITION TO FIBER OPTIC CABLE

NOT TO SCALE

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 75014 TOU VANG

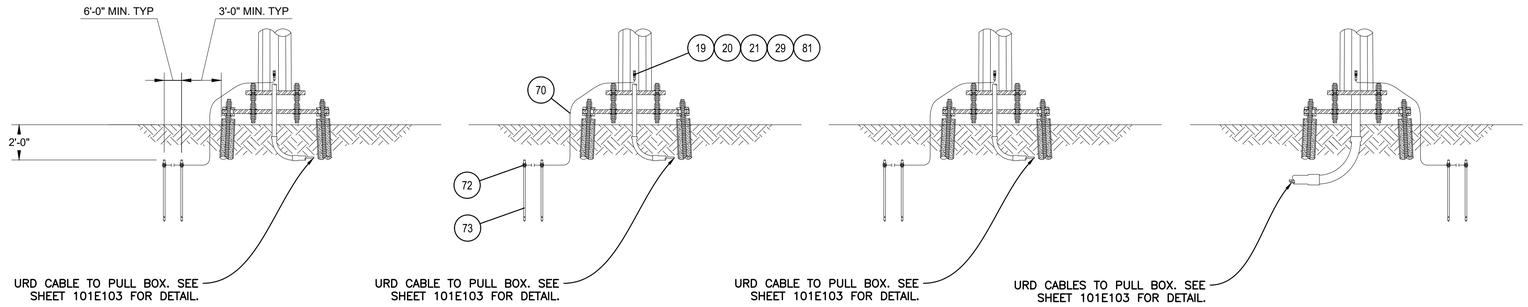
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: B. PLANT DRAWN: C. WALCHUK TECH. REVIEW: J. DOHRENDORF DATE: MAY 2026	SUB SHEET NO. 101 E507	TITLE OF SHEET POWER DISTRIBUTION STRUCTURE DETAILS ASSEMBLY ATTACHMENTS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 23 OF 50

4/17/2026 Walchuk, Chris C:\Users\Cwalchuk\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\ELEC\101-E507.dwg



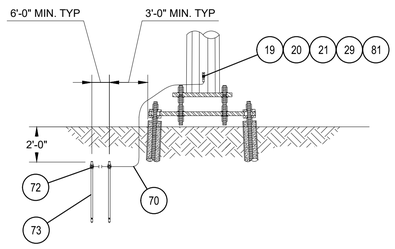
1 101E501 NORTH RIM GROUNDING DETAIL

NOT TO SCALE



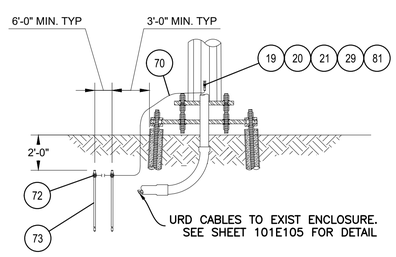
2 101E502 UPPER BENCH GROUNDING DETAIL

NOT TO SCALE



3 101E504 LOWER BENCH GROUNDING DETAIL

NOT TO SCALE



4 101E505 ROARING SPRINGS GROUNDING DETAIL

NOT TO SCALE

GENERAL NOTES

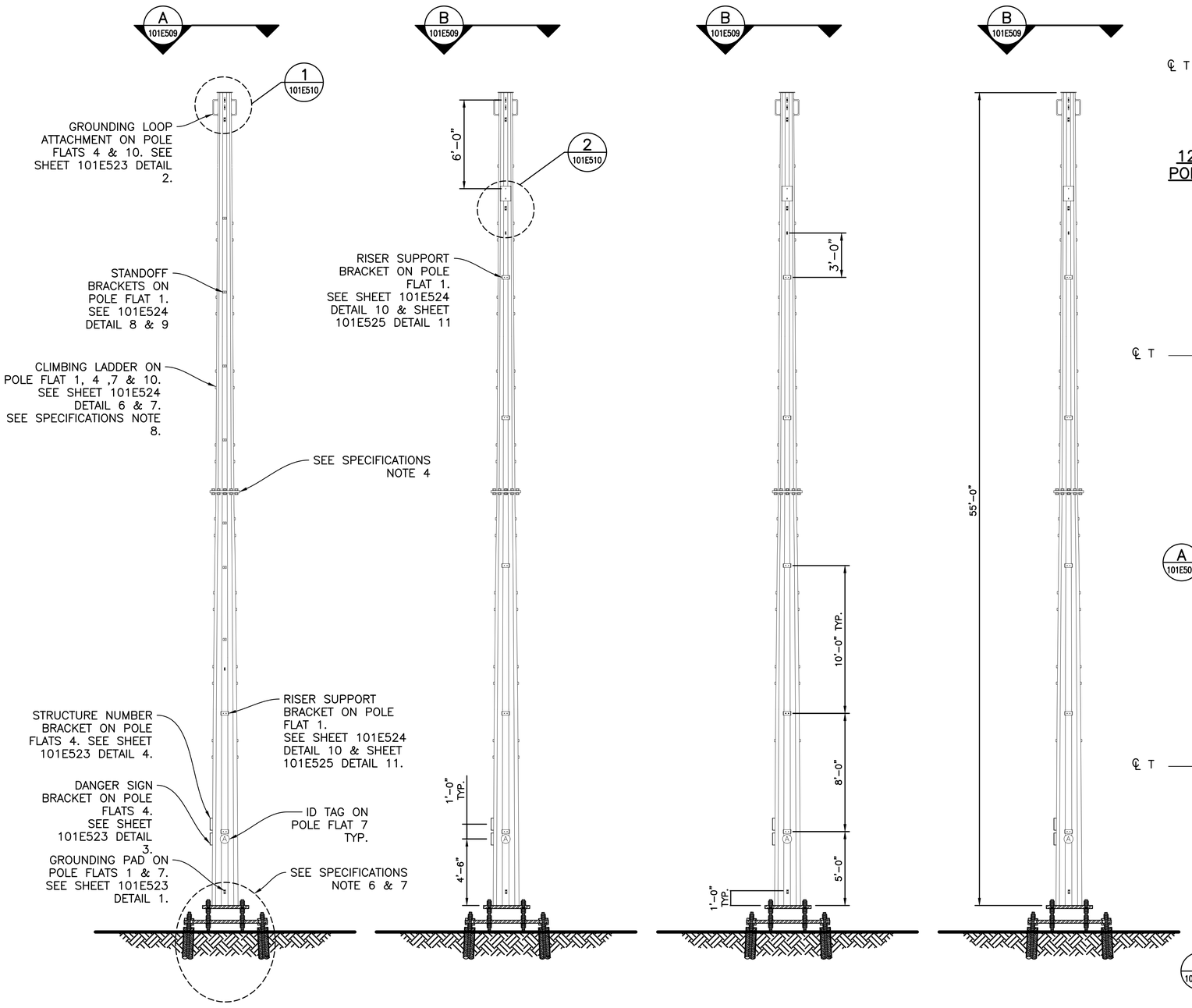
A. INSTALL GROUND RODS PARALLEL WITH CONDUCTORS.

B. STEEL POLES TO BE BONDED TO MICROPILES. MICROPILES TO BE USED AS ADDITIONAL GROUNDING ELECTRODE.

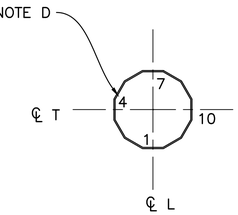
REGISTERED PROFESSIONAL ENGINEER OF COLORADO
CERTIFICATE NO. 75014
TOU VANG

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: B. PLANT DRAFTED C. WALCHUK TECH. REVIEW: J. DOHRENDORF DATE: MAY 2026	SUB SHEET NO. 101 E508	TITLE OF SHEET POWER DISTRIBUTION STRUCTURE DETAIL GROUNDING NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 24 OF 50

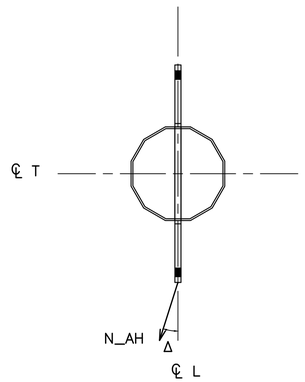
4/17/2026 Walchuk, Chris C:\Users\cwalchuk\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\ELEC\101-E508.dwg



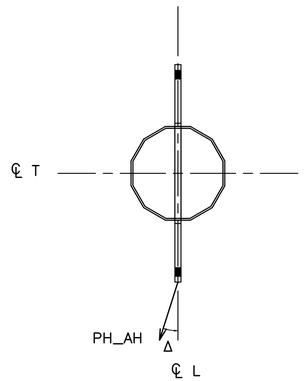
NORTH RIM STRUCTURE
LOOKING NORTHWEST



12 MULTISIDED STEEL POLE FLAT DESIGNATION



A 101E509 VIEW: NEUTRAL ORIENTATION
NOT TO SCALE



B 101E509 VIEW: CONDUCTOR ORIENTATION
NOT TO SCALE

GENERAL NOTES

A. MAXIMUM NEUTRAL WIRE VANG THICKNESS = 3/4"
B. MAXIMUM CONDUCTOR VANG THICKNESS = 3/4", UNLESS NOTED OTHERWISE
C. LOAD TREE VALUES ARE ULTIMATE, INCLUDE OVERLOAD FACTORS, AND ARE SHOWN IN POUNDS.
D. POLE FLAT DESIGNATIONS REFER TO SIDE OF THE POLE WHERE ATTACHMENTS ARE LOCATED

DESIGN CRITERIA

LOADING PARAMETERS FOR: NORTH RIM POLES
(N_AH) NEUTRAL WIRE IS 144 FIBER OPGW, MAXIMUM TENSION AT NESC HEAVY LOADING: 14600# R.S.= 2080'
(PH_AH) CONDUCTOR IS 183–T114 ACCR, MAXIMUM TENSION AT NESC HEAVY LOADING: 13100# R.S.= 2080'

WIND SPAN: 1750' (AHEAD)
MAXIMUM WEIGHT SPAN: 5000' (N_AH)
5250' (PH_AH)
HEIGHT OF STR.: 55'–0"
ACTUAL ANGLE (Δ): 0°00'00"
DESIGN MAXIMUM ANGLE: 3°00'00"

SPECIFICATIONS

1. POLES TO BE DESIGNED PER TECHNICAL SPECIFICATION (33 71 16.23). IF CONFLICT EXIST BETWEEN THIS DRAWING AND THE FABRICATION SPEC, FOLLOW THE DETAILS ON THIS DRAWING. IF CLARIFICATION IS NECESSARY, CONTACT THE CONTRACTING OFFICER REPRESENTATIVE.
2. CAMBERING OF POLES WILL NOT BE PERMITTED.
3. ALL STEEL TO BE WEATHERING STEEL.
4. POLE TO HAVE BOLTED FLANGE TYPE SPLICES.
5. THE (*) LOADINGS SHOWN IN LOAD TABLE CONVEY THE ADDITIONAL VERTICAL LOAD THAT WILL BE APPLIED TO THE STRUCTURE BY LINE PERSONNEL DURING CONSTRUCTION AND MAINTENANCE OPERATIONS.
   A) IT IS REQUIRED THAT ALL CONDUCTOR ATTACHMENT SUPPORTS BE DESIGNED TO MEET THIS REQUIREMENT.
6. BASE DIAMETER OF DESIGN STEEL POLES SHALL BE DESIGNED/PROPOSED IN 6" INCREMENTS TO COINCIDE WITH FOUNDATION DESIGN.
7. STRUCTURE SHALL BE DESIGNED ASSUMING MICROPILE TYPE FOUNDATION.
   A) STEEL VENDOR TO PROVIDE TWO (2) ADDITIONAL GROUNDING PADS FOR EACH POLE TO BE FIELD WELDED ON EACH MICROPILE.
   B) NO FIELD WELDING WILL BE ALLOWED ON THE STRUCTURES EXCEPT TO WELD THE GROUND PADS TO THE MICROPILE FOUNDATION. STEEL VENDOR TO PROVIDE FIELD WELD DESIGN AND DETAILS IN ACCORDANCE WITH AWS D1.1 FOR WELDING GROUND PADS TO MICROPILE FOUNDATIONS.
8. ALL LADDER PROVISIONS SHALL MEET OSHA FALL PROTECTION REQUIREMENTS. VERIFY LADDER CLIPS AND WELDS CAN HANDLE 5000 LBS. OSHA IMPACT LOAD.
9. VENDOR TO VERIFY THAT TOP VANG CAN BE USED FOR LIFTING ASSEMBLED STRUCTURE.
10. VENDOR TO DESIGN STRUCTURE TO HAVE A MAXIMUM UTILIZATION OF 90%.

DEFLECTION

DEFLECTION SHALL BE LIMITED TO 1.5% OR LESS OF THE ABOVE GROUND LENGTH UNDER DEFLECTION LOAD CASE.

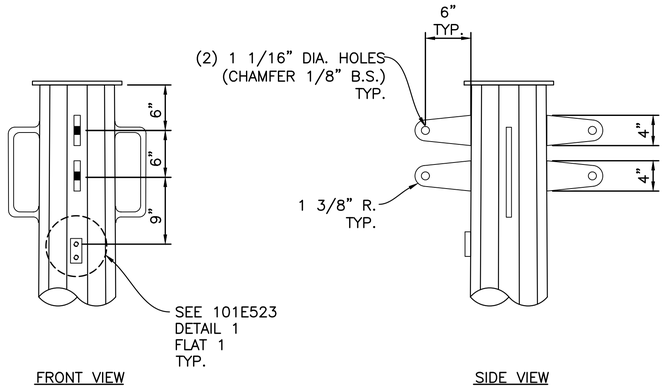
Ⓐ IDENTIFICATION TAG

STR #	NR
STR LENGTH (FEET)	55'
STRUCTURE TYPE	LD–NR
MANUFACTURED DATE	(DATE)
MAX GRD. LINE MOM. (FT–KIPS)	(####)

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 75014 TOU VANG

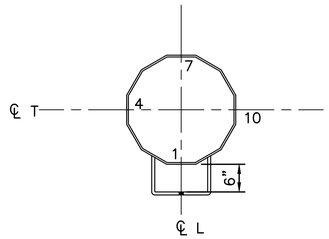
A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: B. PLANT CADD B. PLANT TECH. REVIEW: J. DOHRENDORF DATE: MAY 2026	SUB SHEET NO. 101 E509	TITLE OF SHEET POWER DISTRIBUTION STRUCTURAL & ASSEMBLY LOAD AND DESIGN NORTH RIM NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 25 OF 50

4/17/2026 Walchuk, Chris C:\Users\Cwalchuk\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\ELEC\101-E509.dwg



FRONT VIEW

SIDE VIEW



TOP VIEW

1 / 101E509 DETAIL: NEUTRAL AND CONDUCTOR ATTACHMENT

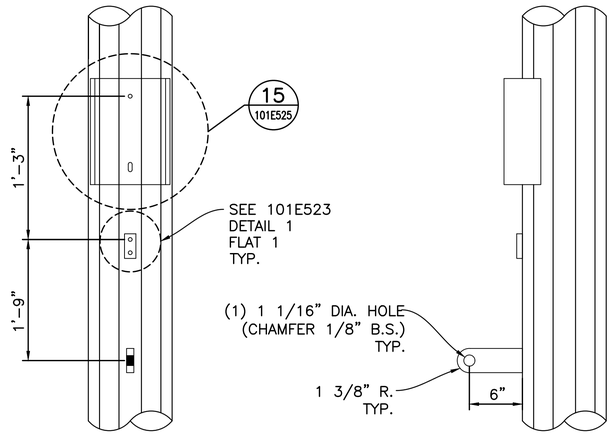
NOT TO SCALE

GENERAL NOTES:

A. SEE SHEET 101E509 FOR STRUCTURE LAYOUT AND DESIGN AT NORTH RIM.

B. SEE SHEET 101E511 FOR STRUCTURE LOADING AT NORTH RIM.

C. SEE SHEET 101E102 FOR NORTH RIM SITE PLAN.



FRONT VIEW

SIDE VIEW

2 / 101E509 DETAIL: RISER BRACKET

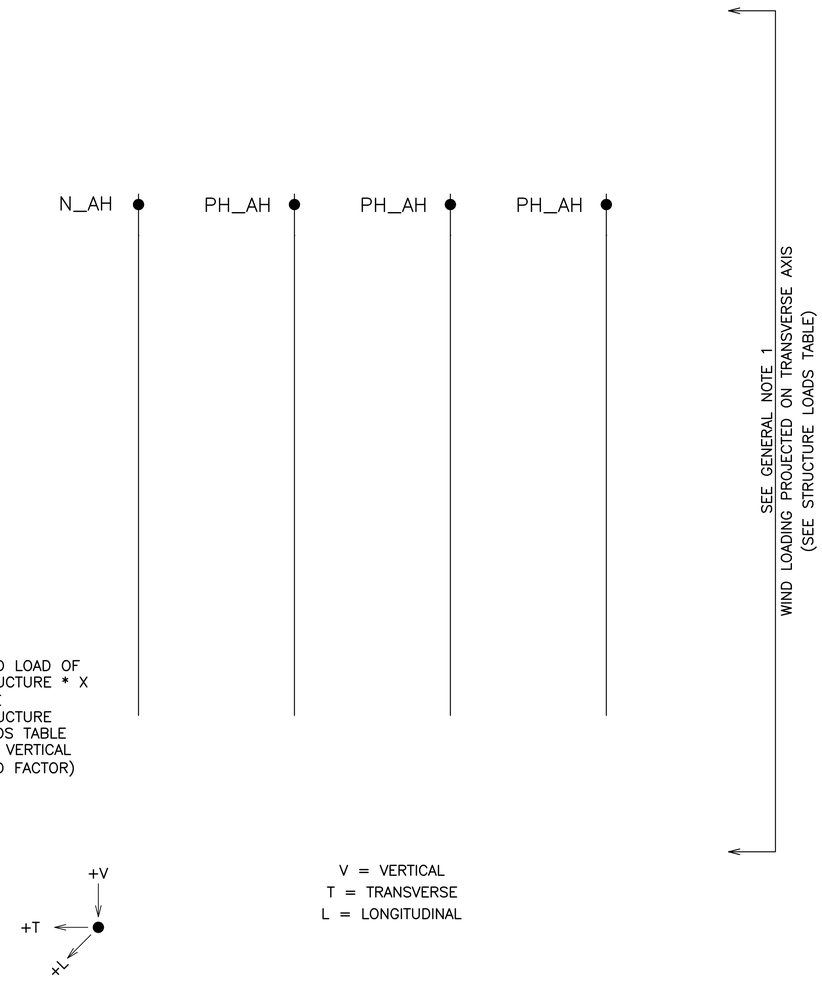
NOT TO SCALE

4/17/2026 Walchuk, Chris C:\Users\cwalchuk\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\ELEC\101-E510.dwg

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 75014 TOU VANG STATE OF COLORADO

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: B. PLANT CADD B. PLANT TECH. REVIEW: J. DOHRENDORF DATE: MAY 2026	SUB SHEET NO. 101 E510	TITLE OF SHEET POWER DISTRIBUTION STRUCTURAL & ASSEMBLY LOAD AND DESIGN NORTH RIM NORTH RIM - GRAND CANYON NATIONAL PARK

DRAWING NO.	113 182752
PMIS/PKG NO.	318719
SHEET	26 OF 50



STRUCTURE LOADS (INCLUDES ALL APPLICABLE LOAD FACTORS)								
LOAD CASE	TEMPERATURE (°F)	RADIAL ICE @ 56 LBS/FT^3 (INCHES)	VERTICAL OVERLOAD FACTOR "X"	WIND LOADING (PSF)	POINT LOAD NAME	LONGITUDINAL (LBS)	TRANSVERSE (LBS)	VERTICAL (LBS)
NESC 250B	0	0.5	1.5	4	N_AH	20700	3700	11000
					PH_AH	19000	3600	9400
NESC 250C	60	0.0	1.1	20.74	N_AH	9000	2100	4000
					PH_AH	8200	2100	2600
NESC 250D	15	0.25	1.1	2.30	N_AH	8900	1000	5600
					PH_AH	6800	900	4300
EXTREME ICE	20	1.0	1	0	N_AH	16600	900	14100
					PH_AH	16300	900	13400
DEFLECTION	40	0.0	1	2	N_AH	6000	600	3600
					PH_AH	3800	500	2400
CONSTRUCTION	−20	0.0	1.1	2	N_AH	9600	800	5200*
					PH_AH	6000	600	3800*
CONSTRUCTION STRINGING	−20	0.0	1.1	2	N_AH	3300	900	10500*
					PH_AH	2000	600	6900*

*MAN LOADED

GENERAL NOTES:

A. TRANSVERSE WIND LOAD IS TO BE APPLIED IN THE DIRECTION RESULTING IN THE GREATEST TRANSVERSE GROUND LINE MOMENT FOR ALL LOAD CASES.

B. SEE SHEET 101E509 FOR STRUCTURE LAYOUT AND DESIGN AT NORTH RIM.

C. SEE SHEET 101E510 FOR STRUCTURE ATTACHMENT DETAILS AT NORTH RIM.

D. SEE SHEET 101E102 FOR NORTH RIM SITE PLAN.

REGISTERED PROFESSIONAL ENGINEER COLORADO CERTIFICATE NO. 75014 TOU VANG

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	B. PLANT	101 E511	POWER DISTRIBUTION STRUCTURAL & ASSEMBLY LOAD AND DESIGN NORTH RIM	113 182752
	CADD: B. PLANT		NORTH RIM - GRAND CANYON NATIONAL PARK	PMIS/PKG NO. 318719
	TECH. REVIEW: J. DOHRENDORF			SHEET 27 OF 50
	DATE: MAY 2026			

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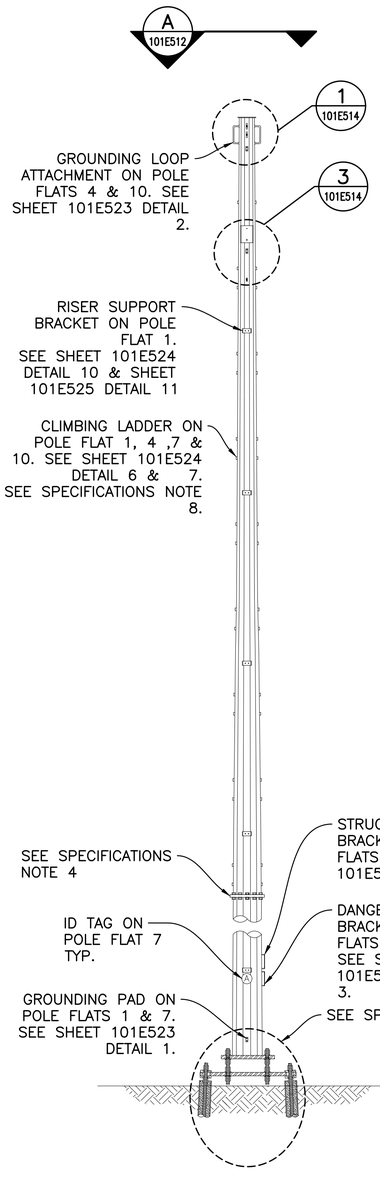
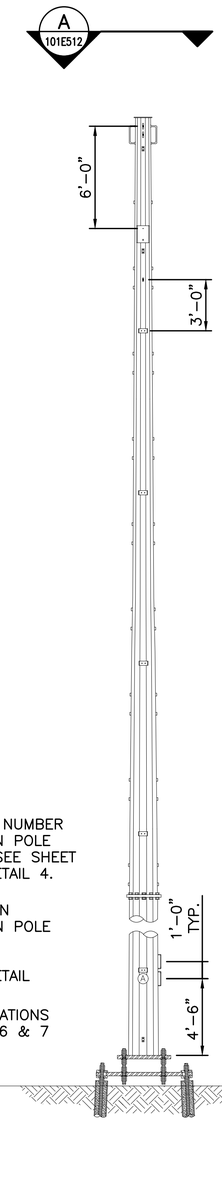
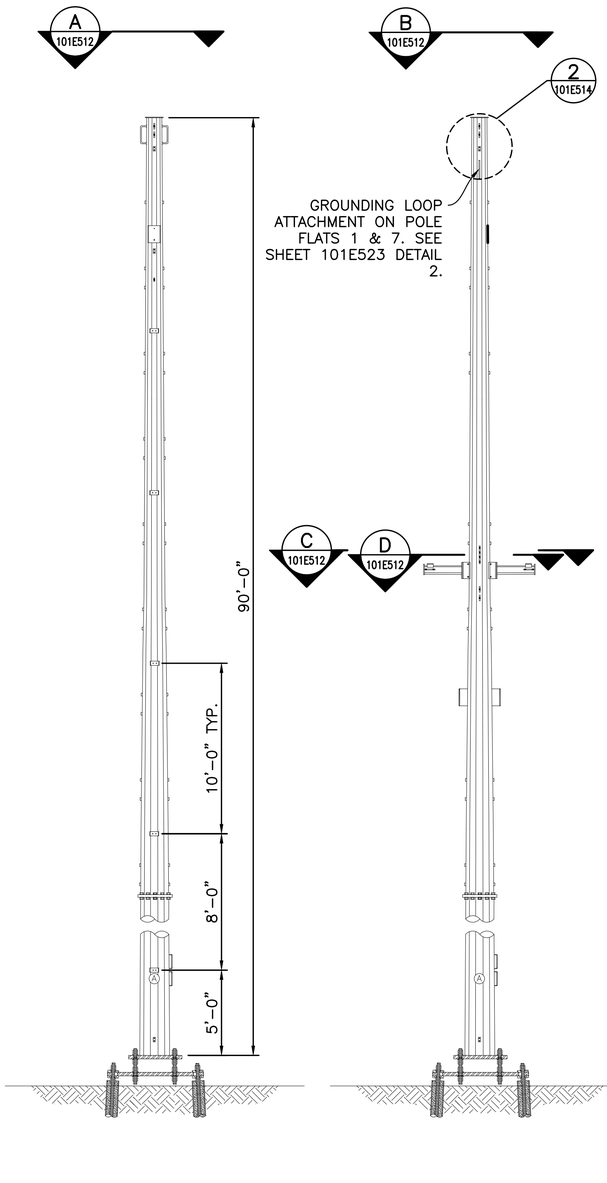
Ⓐ IDENTIFICATION TAG

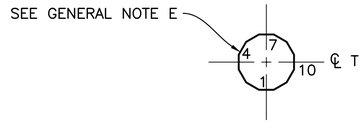
STR #	UB
STR LENGTH (FEET)	90'
STRUCTURE TYPE	LD–UB
MANUFACTURED DATE	(DATE)
MAX GRD. LINE MOM. (FT–KIPS)	(####)



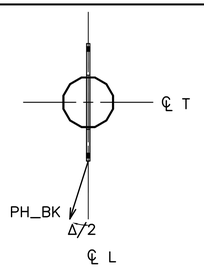




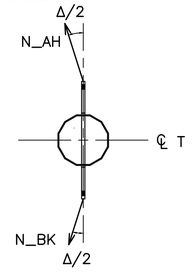
UPPER BENCH STRUCTURE
LOOKING SOUTHEAST



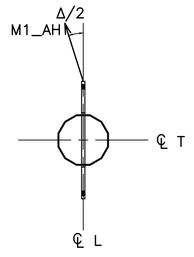
12 MULTISIDED STEEL POLE FLAT DESIGNATION



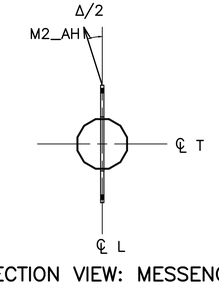
Ⓐ 101E512 VIEW: PHASE CONDUCTOR ORIENTATION
NOT TO SCALE



Ⓑ 101E512 VIEW: NEUTRAL ORIENTATION
NOT TO SCALE



Ⓒ 101E512 SECTION VIEW: MESSENGER ORIENTATION
NOT TO SCALE



Ⓓ 101E512 SECTION VIEW: MESSENGER ORIENTATION
NOT TO SCALE

GENERAL NOTES

A. MAXIMUM NEUTRAL WIRE VANG THICKNESS = 3/4"
B. MAXIMUM CONDUCTOR VANG THICKNESS = 3/4", UNLESS NOTED OTHERWISE
C. MAXIMUM MESSENGER VANG THICKNESS = 1 1/4", UNLESS NOTED OTHERWISE
D. LOAD TREE VALUES ARE ULTIMATE, INCLUDE OVERLOAD FACTORS, AND ARE SHOWN IN POUNDS.
E. POLE FLAT DESIGNATIONS REFER TO SIDE OF THE POLE WHERE ATTACHMENTS ARE LOCATED

DESIGN CRITERIA

LOADING PARAMETERS FOR: UPPER BENCH POLES
(N_BK) NEUTRAL WIRE IS 144 FIBER OPGW, MAXIMUM TENSION AT NESC HEAVY LOADING: 14600# R.S.= 2080'
(N_AH) NEUTRAL WIRE IS 144 FIBER OPGW, MAXIMUM TENSION AT NESC HEAVY LOADING: 14700# R.S. = 1270'
(PH_BK) CONDUCTOR IS 183–T114 ACCR, MAXIMUM TENSION AT NESC HEAVY LOADING: 13100# R.S.= 2080'
(M1_AH) MESSENGER WIRE IS 37 NO. 8 AW, MAXIMUM TENSION AT NESC HEAVY LOADING: 17200# R.S. = 1270'
(M2_AH) MESSENGER WIRE IS 37 NO. 8 AW SUPPORTING (3) 1/0 15 KV INSULATED SPACER CABLES, MAXIMUM TENSION AT NESC HEAVY LOADING: 35500# R.S. = 1270'

WIND SPAN: 1750' (BACK), 1050' (AHEAD)
MAXIMUM WEIGHT SPAN: –2750' (PH_BK)
–2500' (N_BK)
5250' (N_AH)
4350' (M1_AH)
3750' (M2_AH)

HEIGHT OF STR.: 90'–0"
ACTUAL ANGLE (Δ): 9°11'18"
DESIGN MAXIMUM ANGLE: 13°00'00"

SPECIFICATIONS

1. POLES TO BE DESIGNED PER TECHNICAL SPECIFICATION (33 71 16.23). IF CONFLICT EXIST BETWEEN THIS DRAWING AND THE FABRICATION SPEC, FOLLOW THE DETAILS ON THIS DRAWING. IF CLARIFICATION IS NECESSARY, CONTACT THE CONTRACTING OFFICER REPRESENTATIVE.
2. CAMBERING OF POLES WILL NOT BE PERMITTED.
3. ALL STEEL TO BE WEATHERING STEEL.
4. POLE TO HAVE BOLTED FLANGE TYPE SPLICES.
5. THE (*) LOADINGS SHOWN IN LOAD TABLE CONVEY THE ADDITIONAL VERTICAL LOAD THAT WILL BE APPLIED TO THE STRUCTURE BY LINE PERSONNEL DURING CONSTRUCTION AND MAINTENANCE OPERATIONS.
   A) IT IS REQUIRED THAT ALL CONDUCTOR ATTACHMENT SUPPORTS BE DESIGNED TO MEET THIS REQUIREMENT.
6. BASE DIAMETER OF DESIGN STEEL POLES SHALL BE DESIGNED/PROPOSED IN 6" INCREMENTS TO COINCIDE WITH FOUNDATION DESIGN.
7. STRUCTURE SHALL BE DESIGNED ASSUMING MICROPILE TYPE FOUNDATION.
   A) STEEL VENDOR TO PROVIDE TWO (2) ADDITIONAL GROUNDING PADS FOR EACH POLE TO BE FIELD WELDED ON EACH MICROPILE.
   B) NO FIELD WELDING WILL BE ALLOWED ON THE STRUCTURES EXCEPT TO WELD THE GROUND PADS TO THE MICROPILE FOUNDATION. STEEL VENDOR TO PROVIDE FIELD WELD DESIGN AND DETAILS IN ACCORDANCE WITH AWS D1.1 FOR WELDING GROUND PADS TO MICROPILE FOUNDATIONS.
8. ALL LADDER PROVISIONS SHALL MEET OSHA FALL PROTECTION REQUIREMENTS. VERIFY LADDER CLIPS AND WELDS CAN HANDLE 5000 LBS. OSHA IMPACT LOAD.
9. VENDOR TO VERIFY THAT TOP VANG CAN BE USED FOR LIFTING ASSEMBLED STRUCTURE.
10. VENDOR TO DESIGN STRUCTURE TO HAVE A MAXIMUM UTILIZATION OF 90%.

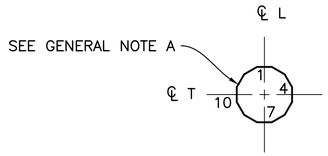
DEFLECTION

DEFLECTION SHALL BE LIMITED TO 1.5% OR LESS OF THE ABOVE GROUND LENGTH UNDER DEFLECTION LOAD CASE.

REGISTERED PROFESSIONAL ENGINEER COLORADO CERTIFICATE NO. 75014 TOU VANG

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: B. PLANT DRAWN: B. PLANT TECH. REVIEW: J. DOHRENDORF DATE: MAY 2026	SUB SHEET NO. 101 E512	TITLE OF SHEET POWER DISTRIBUTION STRUCTURAL & ASSEMBLY LOAD AND DESIGN UPPER BENCH NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 28 OF 50

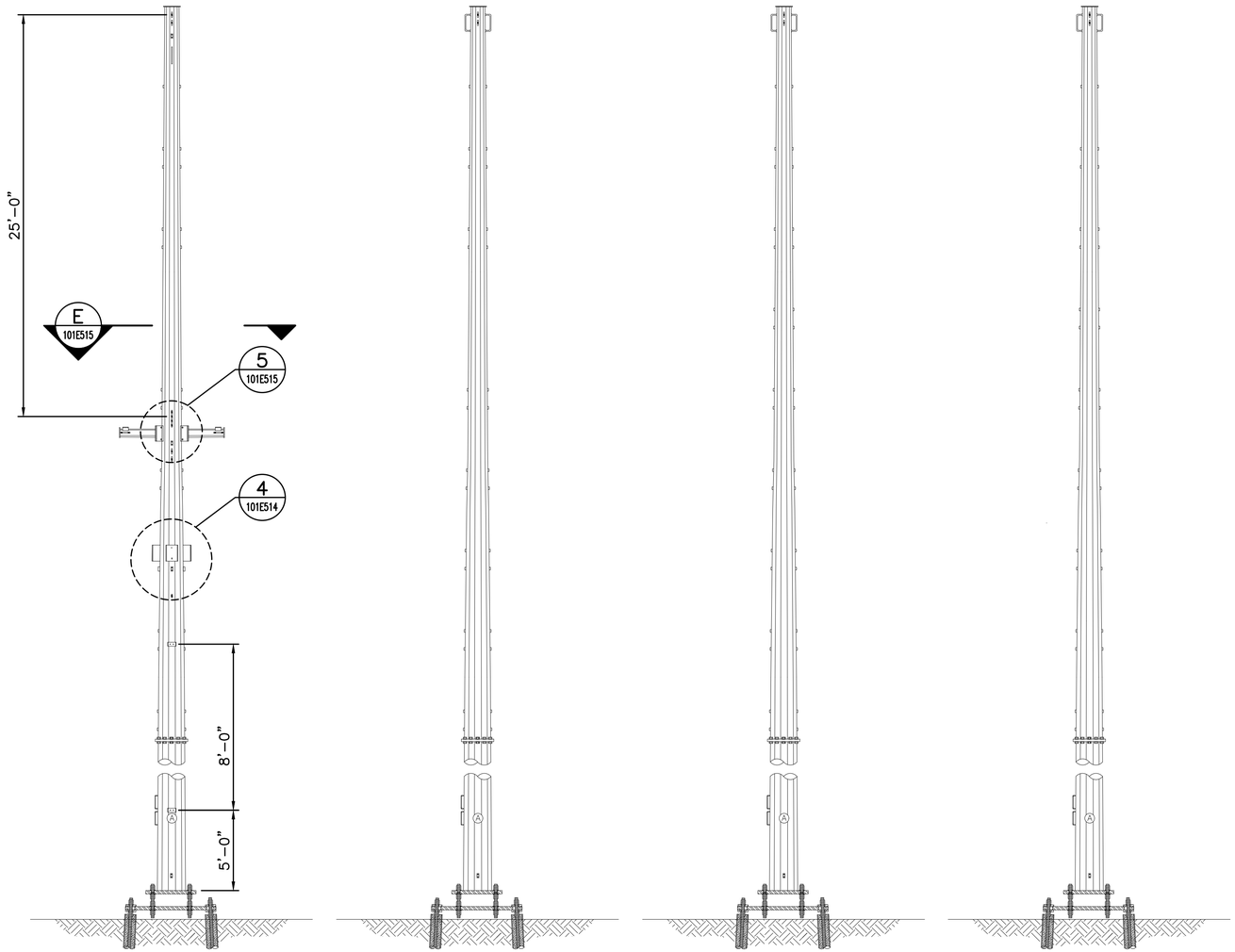
4/17/2026 Walchuk, Chris C:\Users\cwalchuk\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\ELEC\101-E512.dwg



12 MULTISIDED STEEL
POLE FLAT DESIGNATION

GENERAL NOTES

A. POLE FLAT DESIGNATIONS REFER TO SIDE OF THE POLE WHERE ATTACHMENTS ARE LOCATED



UPPER BENCH STRUCTURE
LOOKING NORTHWEST

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 75014 TOU VANG

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: B. PLANT DRAWN: B. PLANT TECH. REVIEW: J. DOHRENDORF DATE: MAY 2026	SUB SHEET NO. 101 E513	TITLE OF SHEET POWER DISTRIBUTION STRUCTURAL & ASSEMBLY LOAD AND DESIGN UPPER BENCH NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 29 OF 50

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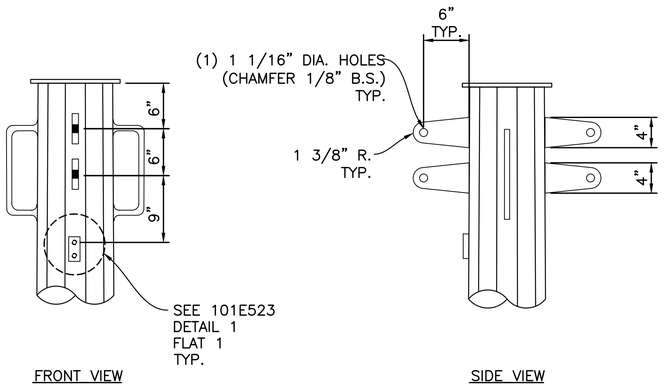
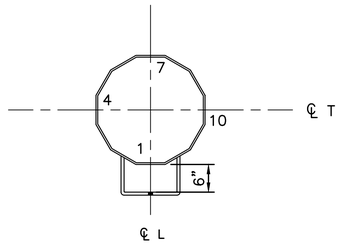
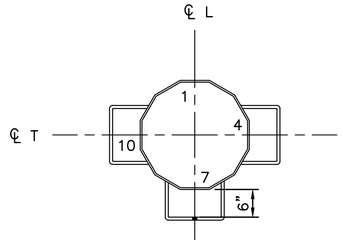


FRONT VIEW SIDE VIEW



TOP VIEW



TOP VIEW

GENERAL NOTES:

A. SEE SHEET 101E512 & 101E513 FOR STRUCTURE SECTION LAYOUT AND DESIGN AT UPPER BENCH.

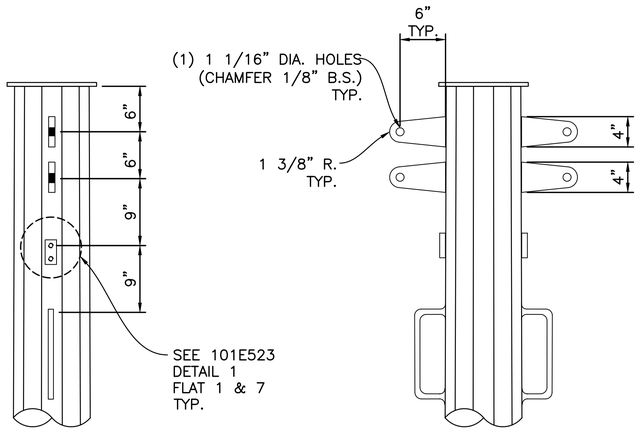
B. SEE SHEET 101E516 FOR STRUCTURE LOADING AT UPPER BENCH.

C. SEE SHEET 101E526 FOR CROSSARM LOADING REQUIREMENTS.

D. SEE SHEET 101E103 FOR UPPER BENCH SITE PLAN.

1 / 101E512 DETAIL: CONDUCTOR ATTACHMENT

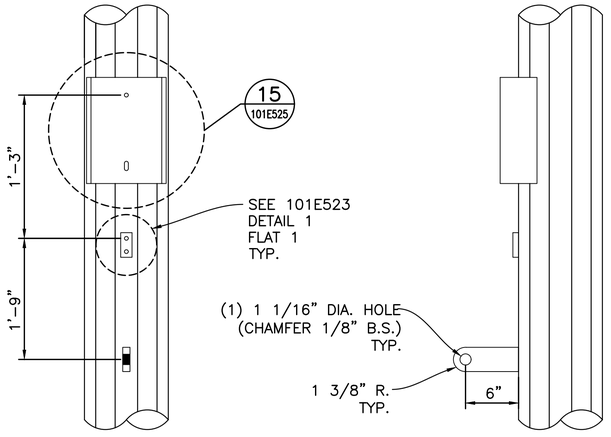
NOT TO SCALE



BACK VIEW SIDE VIEW

2 / 101E512 DETAIL: NEUTRAL ATTACHMENT

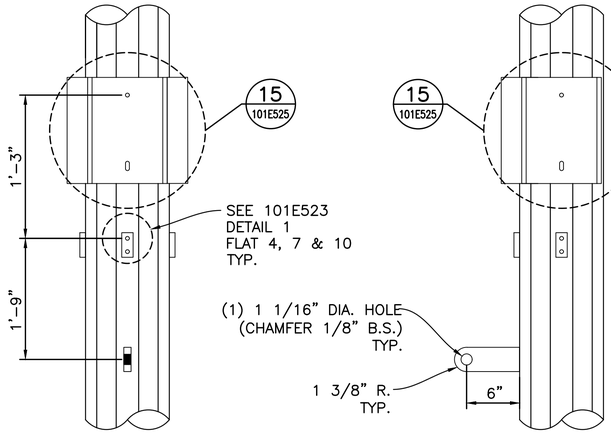
NOT TO SCALE



FRONT VIEW SIDE VIEW

3 / 101E512 DETAIL: RISER BRACKET

NOT TO SCALE



FRONT VIEW SIDE VIEW

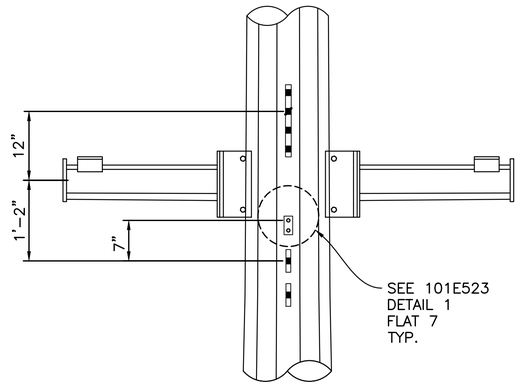
4 / 101E513 DETAIL: THREE–PHASE RISER BRACKET

NOT TO SCALE

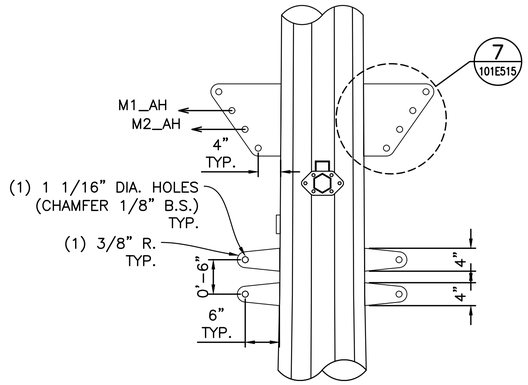
REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 75014 TOU VANG STATE OF COLORADO

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: B. PLANT CADD B. PLANT TECH. REVIEW: J. DOHRENDORF DATE: MAY 2026	SUB SHEET NO. 101 E514	TITLE OF SHEET POWER DISTRIBUTION STRUCTURAL & ASSEMBLY LOAD AND DESIGN UPPER BENCH NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 30 OF 50

4/17/2026 Walchuk, Chris C:\Users\Cwalchuk\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\ELEC\101-E514.dwg



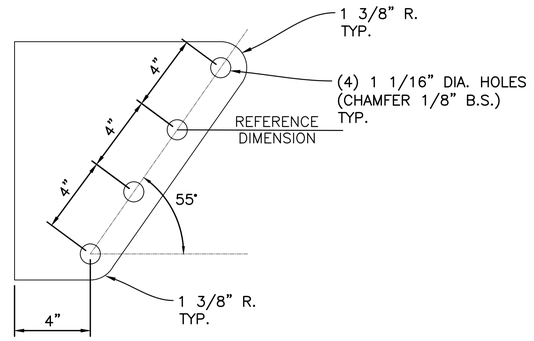
FRONT VIEW



SIDE VIEW

5 / 101E513 DETAIL: MESSENGER & SPACER CABLE ATTACHMENT

NOT TO SCALE



7 / 101E515 DETAIL: MESSENGER ATTACHMENT VANG

NOT TO SCALE

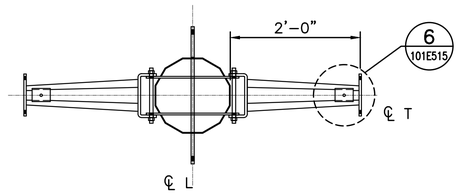
GENERAL NOTES:

A. SEE SHEET 101E512 & 101E513 FOR STRUCTURE SECTION LAYOUT AND DESIGN AT UPPER BENCH.

B. SEE SHEET 101E516 FOR STRUCTURE LOADING AT UPPER BENCH.

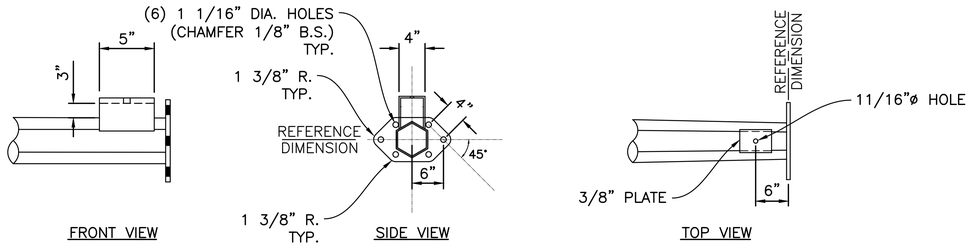
C. SEE SHEET 101E526 FOR CROSSARM LOADING REQUIREMENTS.

D. SEE SHEET 101E103 FOR UPPER BENCH SITE PLAN.



E / 101E513 SECTION VIEW: SPACER CABLE ARM

NOT TO SCALE



FRONT VIEW SIDE VIEW TOP VIEW

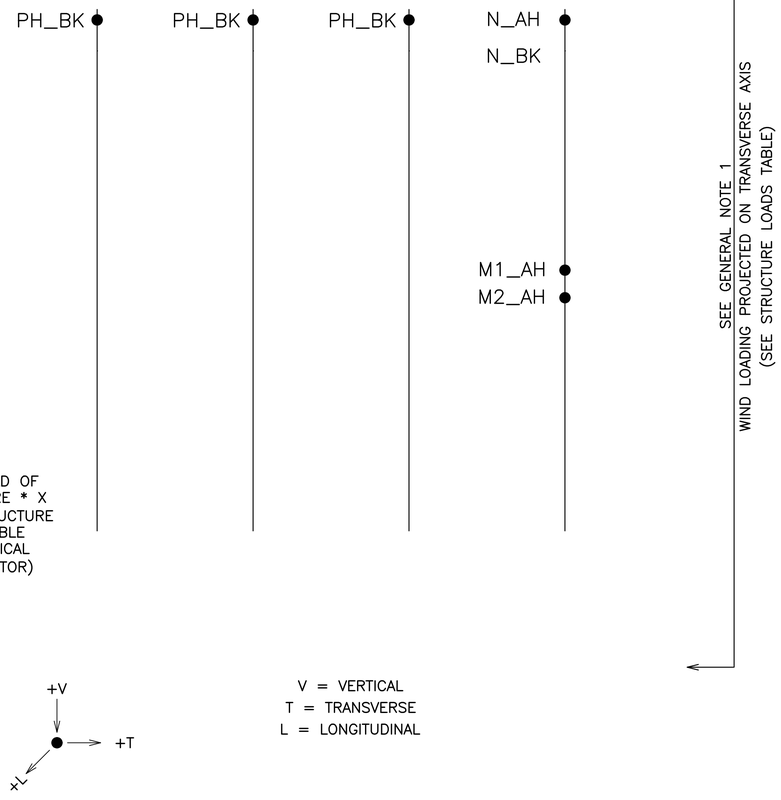
6 / 101E515 DETAIL: CROSSARM ATTACHMENT

NOT TO SCALE

REGISTERED PROFESSIONAL ENGINEER DEPARTMENT OF ARIZONA
CERTIFICATE NO.
75014
TOU
VANG

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: B. PLANT CADD B. PLANT TECH. REVIEW: J. DOHRENDORF DATE: MAY 2026	SUB SHEET NO. 101 E515	TITLE OF SHEET POWER DISTRIBUTION STRUCTURAL & ASSEMBLY LOAD AND DESIGN UPPER BENCH NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 31 OF 50

4/17/2026 Walchuk, Chris C:\Users\Cwalchuk\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\ELEC\101-E515.dwg



STRUCTURE LOADS
(INCLUDES ALL APPLICABLE LOAD FACTORS)

LOAD CASE	TEMPERATURE (°F)	RADIAL ICE @ 56 LBS/FT^3 (INCHES)	VERTICAL OVERLOAD FACTOR "X"	WIND LOADING (PSF)	POINT LOAD NAME	LONGITUDINAL (LBS)	TRANSVERSE (LBS)	VERTICAL (LBS)
NESC 250B	0	0.5	1.5	4	PH_BK	20500	−4900	−4600
					N_BK	22900	−5200	−5500
					N_AH	−20500	−3900	11500
					M1_AH	−23000	−4300	14800
					M2_AH	−48100	−11600	28900
NESC 250C	60	0.0	1.1	20.74	PH_BK	9600	−3000	−1100
					N_BK	10900	−3100	−2000
					N_AH	−10200	−2400	4100
					M1_AH	−10600	−2600	6800
					M2_AH	−22300	−6800	8900
NESC 250D	15	0.25	1.1	2.30	PH_BK	7800	−1400	−2000
					N_BK	10400	−1700	−2800
					N_AH	−10300	−1500	5900
					M1_AH	−11100	−1600	8500
					M2_AH	−19800	−3400	13800
EXTREME ICE	20	1.0	1	0	PH_BK	17300	−2000	−6800
					N_BK	18000	−2100	−7100
					N_AH	−15100	−1800	14800
					M1_AH	−17900	−2100	16300
					M2_AH	−39500	−4500	39700
NESC 250B (NO WIRES AHEAD)	0	0.5	1.5	4	PH_BK	20500	−4900	−4600
					N_BK	22900	−5200	−5500
					N_AH	0	−100	100
					M1_AH	0	−100	100
					M2_AH	0	−100	100
NESC 250B (NO WIRES BACK)	0	0.5	1.5	4	PH_BK	0	−100	300
					N_BK	0	−100	100
					N_AH	−20500	−3900	11500
					M1_AH	−23000	−4300	14800
					M2_AH	−48100	−11600	28900
DEFLECTION	40	0.0	1	2	PH_BK	4400	−800	−1000
					N_BK	7200	−1100	−1800
					N_AH	−7600	−1000	3800
					M1_AH	−8000	−1100	6200
					M2_AH	−12400	−2000	8100
CONSTRUCTION	−20	0.0	1.1	2	PH_BK	7100	−1100	100*
					N_BK	11900	−1600	−700*
					N_AH	−13400	−1700	5400*
					M1_AH	−13200	−1700	8000*
					M2_AH	−20300	−2900	10100*
CONSTRUCTION STRINGING	−20	0.0	1.1	2	PH_BK	2200	−1100	3500*
					N_BK	3600	−1700	5400*
					N_AH	−4300	−1800	12500*
					M1_AH	−4700	−2000	16000*
					M2_AH	−6900	−3300	22000*

*MAN LOADED

GENERAL NOTES:

A. TRANSVERSE WIND LOAD IS TO BE APPLIED IN THE DIRECTION RESULTING IN THE GREATEST TRANSVERSE GROUND LINE MOMENT FOR ALL LOAD CASES.

B. SEE SHEET 101E512 & 513 FOR STRUCTURE LAYOUT AND DESIGN AT UPPER BENCH.

C. SEE SHEET 101E526 FOR CROSSARM LOADING REQUIREMENTS.

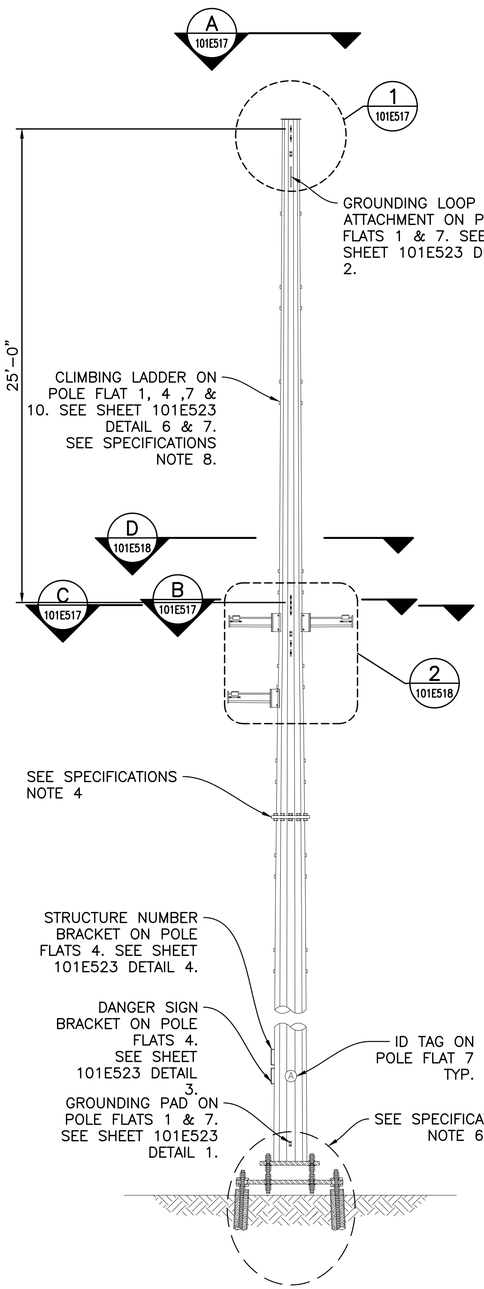
D. SEE SHEET 101E514 & 101E515 FOR STRUCTURE ATTACHMENT DETAILS AT UPPER BENCH.

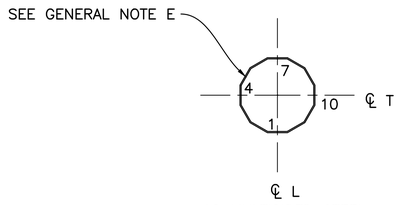
E. SEE SHEET 101E103 FOR UPPER BENCH SITE PLAN.

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 75014 TOU VANG

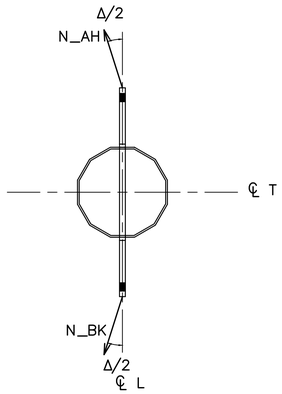
A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	B. PLANT	101 E516	POWER DISTRIBUTION STRUCTURAL & ASSEMBLY LOAD AND DESIGN UPPER BENCH NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752
	CADD: B. PLANT			PMIS/PKG NO. 318719
	TECH. REVIEW: J. DOHRENDORF			SHEET 32 OF 50
	DATE: MAY 2026			

4/17/2026 Walchuk, Chris C:\Users\cwalchuk\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\ELEC\101-E516.dwg

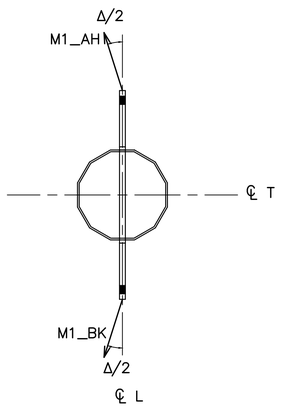




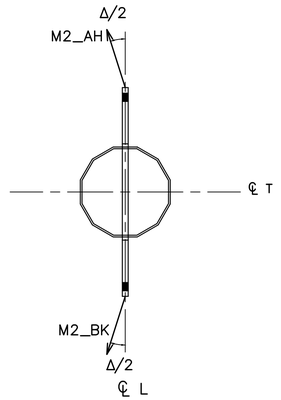
12 MULTISIDED STEEL POLE FLAT DESIGNATION



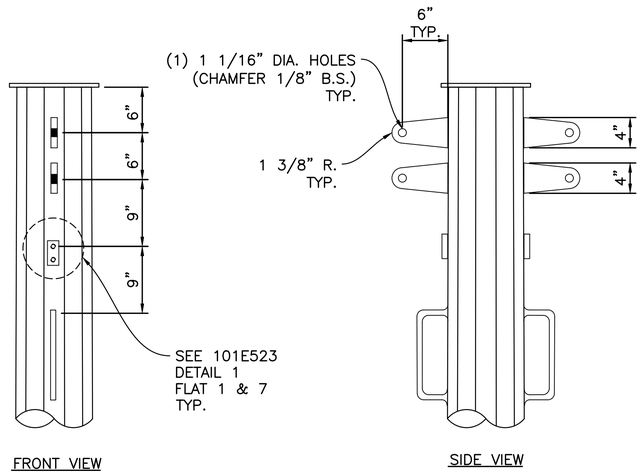
A 101E517 VIEW: NEUTRAL ORIENTATION
NOT TO SCALE



B 101E517 SECTION VIEW: MESSENGER ORIENTATION
NOT TO SCALE



C 101E517 SECTION VIEW: MESSENGER ORIENTATION
NOT TO SCALE



1 101E517 DETAIL: NEUTRAL ATTACHMENT
NOT TO SCALE

Ⓐ IDENTIFICATION TAG

STR #	LB
STR LENGTH (FEET)	100'
STRUCTURE TYPE	LD-LB
MANUFACTURED DATE	(DATE)
MAX GRD. LINE MOM. (FT-KIPS)	(####)

GENERAL NOTES

A. MAXIMUM NEUTRAL WIRE VANG THICKNESS = 3/4"
B. MAXIMUM CONDUCTOR VANG THICKNESS = 3/4", UNLESS NOTED OTHERWISE
C. MAXIMUM MESSENGER VANG THICKNESS = 1 1/4", UNLESS NOTED OTHERWISE
D. LOAD TREE VALUES ARE ULTIMATE, INCLUDE OVERLOAD FACTORS, AND ARE SHOWN IN POUNDS.
E. POLE FLAT DESIGNATIONS REFER TO SIDE OF THE POLE WHERE ATTACHMENTS ARE LOCATED

DESIGN CRITERIA

LOADING PARAMETERS FOR: LOWER BENCH POLE
(N_BK) NEUTRAL WIRE IS 144 FIBER OPGW, MAXIMUM TENSION AT NESC HEAVY LOADING: 14700# R.S.= 1270'
(N_AH) NEUTRAL WIRE IS 144 FIBER OPGW, MAXIMUM TENSION AT NESC HEAVY LOADING: 15400# R.S. = 1340'
(M1_BK) MESSENGER WIRE IS 37 NO. 8 AW, MAXIMUM TENSION AT NESC HEAVY LOADING: 17200# R.S. = 1270'
(M1_AH) MESSENGER WIRE IS 37 NO. 8 AW, MAXIMUM TENSION AT NESC HEAVY LOADING: 17300# R.S. = 1340'
(M2_BK) MESSENGER WIRE IS 37 NO. 8 AW SUPPORTING (3) 1/0 15 KV INSULATED SPACER CABLES, MAXIMUM TENSION AT NESC HEAVY LOADING: 35500# R.S. = 1270'
(M2_AH) MESSENGER WIRE IS 37 NO. 8 AW SUPPORTING (3) 1/0 15 KV INSULATED SPACER CABLES, MAXIMUM TENSION AT NESC HEAVY LOADING: 37900# R.S. = 1340'

WIND SPAN: 1050' (BACK), 1400' (AHEAD)
MAXIMUM WEIGHT SPAN: -3150' (N_BK)
-2250' (M1_BK)
-2100' (M2_BK)
6400' (N_AH)
5200' (M1_AH)
4600' (M2_AH)

HEIGHT OF STR.: 100'-0"
ACTUAL ANGLE (Δ): 6°3'20"
DESIGN MAXIMUM ANGLE: 10°00'00"

SPECIFICATIONS

1. POLES TO BE DESIGNED PER TECHNICAL SPECIFICATION (33 71 16.23). IF CONFLICT EXIST BETWEEN THIS DRAWING AND THE FABRICATION SPEC, FOLLOW THE DETAILS ON THIS DRAWING. IF CLARIFICATION IS NECESSARY, CONTACT THE CONTRACTING OFFICER REPRESENTATIVE.
2. CAMBERING OF POLES WILL NOT BE PERMITTED.
3. ALL STEEL TO BE WEATHERING STEEL.
4. POLE TO HAVE BOLTED FLANGE TYPE SPLICES.
5. THE (*) LOADINGS SHOWN IN LOAD TABLE CONVEY THE ADDITIONAL VERTICAL LOAD THAT WILL BE APPLIED TO THE STRUCTURE BY LINE PERSONNEL DURING CONSTRUCTION AND MAINTENANCE OPERATIONS.
   A) IT IS REQUIRED THAT ALL CONDUCTOR ATTACHMENT SUPPORTS BE DESIGNED TO MEET THIS REQUIREMENT.
6. BASE DIAMETER OF DESIGN STEEL POLES SHALL BE DESIGNED/PROPOSED IN 6" INCREMENTS TO COINCIDE WITH FOUNDATION DESIGN.
7. STRUCTURE SHALL BE DESIGNED ASSUMING MICROPILE TYPE FOUNDATION.
   A) STEEL VENDOR TO PROVIDE TWO (2) ADDITIONAL GROUNDING PADS FOR EACH POLE TO BE FIELD WELDED ON EACH MICROPILE.
   B) NO FIELD WELDING WILL BE ALLOWED ON THE STRUCTURES EXCEPT TO WELD THE GROUND PADS TO THE MICROPILE FOUNDATION. STEEL VENDOR TO PROVIDE FIELD WELD DESIGN AND DETAILS IN ACCORDANCE WITH AWS D1.1 FOR WELDING GROUND PADS TO MICROPILE FOUNDATIONS.
8. ALL LADDER PROVISIONS SHALL MEET OSHA FALL PROTECTION REQUIREMENTS. VERIFY LADDER CLIPS AND WELDS CAN HANDLE 5000 LBS. OSHA IMPACT LOAD.
9. VENDOR TO VERIFY THAT TOP VANG CAN BE USED FOR LIFTING ASSEMBLED STRUCTURE.
10. VENDOR TO DESIGN STRUCTURE TO HAVE A MAXIMUM UTILIZATION OF 90%.

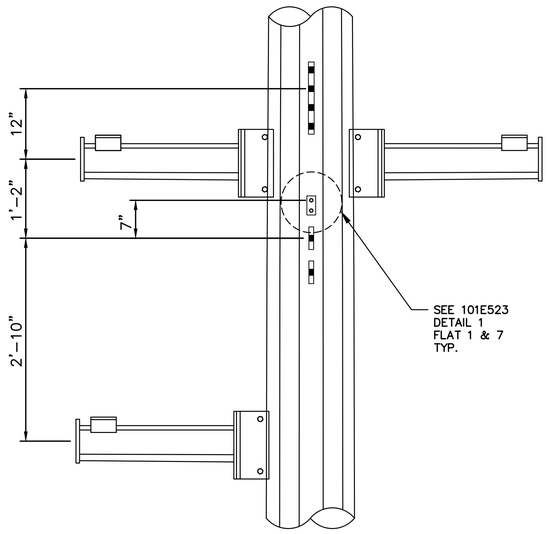
DEFLECTION

DEFLECTION SHALL BE LIMITED TO 1.5% OR LESS OF THE ABOVE GROUND LENGTH UNDER DEFLECTION LOAD CASE.

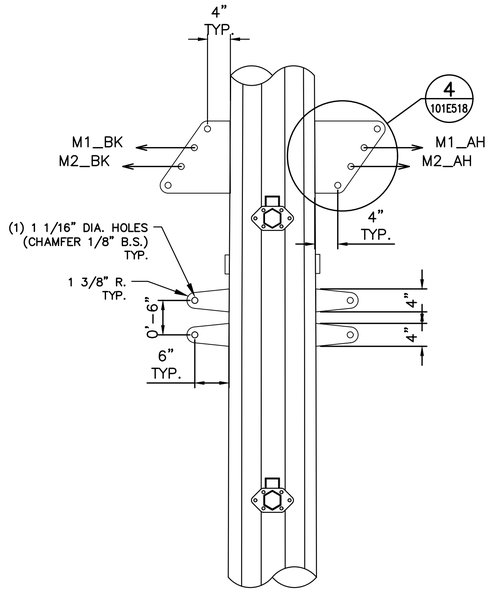
REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 75014 TOU VANG STATE OF COLORADO

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: B. PLANT CADD B. PLANT TECH. REVIEW: J. DOHRENDORF DATE: MAY 2026	SUB SHEET NO. 101 E517	TITLE OF SHEET POWER DISTRIBUTION STRUCTURAL & ASSEMBLY LOAD AND DESIGN LOWER BENCH NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 33 OF 50

4/17/2026 Walchuk, Chris C:\Users\Cwalchuk\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\ELEC\101-E517.dwg



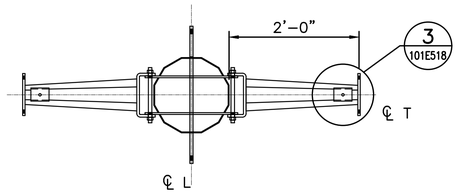
FRONT VIEW



SIDE VIEW

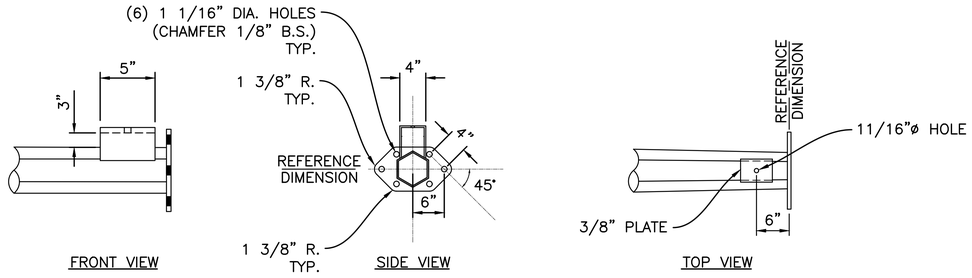
2 101E517 DETAIL: MESSENGER & SPACER CABLE ATTACHMENT

NOT TO SCALE



D 101E517 SECTION VIEW: SPACER CABLE ARM

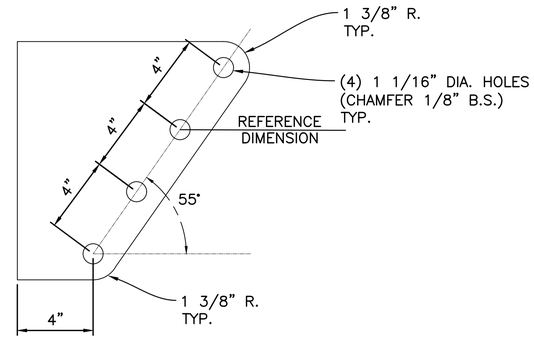
NOT TO SCALE



FRONT VIEW SIDE VIEW TOP VIEW

3 101E518 DETAIL: CROSSARM ATTACHMENT

NOT TO SCALE



4 101E518 DETAIL: MESSENGER ATTACHMENT VANG

NOT TO SCALE

GENERAL NOTES:

A. SEE SHEET 101E517 FOR STRUCTURE LAYOUT AND DESIGN AT LOWER BENCH.

B. SEE SHEET 101E519 FOR STRUCTURE LOADING AT LOWER BENCH.

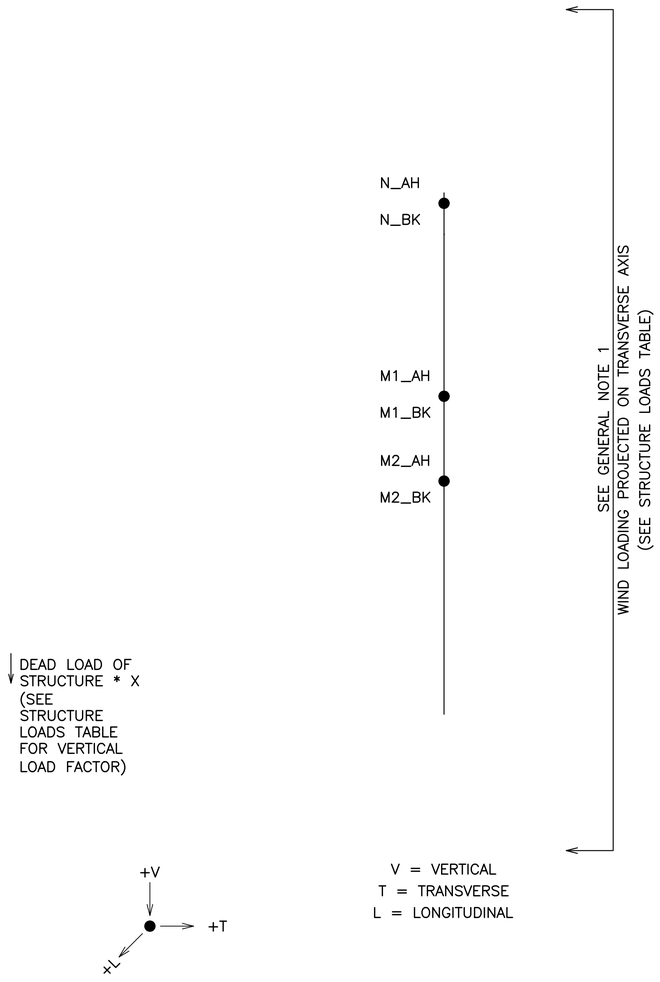
C. SEE SHEET 101E526 FOR CROSSARM LOADING REQUIREMENTS.

D. SEE SHEET 101E104 FOR LOWER BENCH SITE PLAN.

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 75014 TOU VANG STATE OF COLORADO

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: B. PLANT DRAWN: B. PLANT TECH. REVIEW: J. DOHRENDORF DATE: MAY 2026	SUB SHEET NO. 101 E518	TITLE OF SHEET POWER DISTRIBUTION STRUCTURAL & ASSEMBLY LOAD AND DESIGN LOWER BENCH NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 34 OF 50

4/17/2026 Watchuk, Chris C:\Users\Cwatchuk\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\ELEC\101-E518.dwg



STRUCTURE LOADS
(INCLUDES ALL APPLICABLE LOAD FACTORS)

LOAD CASE	TEMPERATURE (°F)	RADIAL ICE @ 56 LBS/FT^3 (INCHES)	VERTICAL OVERLOAD FACTOR "X"	WIND LOADING (PSF)	POINT LOAD NAME	LONGITUDINAL (LBS)	TRANSVERSE (LBS)	VERTICAL (LBS)
NESC 250B	0	0.5	1.5	4	N_BK	22600	−3500	−7500
					N_AH	−20000	−3800	14000
					M1_BK	26500	−4000	−8600
					M1_AH	−20700	−4100	17700
					M2_BK	55000	−10900	−16100
					M2_AH	−47900	−12300	35500
NESC 250C	60	0.0	1.1	20.74	N_BK	11700	−2300	−2700
					N_AH	−9900	−2400	5000
					M1_BK	12500	−2500	−3900
					M1_AH	−9500	−2600	8100
					M2_BK	26500	−6700	−4900
					M2_AH	−22000	−7500	10900
NESC 250D	15	0.25	1.1	2.30	N_BK	11900	−1400	−3800
					N_AH	−9700	−1300	7100
					M1_BK	13100	−1500	−4900
					M1_AH	−9700	−1300	10100
					M2_BK	24200	−3300	−7700
					M2_AH	−19200	−3200	16900
EXTREME ICE	20	1.0	1	0	N_BK	16200	−1500	−9700
					N_AH	−15000	−1400	18100
					M1_BK	20000	−1800	−9500
					M1_AH	−16600	−1500	19400
					M2_BK	43300	−3800	−22200
					M2_AH	−39700	−3500	48700
NESC 250B (NO WIRES AHEAD)	0	0.5	1.5	4	N_BK	22600	−3500	−7500
					N_AH	0	−100	100
					M1_BK	26500	−4000	−8600
					M1_AH	0	−100	100
					M2_BK	55000	−10900	−16100
					M2_AH	0	−100	100
NESC 250B (NO WIRES BACK)	0	0.5	1.5	4	N_BK	0	−100	100
					N_AH	−20000	−3800	14000
					M1_BK	0	−100	100
					M1_AH	−20700	−4100	17700
					M2_BK	0	−100	100
					M2_AH	−47900	−12300	35500
DEFLECTION	40	0.0	1	2	N_BK	9200	−1000	−2500
					N_AH	−7000	−800	4600
					M1_BK	9600	−1000	−3600
					M1_AH	−7100	−900	7400
					M2_BK	16200	−2000	−4500
					M2_AH	−11900	−1800	9900
CONSTRUCTION	−20	0.0	1.1	2	N_BK	16200	−1600	−1500*
					N_AH	−12300	−1300	6300*
					M1_BK	16300	−1600	−2700*
					M1_AH	−11300	−1300	9300*
					M2_BK	27100	−3000	−3700*
					M2_AH	−19200	−2500	12100*
CONSTRUCTION STRINGING	−20	0.0	1.1	2	N_BK	4900	−1600	6900*
					N_AH	−4100	−1500	13100*
					M1_BK	5100	−1700	5900*
					M1_AH	−4600	−1700	17100*
					M2_BK	8200	−3000	10600*
					M2_AH	−7100	−2900	24300*

*MAN LOADED

GENERAL NOTES:

A. TRANSVERSE WIND LOAD IS TO BE APPLIED IN THE DIRECTION RESULTING IN THE GREATEST TRANSVERSE GROUND LINE MOMENT FOR ALL LOAD CASES.

B. SEE SHEET 101E517 & 101E518 FOR STRUCTURE LAYOUT AND DESIGN AT LOWER BENCH.

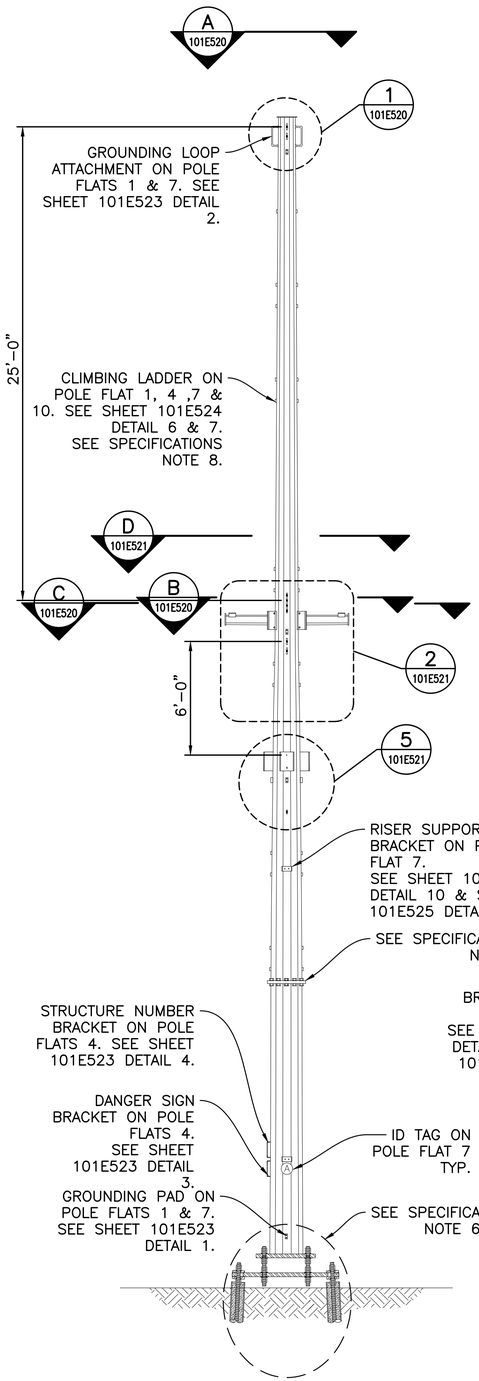
C. SEE SHEET 101E526 FOR CROSSARM LOADING REQUIREMENTS.

D. SEE SHEET 101E104 FOR LOWER BENCH SITE PLAN.

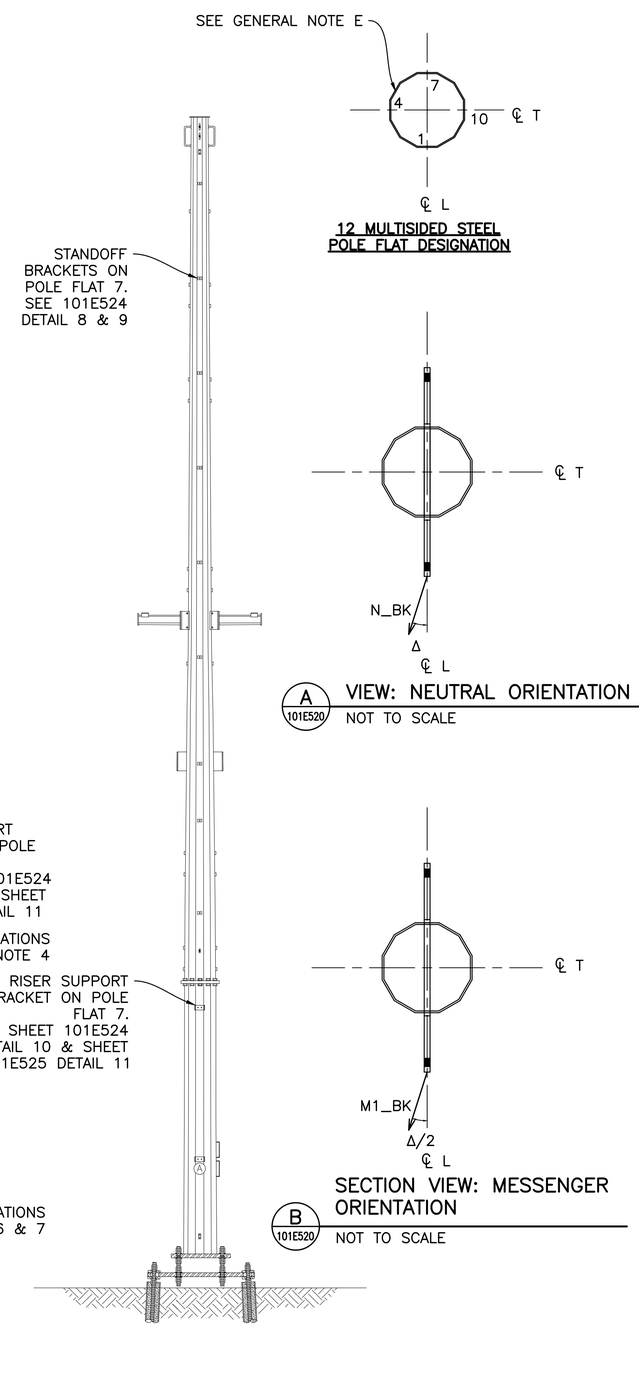
REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 75014 TOU VANG STATE OF COLORADO

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	B. PLANT	101 E519	POWER DISTRIBUTION STRUCTURAL & ASSEMBLY LOAD AND DESIGN LOWER BENCH NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752
	CADD: B. PLANT			PMIS/PKG NO. 318719
	TECH. REVIEW: J. DOHRENDORF			SHEET 35 OF 50
	DATE: MAY 2026			

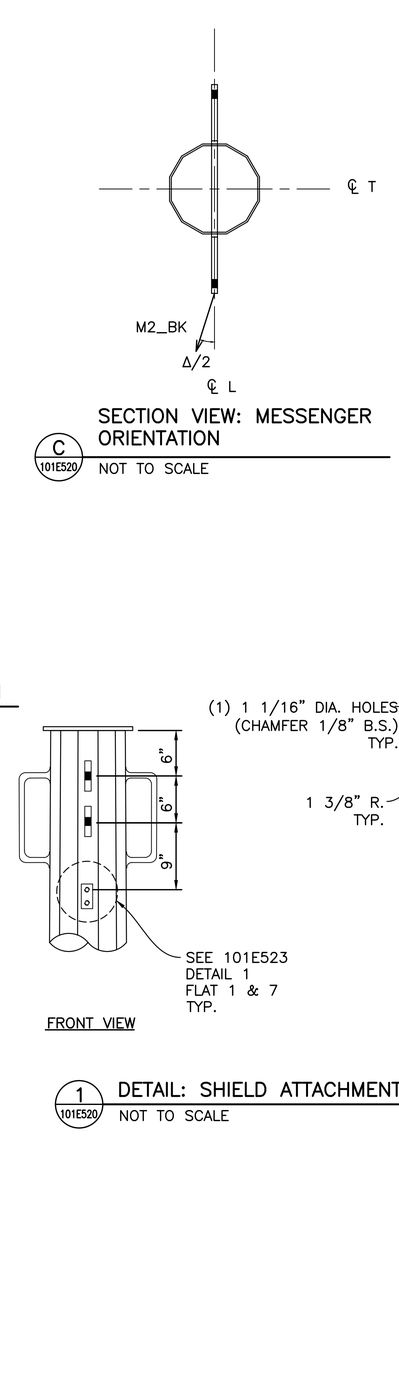
4/17/2026 Walchuk, Chris C:\Users\cwalchuk\DC\ACCDocs\HDR\103505445_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\ELEC\101-E519.dwg

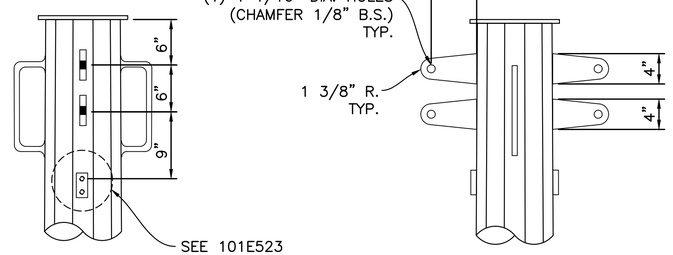


VIEW LOOKING AHEAD



VIEW LOOKING BACK





GENERAL NOTES

A. MAXIMUM NEUTRAL WIRE VANG THICKNESS = 3/4"
B. MAXIMUM CONDUCTOR VANG THICKNESS = 3/4", UNLESS NOTED OTHERWISE
C. MAXIMUM MESSENGER VANG THICKNESS = 1 1/4", UNLESS NOTED OTHERWISE
D. LOAD TREE VALUES ARE ULTIMATE, INCLUDE OVERLOAD FACTORS, AND ARE SHOWN IN POUNDS.
E. POLE FLAT DESIGNATIONS REFER TO SIDE OF THE POLE WHERE ATTACHMENTS ARE LOCATED

DESIGN CRITERIA

LOADING PARAMETERS FOR: ROARING SPRINGS POLE

(N_BK) NEUTRAL WIRE IS 144 FIBER OPGW, MAXIMUM TENSION AT NESC HEAVY LOADING: 15400# R.S. = 1340'
(M1_BK) MESSENGER WIRE IS 37 NO. 8 AW, MAXIMUM TENSION AT NESC HEAVY LOADING: 17300# R.S. = 1340'
(M2_BK) MESSENGER WIRE IS 37 NO. 8 AW SUPPORTING (3) 1/0 15 KV INSULATED SPACER CABLES, MAXIMUM TENSION AT NESC HEAVY LOADING: 37900# R.S. = 1340'

WIND SPAN: 1400' (BACK)
MAXIMUM WEIGHT SPAN: −4200' (N_BK)
−3000' (M1_BK)
−2600' (M2_BK)

HEIGHT OF STR.: 60'−0"
ACTUAL ANGLE (Δ): 0°0'00"
DESIGN MAXIMUM ANGLE: 3°00'00"

SPECIFICATIONS

1. POLES TO BE DESIGNED PER TECHNICAL SPECIFICATION (33 71 16.23). IF CONFLICT EXIST BETWEEN THIS DRAWING AND THE FABRICATION SPEC, FOLLOW THE DETAILS ON THIS DRAWING. IF CLARIFICATION IS NECESSARY, CONTACT THE CONTRACTING OFFICER REPRESENTATIVE.
2. CAMBERING OF POLES WILL NOT BE PERMITTED.
3. ALL STEEL TO BE WEATHERING STEEL.
4. POLE TO HAVE BOLTED FLANGE TYPE SPLICES.
5. THE (*) LOADINGS SHOWN IN LOAD TABLE CONVEY THE ADDITIONAL VERTICAL LOAD THAT WILL BE APPLIED TO THE STRUCTURE BY LINE PERSONNEL DURING CONSTRUCTION AND MAINTENANCE OPERATIONS.
   A) IT IS REQUIRED THAT ALL CONDUCTOR ATTACHMENT SUPPORTS BE DESIGNED TO MEET THIS REQUIREMENT.
6. BASE DIAMETER OF DESIGN STEEL POLES SHALL BE DESIGNED/PROPOSED IN 6" INCREMENTS TO COINCIDE WITH FOUNDATION DESIGN.
7. STRUCTURE SHALL BE DESIGNED ASSUMING MICROPILE TYPE FOUNDATION.
   A) STEEL VENDOR TO PROVIDE TWO (2) ADDITIONAL GROUNDING PADS FOR EACH POLE TO BE FIELD WELDED ON EACH MICROPILE.
   B) NO FIELD WELDING WILL BE ALLOWED ON THE STRUCTURES EXCEPT TO WELD THE GROUND PADS TO THE MICROPILE FOUNDATION. STEEL VENDOR TO PROVIDE FIELD WELD DESIGN AND DETAILS IN ACCORDANCE WITH AWS D1.1 FOR WELDING GROUND PADS TO MICROPILE FOUNDATIONS.
8. ALL LADDER PROVISIONS SHALL MEET OSHA FALL PROTECTION REQUIREMENTS. VERIFY LADDER CLIPS AND WELDS CAN HANDLE 5000 LBS. OSHA IMPACT LOAD.
9. VENDOR TO VERIFY THAT TOP VANG CAN BE USED FOR LIFTING ASSEMBLED STRUCTURE.
10. VENDOR TO DESIGN STRUCTURE TO HAVE A MAXIMUM UTILIZATION OF 90%.

DEFLECTION

DEFLECTION SHALL BE LIMITED TO 1.5% OR LESS OF THE ABOVE GROUND LENGTH UNDER DEFLECTION LOAD CASE.

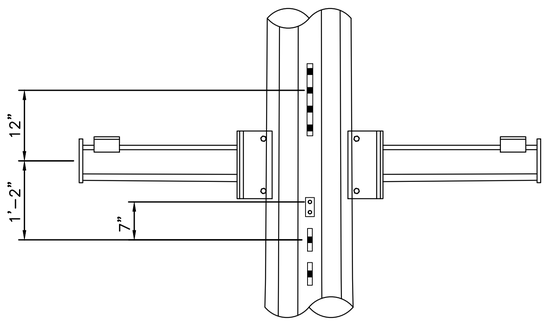
Ⓐ IDENTIFICATION TAG

STR #	RS
STR LENGTH (FEET)	60'
STRUCTURE TYPE	LD−RS
MANUFACTURED DATE	(DATE)
MAX GRD. LINE MOM. (FT−KIPS)	(####)

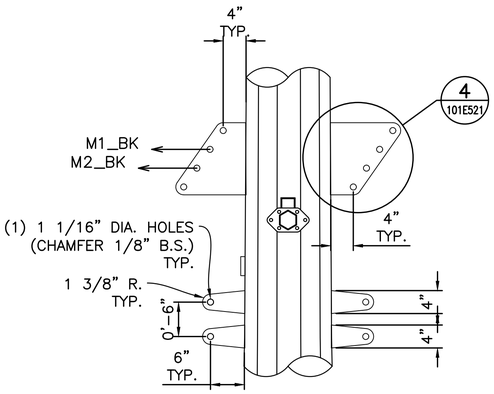
REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 75014 TOU VANG STATE OF COLORADO

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: B. PLANT DRAWN: B. PLANT TECH. REVIEW: J. DOHRENDORF DATE: MAY 2026	SUB SHEET NO. 101 E520	TITLE OF SHEET POWER DISTRIBUTION STRUCTURAL & ASSEMBLY LOAD AND DESIGN ROARING SPRINGS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 36 OF 50

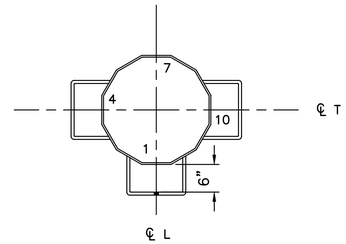
4/17/2026 Walchuk, Chris C:\Users\cwalchuk\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\ELEC\101-E520.dwg



FRONT VIEW



SIDE VIEW



TOP VIEW

GENERAL NOTES:

A. SEE SHEET 101E520 FOR STRUCTURE SECTION LAYOUT AND DESIGN AT ROARING SPRINGS.

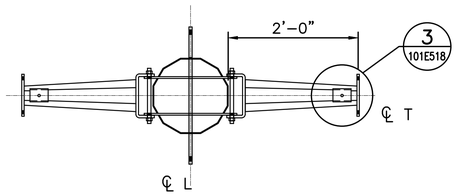
B. SEE SHEET 101E522 FOR STRUCTURE LOADING AT ROARING SPRINGS.

C. SEE SHEET 101E526 FOR CROSSARM LOADING REQUIREMENTS.

D. SEE SHEET 101E105 FOR ROARING SPRINGS SITE PLAN.

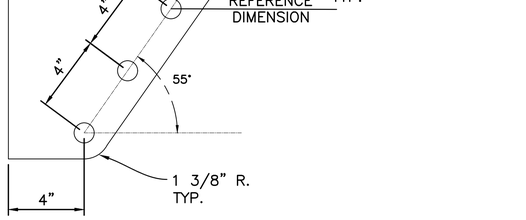
2 / 101E520 DETAIL: MESSENGER & SPACER CABLE ATTACHMENT

NOT TO SCALE



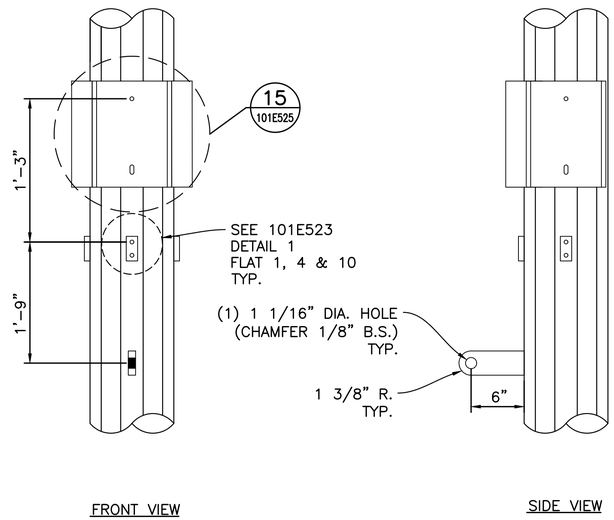
B / 101E520 SECTION VIEW: SPACER CABLE ARM

NOT TO SCALE



4 / 101E521 DETAIL: MESSENGER ATTACHMENT VANG

NOT TO SCALE

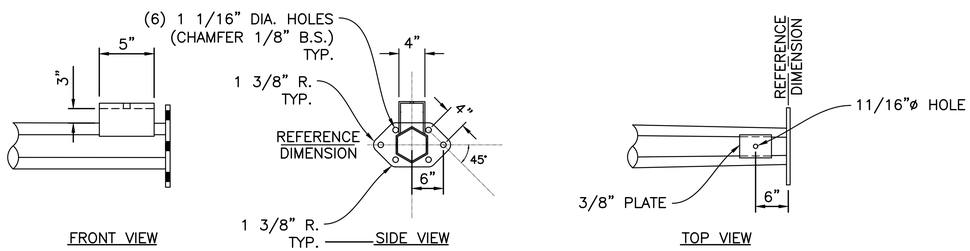


FRONT VIEW

SIDE VIEW

5 / 101E520 DETAIL: THREE–PHASE RISER BRACKET

NOT TO SCALE



FRONT VIEW

SIDE VIEW

TOP VIEW

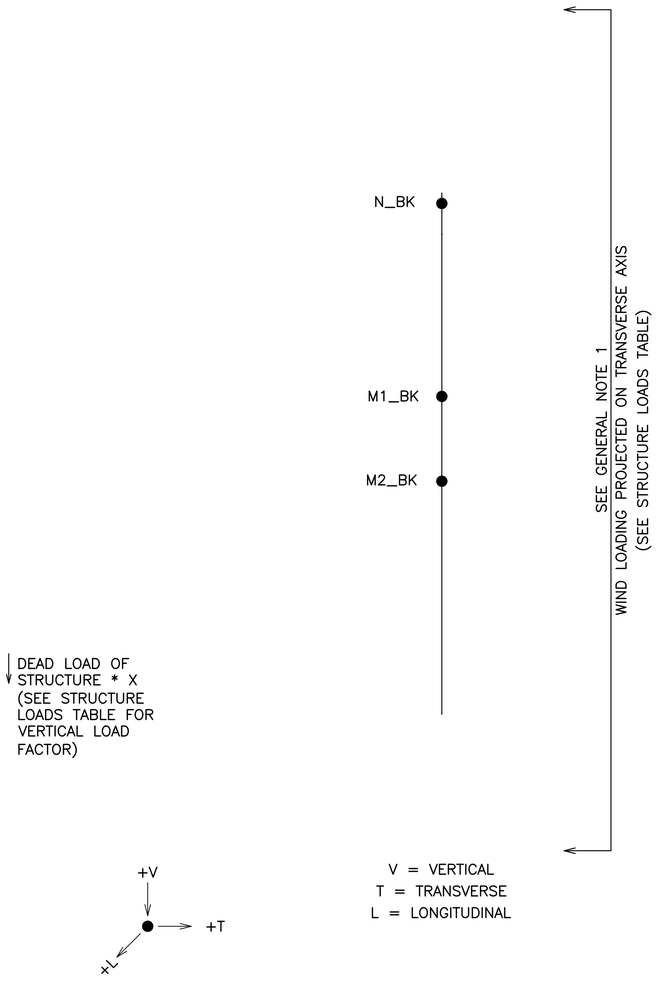
3 / 101E521 DETAIL: CROSSARM ATTACHMENT

NOT TO SCALE

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 75014 TOU VANG STATE OF COLORADO

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: B. PLANT CADD B. PLANT TECH. REVIEW: J. DOHRENDORF DATE: MAY 2026	SUB SHEET NO. 101 E521	TITLE OF SHEET POWER DISTRIBUTION STRUCTURAL & ASSEMBLY LOAD AND DESIGN ROARING SPRINGS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 37 OF 50

4/17/2026 Walchuk, Chris C:\Users\Cwalchuk\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\ELEC\101-E521.dwg



STRUCTURE LOADS (INCLUDES ALL APPLICABLE LOAD FACTORS)								
LOAD CASE	TEMPERATURE (°F)	RADIAL ICE @ 56 LBS/FT^3 (INCHES)	VERTICAL OVERLOAD FACTOR "X"	WIND LOADING (PSF)	POINT LOAD NAME	LONGITUDINAL (LBS)	TRANSVERSE (LBS)	VERTICAL (LBS)
NESC 250B	0	0.5	1.5	4	N_BK	23200	−3300	−9200
					M1_BK	26100	−3600	−10100
					M2_BK	57900	−11100	−19900
NESC 250C	60	0.0	1.1	20.74	N_BK	11600	−2000	−3300
					M1_BK	11700	−2100	−4600
					M2_BK	26100	−6100	−6100
NESC 250D	15	0.25	1.1	2.30	N_BK	12200	−1100	−4700
					M1_BK	12800	−1100	−5800
					M2_BK	25700	−2800	−9500
EXTREME ICE	20	1.0	1	0	N_BK	16600	−900	−11800
					M1_BK	19900	−1100	−11200
					M2_BK	45300	−2400	−27500
DEFLECTION	40	0.0	1	2	N_BK	9400	−700	−3000
					M1_BK	9300	−700	−4200
					M2_BK	17300	−1600	−5500
CONSTRUCTION	−20	0.0	1.1	2	N_BK	16400	−1100	−2100*
					M1_BK	15600	−1100	−3400*
					M2_BK	28900	−2300	−4900*
CONSTRUCTION STRINGING	−20	0.0	1.1	2	N_BK	5000	−1100	6500*
					M1_BK	5000	−1200	5000*
					M2_BK	8900	−2400	10500*

*MAN LOADED

GENERAL NOTES:

A. TRANSVERSE WIND LOAD IS TO BE APPLIED IN THE DIRECTION RESULTING IN THE GREATEST TRANSVERSE GROUND LINE MOMENT FOR ALL LOAD CASES.

B. SEE SHEET 101E520 & 101E521 FOR STRUCTURE LAYOUT AND DESIGN AT ROARING SPRINGS

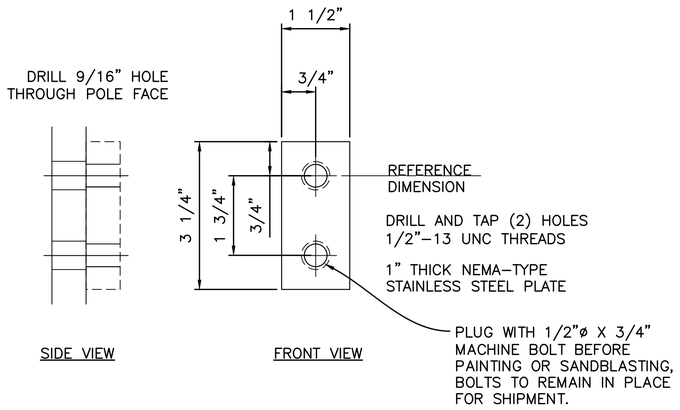
C. SEE SHEET 101E526 FOR CROSSARM LOADING REQUIREMENTS.

D. SEE SHEET 101E105 FOR ROARING SPRINGS SITE PLAN.

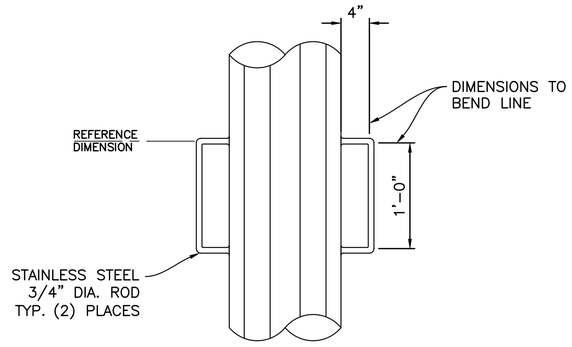
4/17/2026 Walchuk, Chris C:\Users\Cwalchuk\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\ELEC\101-E522.dwg

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 75014 TOU VANG

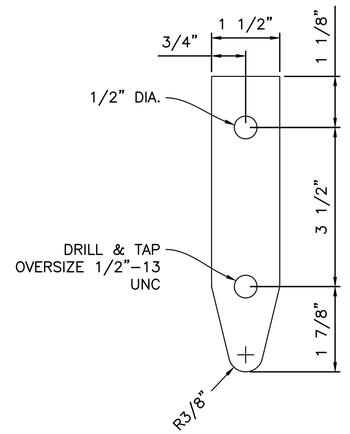
A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	B. PLANT	101 E522	POWER DISTRIBUTION STRUCTURAL & ASSEMBLY LOAD AND DESIGN ROARING SPRINGS NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752
	CADD: B. PLANT			PMIS/PKG NO. 318719
	TECH. REVIEW: J. DOHRENDORF			SHEET 38 OF 50
	DATE: MAY 2026			



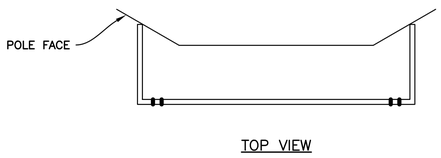
1 / 101E523 DETAIL: GROUNDING PAD
NOT TO SCALE

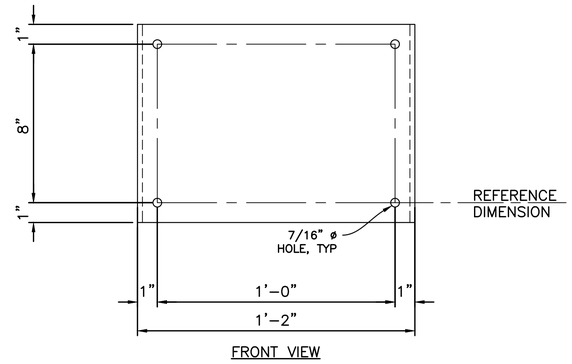


2 / 101E523 DETAIL: GROUNDING LOOP
NOT TO SCALE

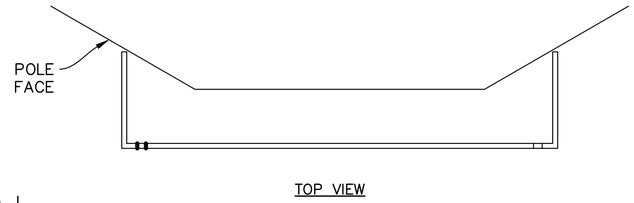


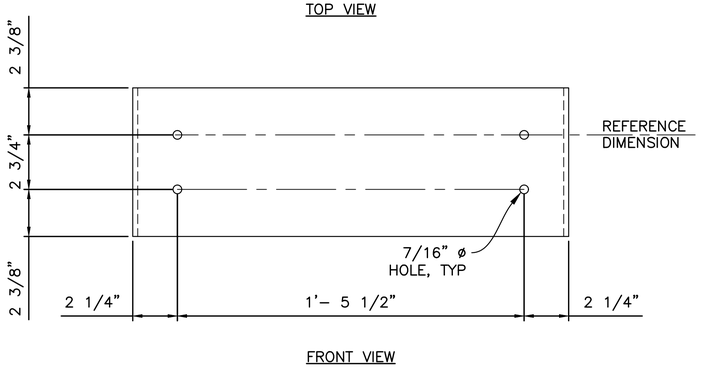
A / 101E523 VIEW
NOT TO SCALE



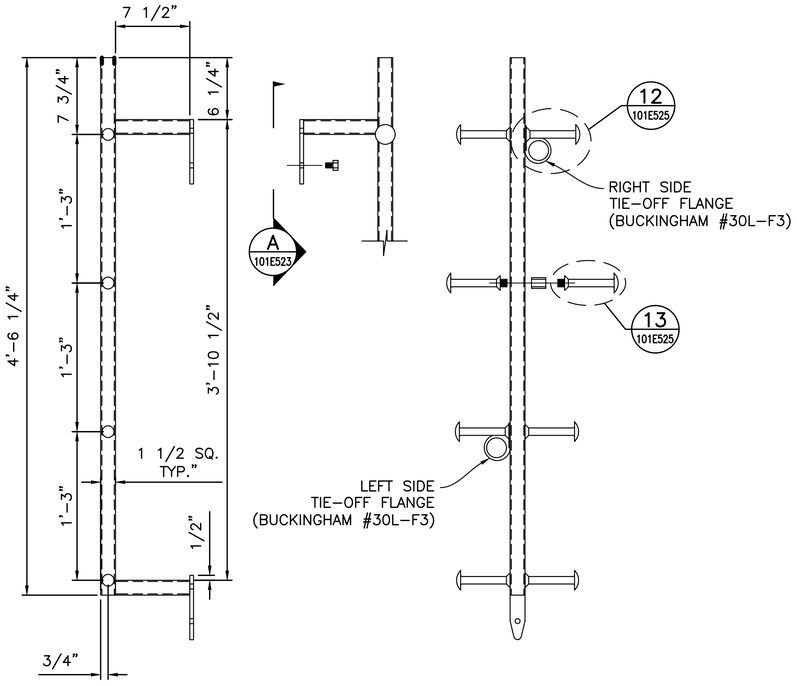


3 / 101E523 DETAIL: DANGER SIGN BRACKET
NOT TO SCALE





4 / 101E523 DETAIL: STRUCTURE NUMBER BRACKET
NOT TO SCALE

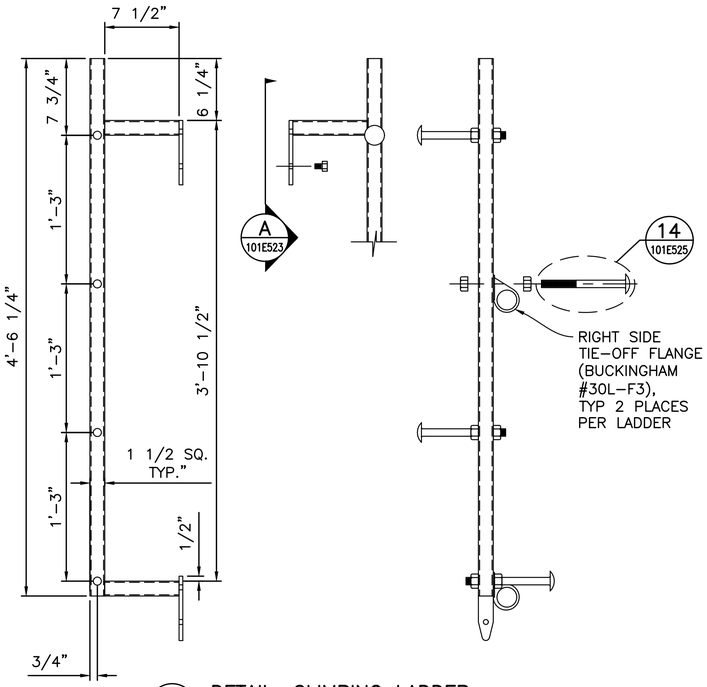


5 / 101E523 DETAIL: WORKING LADDER
NOT TO SCALE

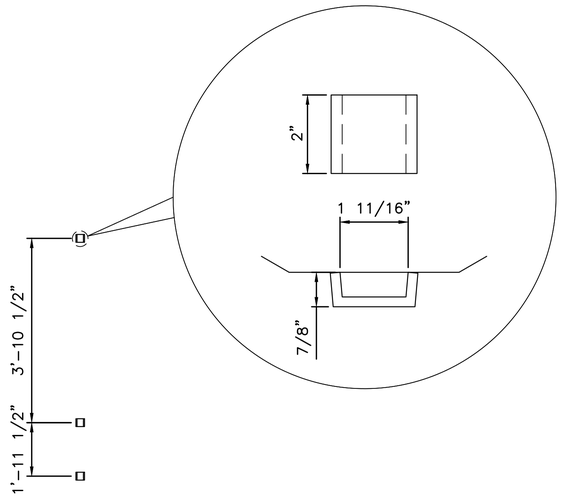
REGISTERED PROFESSIONAL ENGINEER
CERTIFICATE NO. 75014
TOU VANG
STATE OF COLORADO

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: B. PLANT CADD B. PLANT TECH. REVIEW: J. DOHRENDORF DATE: MAY 2026	SUB SHEET NO. 101 E523	TITLE OF SHEET POWER DISTRIBUTION STRUCTURAL & ASSEMBLY LOAD AND DESIGN DETAILS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 39 OF 50

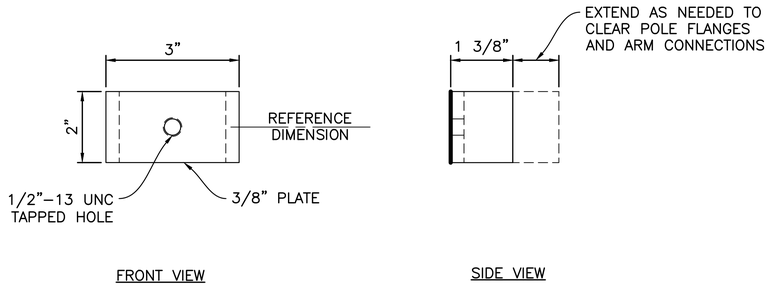
4/17/2026 Walchuk, Chris C:\Users\cwalchuk\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\ELEC\101-E523.dwg



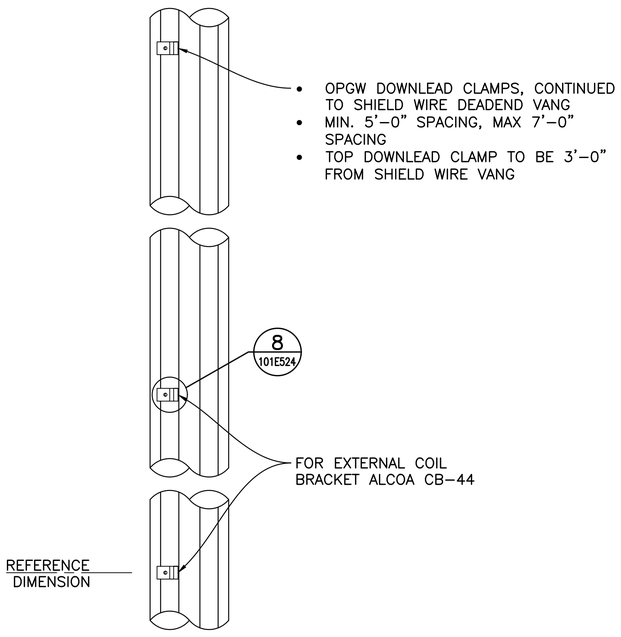
6 101E524 DETAIL: CLIMBING LADDER
NOT TO SCALE



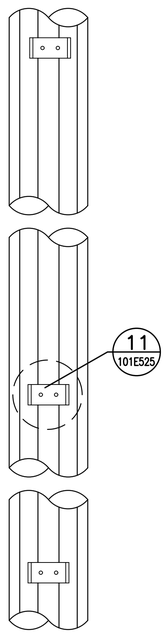
7 101E524 DETAIL: LADDER CLIPS FOR CLIMBING AND WORKING PROVISIONS
NOT TO SCALE



8 101E524 DETAIL: OPGW & NEUTRAL STANDOFF BRACKETS
NOT TO SCALE



9 101E524 DETAIL: OPGW & NEUTRAL BRACKET & DOWNLOAD ATTACHMENT
NOT TO SCALE



10 101E524 DETAIL: RISER SUPPORT BRACKET
NOT TO SCALE

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 75014 TOU VANG STATE OF OHIO

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: B. PLANT DRAWN: B. PLANT TECH. REVIEW: J. DOHRENDORF DATE: MAY 2026	SUB SHEET NO. 101 E524	TITLE OF SHEET POWER DISTRIBUTION STRUCTURAL & ASSEMBLY LOAD AND DESIGN DETAILS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 40 OF 50

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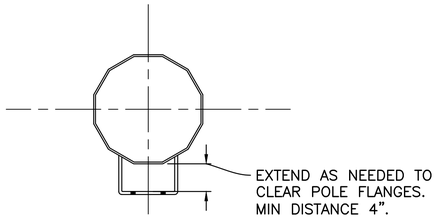
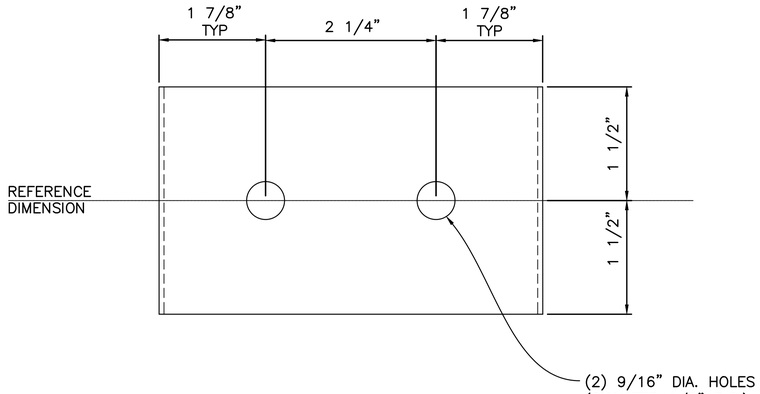
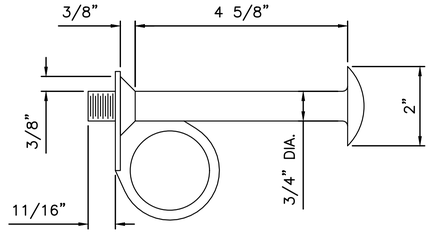


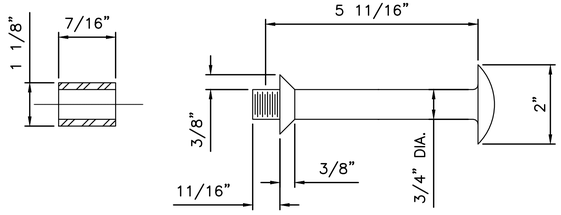
TOP VIEW



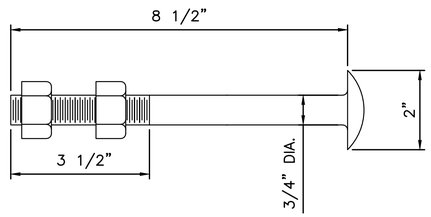
11 / 101E525 VIEW: RISER SUPPORT BRACKET
NOT TO SCALE



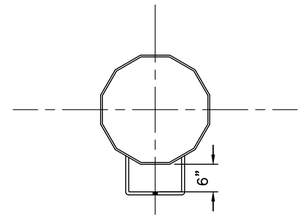
12 / 101E525 DETAIL: WORKING LADDER STEP BOLT WITH TIE–OFF FLANGE
NOT TO SCALE



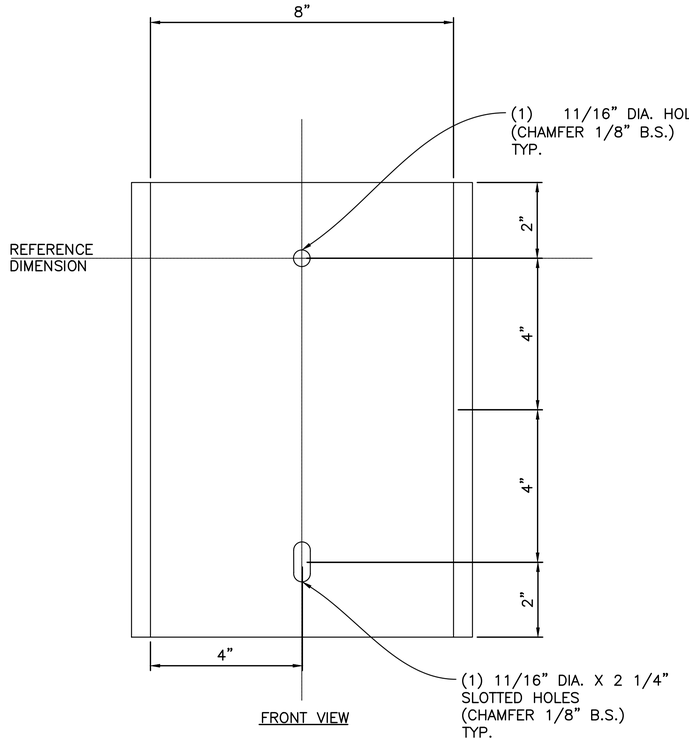
13 / 101E525 DETAIL: WORKING LADDER STEP BOLT WITHOUT TIE–OFF FLANGE
NOT TO SCALE



14 / 101E525 DETAIL: CLIMBING LADDER STEP BOLT
NOT TO SCALE



TOP VIEW



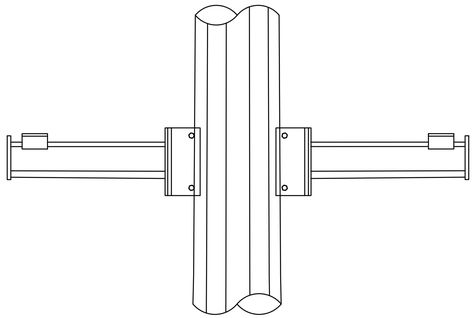
FRONT VIEW

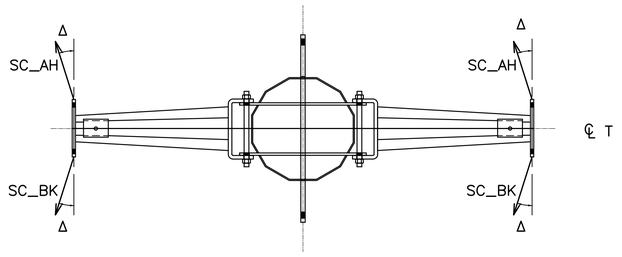
15 / 101E525 DETAIL: CABLE TERMINATION BRACKET
NOT TO SCALE

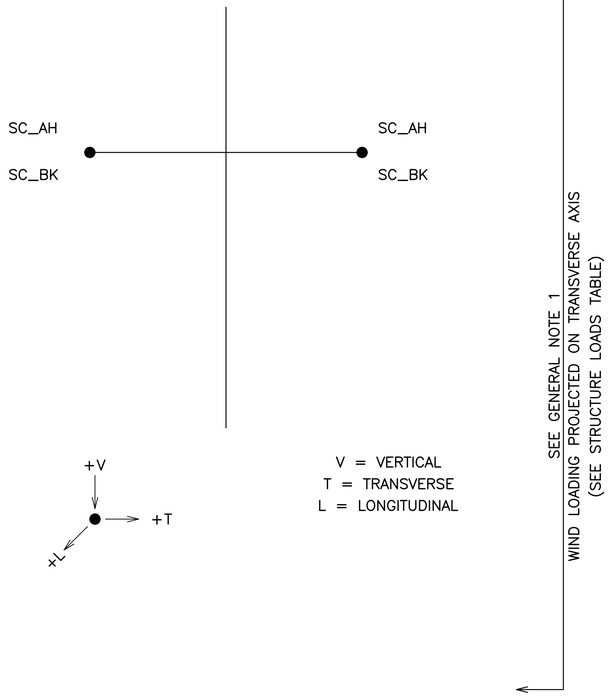
REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 75014 TOU VANG STATE OF COLORADO

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: B. PLANT CADD B. PLANT TECH. REVIEW: J. DOHRENDORF DATE: MAY 2026	SUB SHEET NO. 101 E525	TITLE OF SHEET POWER DISTRIBUTION STRUCTURAL & ASSEMBLY LOAD AND DESIGN DETAILS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 41 OF 50

4/17/2026 Watchuk, Chris C:\Users\cwatchuk\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\ELEC\101-E525.dwg







UPPER BENCH CROSSARM LOADS
(INCLUDES ALL APPLICABLE LOAD FACTORS)

LOAD CASE	TEMPERATURE (°F)	RADIAL ICE @ 56 LBS/FT^3 (INCHES)	WIND LOADING (PSF)	POINT LOAD NAME	LONGITUDINAL (LBS)	TRANSVERSE (LBS)	VERTICAL (LBS)
NESC 250B	0	0.5	4	SC_BK	0	0	0
				SC_AH	−900	−1600	5300
NESC 250C	60	0.0	20.74	SC_BK	0	0	0
				SC_AH	−400	−1100	1000
NESC 250D	15	0.25	2.30	SC_BK	0	0	0
				SC_AH	−400	300	2100
EXTREME ICE	20	1.0	0	SC_BK	0	0	0
				SC_AH	−1100	−200	8500
DEFLECTION	40	0.0	2	SC_BK	0	0	0
				SC_AH	−200	200	900
CONSTRUCTION	−20	0.0	2	SC_BK	0	0	0
				SC_AH	−200	−200	2200*
CONSTRUCTION STRINGING	−20	0.0	2	SC_BK	0	0	0
				SC_AH	−400	−300	2500*

*MAN LOADED

LOWER BENCH CROSSARM LOADS
(INCLUDES ALL APPLICABLE LOAD FACTORS)

LOAD CASE	TEMPERATURE (°F)	RADIAL ICE @ 56 LBS/FT^3 (INCHES)	WIND LOADING (PSF)	POINT LOAD NAME	LONGITUDINAL (LBS)	TRANSVERSE (LBS)	VERTICAL (LBS)
NESC 250B	0	0.5	4	SC_BK	1000	−1600	−2900
				SC_AH	−900	−2100	6500
NESC 250C	60	0.0	20.74	SC_BK	400	−1100	−500
				SC_AH	−400	−1400	1200
NESC 250D	15	0.25	2.30	SC_BK	300	−300	−1100
				SC_AH	−300	−400	2600
EXTREME ICE	20	1.0	0	SC_BK	1100	−100	−4700
				SC_AH	−1100	−100	10800
NESC 250B (NO WIRES AHEAD)	0	0.5	4	SC_BK	1000	−1600	−2900
				SC_AH	0	−100	100
NESC 250B (NO WIRES BACK)	0	0.5	4	SC_BK	0	−100	100
				SC_AH	−900	−2100	6500
DEFLECTION	40	0.0	2	SC_BK	200	−200	−500
				SC_AH	−100	−200	1100
CONSTRUCTION	−20	0.0	2	SC_BK	200	−200	700*
				SC_AH	−200	−200	2400*
CONSTRUCTION STRINGING	−20	0.0	2	SC_BK	300	−200	700*
				SC_AH	−500	−400	2800*

*MAN LOADED

ROARING SPRINGS BENCH CROSSARM LOADS
(INCLUDES ALL APPLICABLE LOAD FACTORS)

LOAD CASE	TEMPERATURE (°F)	RADIAL ICE @ 56 LBS/FT^3 (INCHES)	WIND LOADING (PSF)	POINT LOAD NAME	LONGITUDINAL (LBS)	TRANSVERSE (LBS)	VERTICAL (LBS)
NESC 250B	0	0.5	4	SC_BK	1000	−2000	−3600
				SC_AH	0	0	0
NESC 250C	60	0.0	20.74	SC_BK	500	1100	−600
				SC_AH	0	0	0
NESC 250D	15	0.25	2.30	SC_BK	300	−400	−1400
				SC_AH	0	0	0
EXTREME ICE	20	1.0	0	SC_BK	1100	−100	−5900
				SC_AH	0	0	0
DEFLECTION	40	0.0	2	SC_BK	200	−200	−300
				SC_AH	0	0	0
CONSTRUCTION	−20	0.0	2	SC_BK	200	−200	600*
				SC_AH	0	0	0
CONSTRUCTION STRINGING	−20	0.0	2	SC_BK	300	−300	700*
				SC_AH	0	0	0

*MAN LOADED

GENERAL NOTES

A. MAXIMUM CONDUCTOR VANG THICKNESS = 3/4", UNLESS NOTED OTHERWISE LOAD TREE VALUES ARE ULTIMATE, INCLUDE OVERLOAD FACTORS, AND ARE SHOWN IN POUNDS.

DESIGN CRITERIA

(SC_AH & SC_BK) CONDUCTOR IS 1/0 KCMIL AAC 15KV INSULATED SPACER CABLE," HORZONTAL TENSION AT 60°F INITIAL: 100 LBS

UPPER BENCH:

RS: 0' (SC_BK), 1270' (SC_AH)

DESIGN MAXIMUM WIND SPAN:0' (SC_BK), 1050' (SC_AH)

DESIGN MAXIMUM WEIGHT SPAN:0' (SC_BK), 3750' (SC_AH)

DESIGN MAXIMUM ANGLE (Δ):6°30'00"

LOWER BENCH:

RS:1270' (SC_BK), 1340' (SC_AH)

DESIGN MAXIMUM WIND SPAN:1050' (SC_BK), 1400' (SC_AH)

DESIGN MAXIMUM WEIGHT SPAN:−2100' (SC_BK), 4600' (SC_AH)

DESIGN MAXIMUM ANGLE (Δ):5°00'00"

LOWER BENCH:

RS:1270' (SC_BK), 0' (SC_AH)

DESIGN MAXIMUM WIND SPAN:1400' (SC_BK), 0' (SC_AH)

DESIGN MAXIMUM WEIGHT SPAN:−2600' (SC_BK), 0' (SC_AH)

DESIGN MAXIMUM ANGLE (Δ):3°00'00"

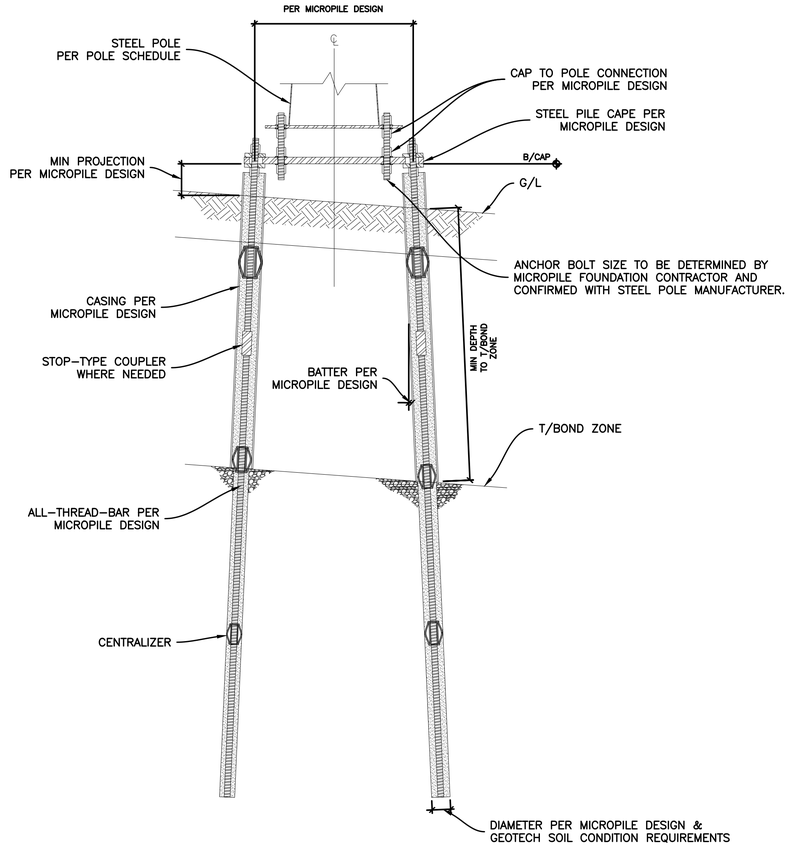
SPECIFICATIONS

1. ARMS TO BE DESIGNED PER TECHNICAL SPECIFICATION (33 71 16.23). IF CONFLICT EXIST BETWEEN THIS DRAWING AND THE FABRICATION SPEC, FOLLOW THE DETAILS ON THIS DRAWING. IF CLARIFICATION IS NECESSARY, CONTACT THE OWNER'S REPRESENTATIVE.
2. ALL STEEL TO BE WEATHERING STEEL.
3. THE (*) LOADINGS SHOWN IN LOAD TABLES CONVEY THE ADDITIONAL VERTICAL LOAD THAT WILL BE APPLIED TO THE STRUCTURE BY LINE PERSONNEL DURING CONSTRUCTION AND MAINTENANCE OPERATIONS.
   A) IT IS REQUIRED THAT ALL CONDUCTOR ATTACHMENT SUPPORTS BE DESIGNED TO MEET THIS REQUIREMENT. HOWEVER, AS THE LOADING WILL BE APPLIED TO ONLY ONE SHIELD AND ONE PHASE AT ANY GIVEN TIME, THE ARM MAY BE DESIGNED FOR ITS APPLICATION AT A SINGLE CONTROLLING PHASE LOCATION TO BE DETERMINED BY THE STEEL MANUFACTURER.
4. ARM SHALL BE HERMETICALLY SEALED.

REGISTERED PROFESSIONAL ENGINEER COLORADO
CERTIFICATE NO. 75014
TOU VANG

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	B. PLANT	101 E526	POWER DISTRIBUTION STRUCTURAL & ASSEMBLY LOAD AND DESIGN DETAILS	113 182752
	CADD: B. PLANT			PMIS/PKG NO. 318719
	TECH. REVIEW: J. DOHRENDORF		NORTH RIM - GRAND CANYON NATIONAL PARK	SHEET 42 OF 50
	DATE: MAY 2026			

4/17/2026 Walchuk, Chris C:\Users\Cwalchuk\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\ELEC\101-E526.dwg



1 / 101E527 MICROPILE FOUNDATION DETAIL

NOT TO SCALE

GENERAL NOTES:

A. ALL WORK SHALL BE IN ACCORDANCE WITH THE PROJECT MICROPILE FOUNDATION PERFORMANCE SPECIFICATION.

B. CONTRACTOR SHALL BE RESPONSIBLE FOR DESIGN AND INSTALLATION OF MICROPILE FOUNDATIONS. FOUNDATIONS TO BE DESIGNED TO SUPPORT STRUCTURE DESIGN LOADS SHOWN IN DRAWINGS. GROUND LINE REACTIONS AND ANCHOR BOLT DIMENSIONS TO BE DETERMINED AND COORDINATED BY MICROPILE FOUNDATION CONTRACTOR AND STEEL POLE MANUFACTURER.

C. FOUNDATION PERFORMANCE CRITERIA:
   - TOTAL DEFLECTION: 3.0 IN
   - TOTAL ROTATION 1.5 DEG
   - NON–RECOVERABLE DEFLECTION: 1.5 IN
   - NON–RECOVERABLE ROTATION: 0.75 DE

D. CONTRACTOR TO PERFORM ON–SITE GEOTECHNICAL INVESTIGATION DURING INSTALLATION.

E. CONTRACTOR TO BE ABLE TO TRANSPORT ALL REQUIRED EQUIPMENT, MATERIAL, AND PERSONNEL BY HELICOPTER TO ALL SITES SHOWN IN THIS PACKAGE.

F. THE CONTRACTOR SHALL PROVIDE INDIVIDUALS EXPERIENCED IN MICROPILE CONSTRUCTION AND FAMILIAR WITH THE DESIGN ASSUMPTIONS FOR THESE MICROPILES TO MONITOR ALL ASPECTS OF THE MICROPILE INSTALLATION AND TESTING. THIS INDIVIDUAL SHALL CONSULT WITH THE MICROPILE DESIGNER IF CONDITIONS ARE DIFFERENT THAN THOSE DESIGNED FOR.

G. CONTRACTOR TO PROVIDE ALL MEANS AND METHODS FOR PERFORMING WORK.

H. EMBEDMENT DEPTH OF MICROPILE PIERS SHALL BE DETERMINED BY CONTRACTOR DURING THEIR ONSITE GEOTECHNICAL INVESTIGATION DURING INSTALLATION.

I. THE CONTRACTOR SHALL SUBMIT DESIGN AND CALCULATIONS SEALED BY A PROFESSIONAL ENGINEER REGISTERED IN THE STATE OF ARIZONA.

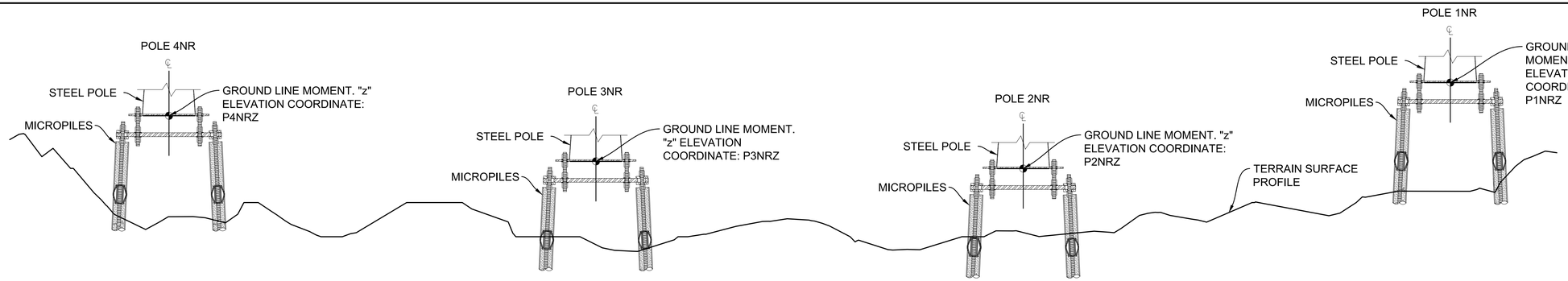
J. ALL PROVIDED LOADS ARE PRELIMINARY AND NOT FOR CONSTRUCTION. FINAL LOADS TO BE BASED ON STEEL VENDOR CALCULATIONS.

K. CONTRACTOR TO COORDINATE AND PROVIDE GROUNDING PREVISIONS WITH MICROPILES. STEEL POLES TO BE BONDED TO MICROPILES. MICROPILES TO BE USED AS ADDITIONAL GROUNDING ELECTRODE.

REGISTERED PROFESSIONAL ENGINEER CERTIFICATE NO. 75014 TOU VANG STATE OF ARIZONA

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: B. PLANT DRAWN: C. WALCHUK TECH. REVIEW: J. DOHRENDORF DATE: MAY 2026	SUB SHEET NO. 101 E527	TITLE OF SHEET POWER DISTRIBUTION MICROPILE DETAILS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 43 OF 50

4/17/2026 Walchuk, Chris C:\Users\Cwalchuk\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\ELEC\101-E527.dwg



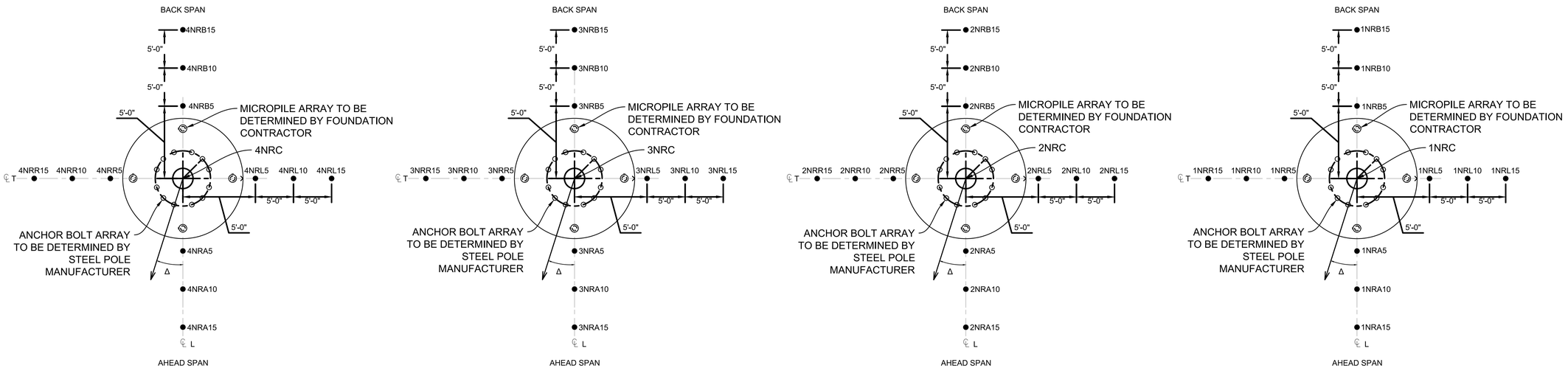
1 / 101E528 NORTH RIM FOUNDATION COORDINATE DETAIL ELEVATION VIEW

NOT TO SCALE

GENERAL NOTES:

A. GROUND LINE MOMENT IS LOCATED AT BOTTOM OF POLE BASE PLATE.

LINE ANGLE
ACTUAL ANGLE (Δ): 0°00'00"
DESIGN MAXIMUM ANGLE: 3°00'00"



POLE 4NR COORDINATES			
POINT	X COORDINATE	Y COORDINATE	Z COORDINATE
P4NRZ	1330324.529	13144814.14	8181.088
4NRC	1330324.529	13144814.14	8175.17
4NRA5	1330328.822	13144811.57	8172.743
4NRA10	1330333.114	13144809.01	8170
4NRA15	1330337.407	13144806.44	8170
4NRB5	1330320.237	13144816.7	8179.088
4NRB10	1330315.944	13144819.27	8184.506
4NRB15	1330311.652	13144821.83	8188.244
4NRL5	1330327.093	13144818.4300	8176
4NRL10	1330329.657	13144822.72	8174
4NRL15	1330332.222	13144827.01	8176
4NRR5	1330321.965	13144809.84	8177.277
4NRR10	1330319.401	13144805.55	8179.456
4NRR15	1330316.837	13144801.259	8176

POLE 3NR COORDINATES			
POINT	X COORDINATE	Y COORDINATE	Z COORDINATE
P3NRZ	1330337.35	13144835.5990	8178.433
3NRC	1330337.35	13144835.5990	8173.527
3NRA5	1330341.642	13144833.04	8170.757
3NRA10	1330345.935	13144830.47	8167.416
3NRA15	1330350.227	13144827.91	8164.217
3NRB5	1330333.057	13144838.16	8176.433
3NRB10	1330328.765	13144840.73	8179.473
3NRB15	1330324.472	13144843.29	8184
3NRL5	1330339.914	13144839.89	8174
3NRL10	1330342.478	13144844.18	8174.9
3NRL15	1330345.042	13144848.48	8173.864
3NRR5	1330334.786	13144831.31	8174.498
3NRR10	1330332.222	13144827.01	8176
3NRR15	1330329.657	13144822.72	8174

POLE 2NR COORDINATES			
POINT	X COORDINATE	Y COORDINATE	Z COORDINATE
P2NRZ	1330350.17	13144857.06	8178
2NRC	1330350.17	13144857.06	8174.245
2NRA5	1330354.463	13144854.5	8170.478
2NRA10	1330358.755	13144851.93	8168.324
2NRA15	1330363.048	13144849.37	8168
2NRB5	1330345.878	13144859.63	8176
2NRB10	1330341.585	13144862.19	8179.542
2NRB15	1330337.293	13144864.75	8184.655
2NRL5	1330352.734	13144861.35	8174.265
2NRL10	1330355.298	13144865.65	8175.011
2NRL15	1330357.863	13144869.94	8175.772
2NRR5	1330347.606	13144852.77	8173.563
2NRR10	1330345.042	13144848.48	8173.864
2NRR15	1330342.478	13144844.18	8174.9

POLE 1NR COORDINATES			
POINT	X COORDINATE	Y COORDINATE	Z COORDINATE
P1NRZ	1330362.991	13144878.52	8183.037
1NRC	1330362.991	13144878.52	8176.662
1NRA5	1330367.283	13144875.96	8174.225
1NRA10	1330371.576	13144873.4	8170.547
1NRA15	1330375.868	13144870.83	8167.162
1NRB5	1330358.698	13144881.09	8181.037
1NRB10	1330354.406	13144883.65	8183.08
1NRB15	1330350.113	13144886.22	8186.805
1NRL5	1330365.555	13144882.82	8178.773
1NRL10	1330368.119	13144887.11	8178.471
1NRL15	1330370.683	13144891.4	8178
1NRR5	1330360.427	13144874.23	8176.315
1NRR10	1330357.863	13144869.94	8175.772
1NRR15	1330355.298	13144865.65	8175.011

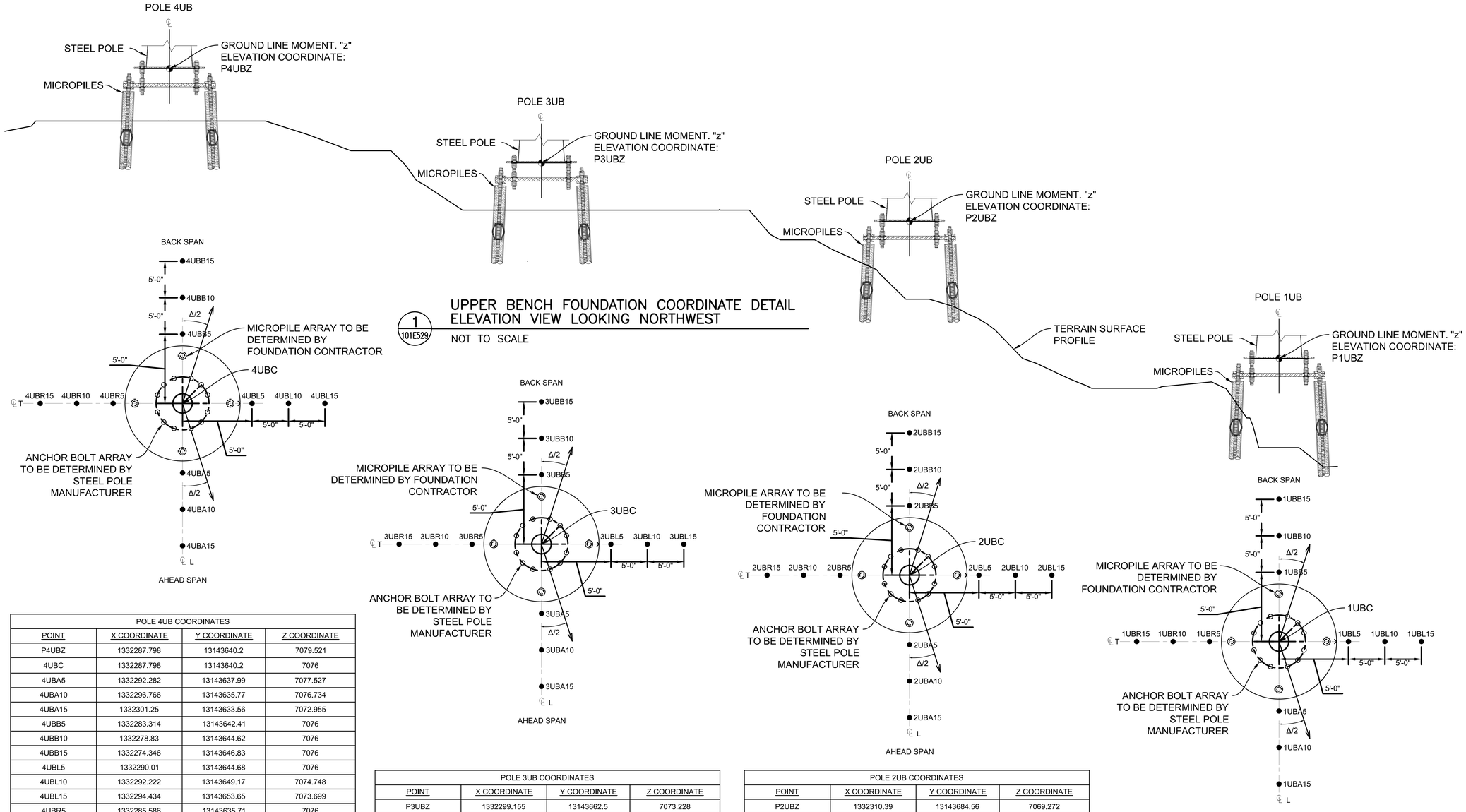
2 / 101E528 NORTH RIM FOUNDATION COORDINATE DETAIL PLAN VIEW

NOT TO SCALE

REGISTERED PROFESSIONAL ENGINEER (ELECTRICAL) CERTIFICATE NO. 75014 TOU VANG

A/E FIRM: PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: B. PLANT CADD: C. WALCHUK TECH. REVIEW: J. DOHRENDORF DATE: MAY 2026	SUB SHEET NO. 101 E528	TITLE OF SHEET POWER DISTRIBUTION MICROPILE FOUNDATION DETAILS NORTH RIM NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 44 OF 50

4/29/2026 Lynch, Nicole C:\Users\nlynch\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\ELEC\101-E528-530.dwg



1 / 101E529 UPPER BENCH FOUNDATION COORDINATE DETAIL ELEVATION VIEW LOOKING NORTHWEST

NOT TO SCALE

GENERAL NOTES:

A. GROUND LINE MOMENT IS LOCATED AT BOTTOM OF POLE BASE PLATE.

LINE ANGLE
ACTUAL ANGLE (Δ): 9°11'18"
DESIGN MAXIMUM ANGLE: 13°00'00"

POLE 4UB COORDINATES			
POINT	X COORDINATE	Y COORDINATE	Z COORDINATE
P4UBZ	1332287.798	13143640.2	7079.521
4UBC	1332287.798	13143640.2	7076
4UBA5	1332292.282	13143637.99	7077.527
4UBA10	1332296.766	13143635.77	7076.734
4UBA15	1332301.25	13143633.56	7072.955
4UBB5	1332283.314	13143642.41	7076
4UBB10	1332278.83	13143644.62	7076
4UBB15	1332274.346	13143646.83	7076
4UBL5	1332290.01	13143644.68	7076
4UBL10	1332292.222	13143649.17	7074.748
4UBL15	1332294.434	13143653.65	7073.699
4UBR5	1332285.586	13143635.71	7076
4UBR10	1332283.374	13143631.23	7075.169
4UBR15	1332281.162	13143626.75	7076

POLE 1UB COORDINATES			
POINT	X COORDINATE	Y COORDINATE	Z COORDINATE
P1UBZ	1332321.691	13143706.74	7060.094
1UBC	1332321.691	13143706.74	7054
1UBA5	1332326.175	13143704.53	7055.541
1UBA10	1332330.659	13143702.32	7058
1UBA15	1332335.143	13143700.11	7054.375
1UBB5	1332317.207	13143708.95	7056
1UBB10	1332312.723	13143711.17	7055.752
1UBB15	1332308.239	13143713.38	7056.53
1UBL5	1332323.903	13143711.23	7053.185
1UBL10	1332326.115	13143715.71	7052.382
1UBL15	1332328.327	13143720.19	7050
1UBR5	1332319.479	13143702.26	7058.097
1UBR10	1332317.267	13143697.77	7058
1UBR15	1332315.055	13143693.29	7059.046

POLE 3UB COORDINATES			
POINT	X COORDINATE	Y COORDINATE	Z COORDINATE
P3UBZ	1332299.155	13143662.5	7073.228
3UBC	1332299.155	13143662.5	7070
3UBA5	1332303.639	13143660.28	7070.991
3UBA10	1332308.123	13143658.07	7071.133
3UBA15	1332312.607	13143655.86	7069.371
3UBB5	1332294.671	13143664.71	7071.226
3UBB10	1332290.187	13143666.92	7072.09
3UBB15	1332285.702	13143669.13	7074.008
3UBL5	1332301.367	13143666.98	7070
3UBL10	1332303.579	13143671.46	7070
3UBL15	1332305.791	13143675.95	7068.809
3UBR5	1332296.943	13143658.01	7070
3UBR10	1332294.731	13143653.53	7073.793
3UBR15	1332292.519	13143649.04	7074.784

POLE 2UB COORDINATES			
POINT	X COORDINATE	Y COORDINATE	Z COORDINATE
P2UBZ	1332310.39	13143684.56	7069.272
2UBC	1332310.39	13143684.56	7064
2UBA5	1332314.874	13143682.34	7065.511
2UBA10	1332319.358	13143680.13	7065.542
2UBA15	1332323.843	13143677.92	7065.288
2UBB5	1332305.906	13143686.77	7067.123
2UBB10	1332301.422	13143688.98	7068
2UBB15	1332296.938	13143691.19	7066.733
2UBL5	1332312.602	13143689.04	7062.565
2UBL10	1332314.814	13143693.52	7059.006
2UBL15	1332317.026	13143698.01	7058
2UBR5	1332308.178	13143680.07	7067.272
2UBR10	1332305.966	13143675.59	7069.125
2UBR15	1332303.754	13143671.1	7070

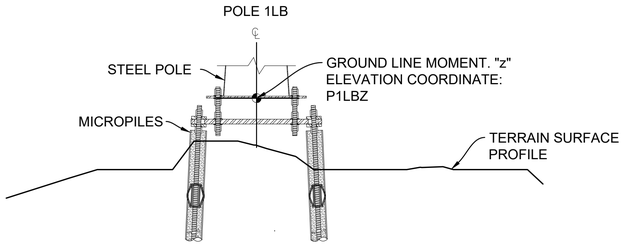
2 / 101E529 UPPER BENCH FOUNDATION COORDINATE DETAIL PLAN VIEW

NOT TO SCALE

REGISTERED PROFESSIONAL ENGINEER COLORADO CERTIFICATE NO. 75014 TOU VANG

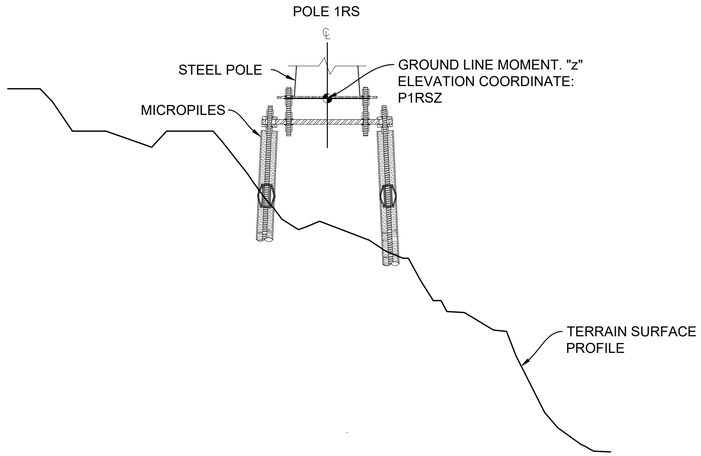
A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	B. PLANT C. WALCHUK TECH. REVIEW: J. DOHRENDORF DATE: MAY 2026	101 E529	POWER DISTRIBUTION MICROPILE FOUNDATION DETAILS UPPER BENCH NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752 PMIS/PKG NO. 318719 SHEET 45 OF 50

4/29/2026 Lynch, Nicole C:\Users\nlynch\DC\ACCDocs\HDR\103505545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\ELEC\101-E528-530.dwg



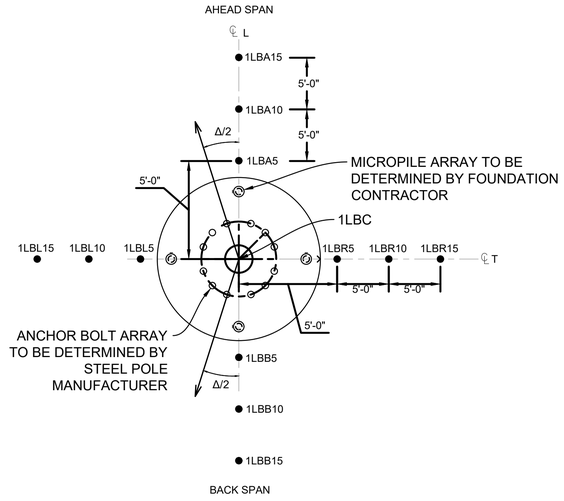
1 / 101E530 LOWER BENCH FOUNDATION COORDINATE DETAIL
LOOKING SOUTH EAST

NOT TO SCALE



3 / 101E530 ROARING SPRINGS FOUNDATION COORDINATE DETAIL
LOOKING NORTH WEST

NOT TO SCALE



POLE 1LB COORDINATES			
POINT	X COORDINATE	Y COORDINATE	Z COORDINATE
P1LBZ	1333671.084	13143132.11	6184
1LBC	1333671.084	13143132.11	6181.701
1LBA5	1333675.822	13143130.51	6178.76
1LBA10	1333680.56	13143128.91	6177.309
1LBA15	1333685.298	13143127.32	6173.807
1LBB5	1333666.346	13143133.71	6182
1LBB10	1333661.608	13143135.3	6180
1LBB15	1333656.87	13143136.9	6181.613
1LBL5	1333672.682	13143136.85	6181.151
1LBL10	1333674.279	13143141.58	6180
1LBL15	1333675.877	13143146.32	6178.801
1LBR5	1333669.486	13143127.37	6180
1LBR10	1333667.889	13143122.63	6180
1LBR15	1333666.291	13143117.89	6180

2 / 101E530 LOWER BENCH FOUNDATION COORDINATE DETAIL PLAN VIEW

NOT TO SCALE

POLE 1RS COORDINATES			
POINT	X COORDINATE	Y COORDINATE	Z COORDINATE
P1RSZ	1335296.862	13142677.94	4913.545
1RSC	1335296.862	13142677.94	4907.605
1RSA5	1335301.678	13142676.6	4903.718
1RSA10	1335306.493	13142675.25	4903.299
1RSA15	1335311.309	13142673.91	4903.033
1RSB5	1335292.046	13142679.29	4910
1RSB10	1335287.231	13142680.63	4910.215
1RSB15	1335282.415	13142681.98	4912.195
1RSL5	1335298.207	13142682.76	4904
1RSL10	1335299.552	13142687.57	4899.221
1RSL15	1335300.898	13142692.39	4896.453
1RSR5	1335295.517	13142673.13	4911.545
1RSR10	1335294.171	13142668.31	4911.84
1RSR15	1335292.826	13142663.49	4914

4 / 101E530 ROARING SPRINGS FOUNDATION COORDINATE DETAIL PLAN VIEW

NOT TO SCALE

GENERAL NOTES:

A. GROUND LINE MOMENT IS LOCATED AT BOTTOM OF POLE BASE PLATE.

LOWER BENCH LINE ANGLE
ACTUAL ANGLE (Δ): 6°3'20"
DESIGN MAXIMUM ANGLE: 10°00'00"

ROARING SPRINGS LINE ANGLE
ACTUAL ANGLE (Δ): 0°0'00"
DESIGN MAXIMUM ANGLE: 3°00'00"

REGISTERED PROFESSIONAL ENGINEER OF ARIZONA
CERTIFICATE NO. 75014
TOU VANG
Date Signed

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	B. PLANT CADD C. WALCHUK TECH. REVIEW: J. DOHRENDORF DATE: MAY 2026	101 E530	POWER DISTRIBUTION MICROPILE FOUNDATION DETAILS, LOWER BENCH & ROARING SPRINGS NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752 PMIS/PKG NO. 318719 SHEET 46 OF 50

4/17/2026 Walchuk, Chris C:\Users\cwalchuk\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\ELEC\101-E528-530.dwg

GENERAL NOTES:

A. ALL WORK SHALL BE IN ACCORDANCE WITH THE PROJECT MICROPILE FOUNDATION PERFORMANCE SPECIFICATION.

B. ALL PROVIDED LOADS ARE PRELIMINARY AND NOT FOR CONSTRUCTION. FINAL LOADS TO BE BASED ON STEEL VENDOR CALCULATIONS.

MICROPILE LOADS – NORTH RIM
(INCLUDES ALL APPLICABLE LOAD FACTORS)

LOAD CASE	POINT	LONG. FORCE (KIPS)	TRAN. FORCE (KIPS)	VERT. FORCE (KIPS)	SHEAR FORCE (KIPS)	TRAN. MOMENT (FT–K)	LONG. MOMENT (FT–K)	BENDING MOMENT (FT–K)
NESC 250B	P1NRZ P2NRZ P3NRZ	18.9	–4.63	–14.79	19.45	222.38	1063.78	1086.77
	P4NRZ	20.65	–4.71	–16.3	21.18	227.65	1167.21	1189.21
NESC 250C	P1NRZ P2NRZ P3NRZ	8.09	–4.34	–6.55	9.18	166.8	447.37	477.46
	P4NRZ	8.96	–4.34	–7.85	9.95	167.3	497.61	524.98
NESC 250D	P1NRZ P2NRZ P3NRZ	6.78	–1.09	–8.25	6.87	51.33	378.58	382.05
	P4NRZ	8.84	–1.2	–9.47	8.92	57.29	494.9	498.2
EXTREME ICE	P1NRZ P2NRZ P3NRZ	16.21	–0.86	–16.97	16.24	48.1	926.66	927.9
	P4NRZ	16.59	–0.88	–17.65	16.61	49.27	949.72	950.99
CONSTRUCTION	P1NRZ P2NRZ P3NRZ	5.94	–0.79	–6.87	5.99	35.76	329.84	331.77
	P4NRZ	9.57	–0.98	–8.17	9.62	46.2	532.03	534.03

MICROPILE LOADS – UPPER BENCH
(INCLUDES ALL APPLICABLE LOAD FACTORS)

LOAD CASE	POINT	LONG. FORCE (KIPS)	TRAN. FORCE (KIPS)	VERT. FORCE (KIPS)	SHEAR FORCE (KIPS)	TRAN. MOMENT (FT–K)	LONG. MOMENT (FT–K)	BENDING MOMENT (FT–K)
NESC 250B	P1UBZ P2UBZ P3UBZ	–20.5	7.02	–6.38	21.67	–509.67	–1809.06	1879.48
	P4UBZ	68.6	27.6	–76.58	73.94	–1965.49	4531.59	4939.49
NESC 250C	P1UBZ P2UBZ P3UBZ	–9.57	7.73	–6.96	12.3	–438.4	–855.54	961.32
	P4UBZ	32.1	20.93	–37.48	38.32	–1319.33	2099.79	2479.87
NESC 250D	P1UBZ P2UBZ P3UBZ	–7.72	1.88	–6.02	7.95	–139.29	–687.25	701.22
	P4UBZ	30.52	8.68	–44.93	31.73	–623.35	2029.86	2123.42
EXTREME ICE	P1UBZ P2UBZ P3UBZ	–17.32	1.95	–0.51	17.43	–172.12	–1512.48	1522.25
	P4UBZ	54.38	10.25	–81.53	55.34	–774.64	3627.51	3709.3
NESC 250B (NO WIRES AHEAD)	P1UBZ P2UBZ P3UBZ	–20.5	7.02	–6.38	21.67	–509.67	–1809.06	1879.48
	P4UBZ	–22.87	7.92	–21.61	24.2	–560.48	–2037.88	2113.55
NESC 250B (NO WIRES BACK)	P1UBZ P2UBZ P3UBZ	–0.02	2.17	–11.14	2.17	–80.71	–1.18	80.72
	P4UBZ	91.43	22.47	–82.02	94.15	–1498.68	6641.09	6808.09
CONSTRUCTION	P1UBZ P2UBZ P3UBZ	–7.09	1.51	–7.24	7.25	–110.81	–634.48	644.08
	P4UBZ	34.85	8.4	–38.76	35.85	–610.81	2335.89	2414.43

MICROPILE LOADS – LOWER BENCH
(INCLUDES ALL APPLICABLE LOAD FACTORS)

LOAD CASE	POINT	LONG. FORCE (KIPS)	TRAN. FORCE (KIPS)	VERT. FORCE (KIPS)	SHEAR FORCE (KIPS)	TRAN. MOMENT (FT–K)	LONG. MOMENT (FT–K)	BENDING MOMENT (FT–K)
NESC 250B	P1LBZ	–15.51	41.76	–64.39	44.55	–3229.02	–1073.78	3402.88
NESC 250C	P1LBZ	–9.32	31.18	–34.11	32.54	–2189.80	–686.67	2294.94
NESC 250D	P1LBZ	–10.6	12.43	–39.3	16.34	–968.11	–769.59	1236.73
EXTREME ICE	P1LBZ	–8.22	13.17	–64.38	15.52	–1069.79	–437.86	1155.93
NESC 250B (NO WIRES AHEAD)	P1LBZ	–103.96	21.70	2.49	106.2	–1569.34	–8155.45	8305.07
NESC 250B (NO WIRES BACK)	P1LBZ	88.45	23.42	–96.64	91.5	–1774.74	7420.51	7629.79
CONSTRUCTION	P1LBZ	–16.77	11.71	–36.02	20.45	–924.41	–1298.41	1593.87

MICROPILE LOADS – ROARING SPRINGS
(INCLUDES ALL APPLICABLE LOAD FACTORS)

LOAD CASE	POINT	LONG. FORCE (KIPS)	TRAN. FORCE (KIPS)	VERT. FORCE (KIPS)	SHEAR FORCE (KIPS)	TRAN. MOMENT (FT–K)	LONG. MOMENT (FT–K)	BENDING MOMENT (FT–K)
NESC 250B	P2RSZ	–107.03	19.46	27.55	108.78	–727.66	–4146.72	4210.08
NESC 250C	P2RSZ	–49.21	13.27	5.37	50.97	–474.77	–1951.8	2008.72
NESC 250D	P2RSZ	–50.48	5.27	11.32	50.75	–201.48	–1995.7	2005.84
EXTREME ICE	P2RSZ	–81.76	4.28	42.61	81.87	–166.55	–3095.61	3100.09
CONSTRUCTION	P2RSZ	–60.74	4.69	4.39	60.92	–183.61	–2467.6	2474.42

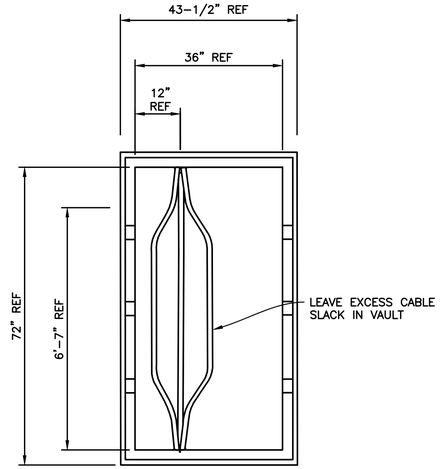
1 / 101E531 DETAIL: MICROPILE FOUNDATION LOADS

NOT TO SCALE

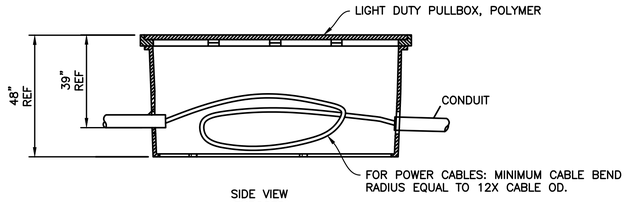
REGISTERED PROFESSIONAL ENGINEER COLORADO CERTIFICATE NO. 75014 TOU VANG

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	B. PLANT CADD C. WALCHUK TECH. REVIEW: J. DOHRENDORF DATE: MAY 2026	101 E531	POWER DISTRIBUTION MICROPILE FOUNDATION LOADS NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752 PMIS/PKG NO. 318719 SHEET 47 OF 50

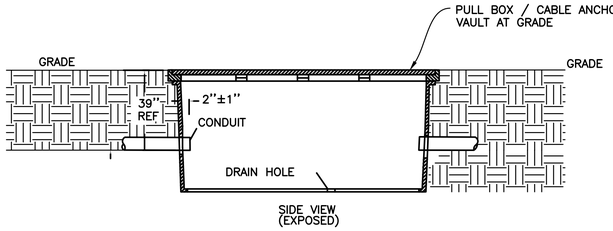
4/17/2026 Walchuk, Chris C:\Users\Cwalchuk\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\ELEC\101-E531.dwg



TOP VIEW
15kV (3) 1/C 4/0 AWG URD POWER CABLE



SIDE VIEW
(EXPOSED)

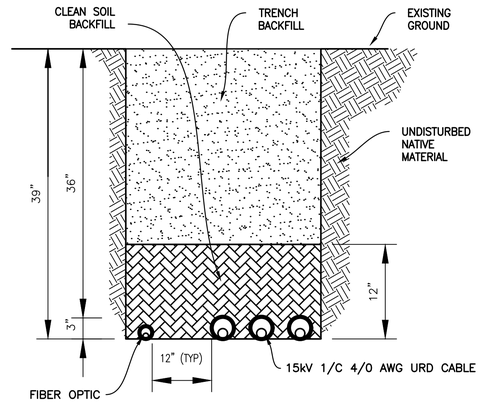


SIDE VIEW
(EXPOSED)

NOTES:
1. FOR AREAS OF SHARED TRENCH DETAIL 1 SHEET 111E502
2. PROVIDE VAULT KNOCKOUTS FOR CABLE AND CONDUIT ENTRANCES.

3 / 101E532 CABLE VAULT

NOT TO SCALE

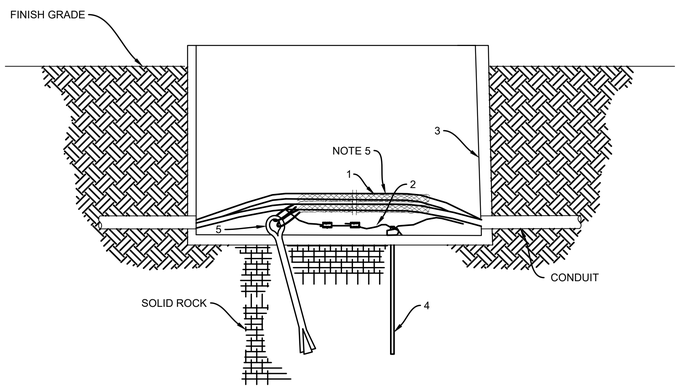


36" DEPTH TRENCH
N.T.S.

4 / 101E532 TYPICAL JOINT-USE TRENCH DETAILS

NOT TO SCALE

NOTES:
1. INSTALL UNDERGROUND PRIMARY CABLE ROUTER MARKERS AS NEEDED.
2. COMMUNICATION CONDUIT TO BE PLACED 12" FROM POWER CABLE FOR ALL NEW INSTALLATIONS.
3. SEE VOLUME 0 FOR ADDITIONAL UTILITY TRENCHING DETAILS.
4. TRENCH BACKFILL SHALL BE FREE FROM ROCK, GRAVEL, OR OTHER MATERIAL WHICH MIGHT DAMAGE THE CABLE JACKET. CLEANED SOIL BACKFILL WHEN USED SHALL CONTAIN NO SOLID MATERIAL LARGER THAN 1 INCH. THIS SOIL LAYER SHALL BE CAREFULLY COMPACTED SO THAT THE CABLE WILL NOT BE DAMAGED.

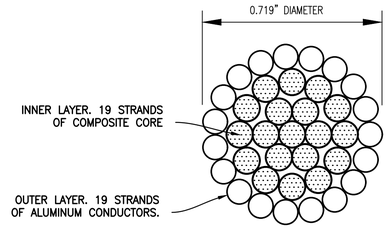


ITEM	DESCRIPTION*	QTY
1	CABLE SUPPORT GRIPS	3
2	CABLE CU GROUND WIRE	1
3	GROUND VAULT	1
4	GROUND ROD	2
5	EXPANDING ROCK ANCHOR	1

NOTES:
1. INSTALL ONE SUPPORT GRIP PER CONDUCTOR
2. ATTACH SUPPORT GRIP TO PULL EYE IN BASE OF VAULT WHERE APPLICABLE TO BE UTILIZED FOR HOLDING CABLE BACK.
3. INSTALL ANCHOR RODS PER MANUFACTURER SPECIFICATIONS.
4. QUANTITIES ARE PER VAULT AND ONLY FOR REFERENCE. INSTALLATION TO VARY BY VAULT/BASEMENT.
5. SPLIT CABLE GRIPS ARE UTILIZED TO RESTRAIN DUE TO ELEVATION CHANGES.

5 / 101E532 ROCK ANCHOR DETAIL

NOT TO SCALE



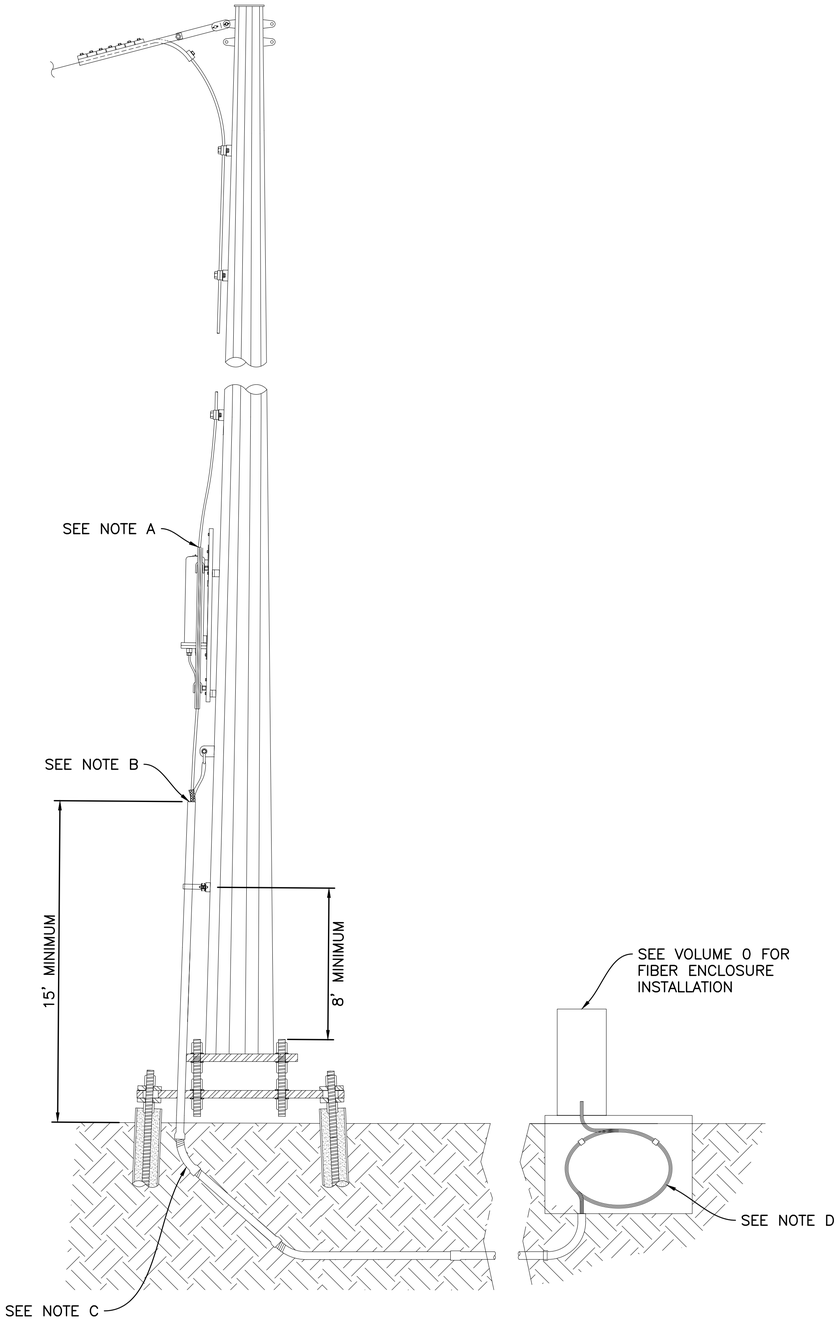
6 / 101E532 CROSS SECTION OF 183 KCMIL ACCR CONDUCTOR

NOT TO SCALE

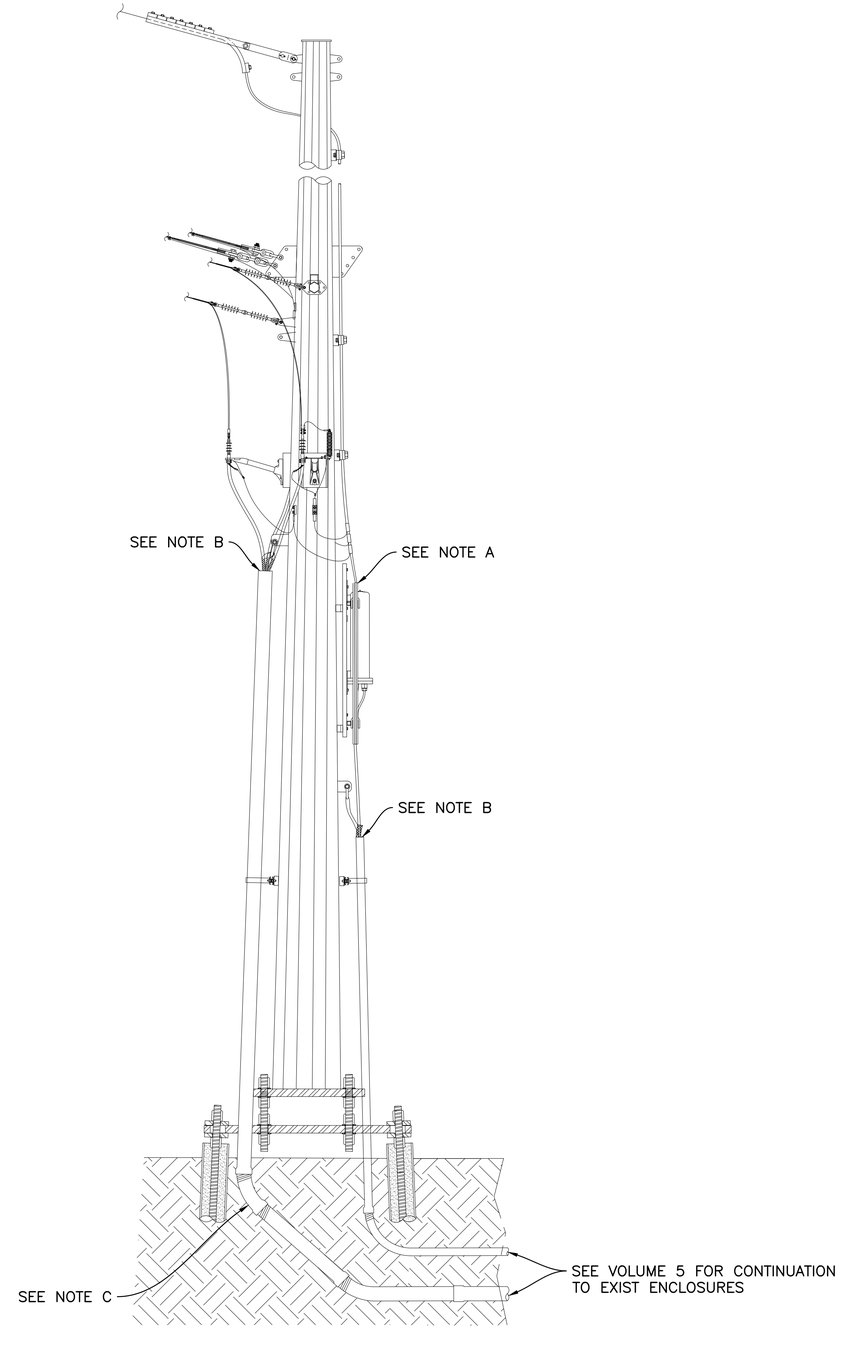
REGISTERED PROFESSIONAL ENGINEER COLORADO CERTIFICATE NO. 75014 TOU VANG

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: B. PLANT DRAWN: C. WALCHUK TECH. REVIEW: J. DOHRENDORF DATE: JUNE 2026	SUB SHEET NO. 101 E532	TITLE OF SHEET POWER DISTRIBUTION DETAILS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 48 OF 50

8/18/2026 Vang, Tou C:\Users\TVANG\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\ELEC\101-E532.dwg



NORTH RIM



ROARING SPRINGS

GENERAL NOTES:

A. COIL 50'–0" OF OPGW AROUND COIL BRACKET. TYPICAL.

B. SEAL CONDUIT SO THAT NO MOISTURE MAY ENTER. TYPICAL.

C. DRAWING OF CONDUIT ROUTING IS FOR ILLUSTRATION ONLY. ACTUAL ROUTING MAY VARY TO ACCOMMODATE MICROPILE. TYPICAL.

D. COIL 20'–0" OF OPGW IN VAULT. TYPICAL.

1 / 101E533 DETAIL: OPGW VAULT TERNIMATION

NOT TO SCALE

REGISTERED PROFESSIONAL ENGINEER OF COLORADO
CERTIFICATE NO. 75014
TOU VANG

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: B. PLANT CADD C. WALCHUK TECH. REVIEW: J. DOHRENDORF DATE: MAY 2026	SUB SHEET NO. 101 E533	TITLE OF SHEET POWER DISTRIBUTION OPGW DETAILS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 49 OF 50

4/17/2026 Vang, Tou C:\Users\TVANG\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\ELEC\101-E533.dwg

ITEM #	DESCRIPTION
1	Y–CLEVIS, CLEVIS, 3 1/4" LONG, 5/8" PIN, 30K LBS
2	DEAD END LINK PLATE, 10" SPACING, 3/4" THICK, 7/8" DIA HOLES, 40K LBS
3	BOLTED DEAD END, OPGW, 40K LBS
4	FIBER DEAD END GROUND WIRE, 4/0 18 STR CABLE, 60" LENGTH, 1/2" HARDWARE
5	FIBER DOWN LEAD GUIDE CLAMP, GROOVE FOR .724" DIA OPGW, 1/2" HARDWARE
6	EXTERNAL COIL BRACKET, 400LB LOADING CAPACITY
7	SPLICE ENCLOSURE WITH SPLICE TRAYS
8	PARALLEL GROVE CLAMP (AL TO CU) , OPGW – 1/0 CU
9	ANCHOR SHACKLE, 2 3/8" LONG, 5/8" PIN , 30K LBS
10	CABLE SUPPORT GRIP – TIN COATED BRONZE, SINGLE EYE, CLOSED MESH, 1/0 NEUTRAL, 50 LB WORKING LOAD
11	CABLE SUPPORT GRIP – OFFSET EYE, SPLIT MESH, ROD CLOSING – FIBER OPTIC CABLE, 161 LB WORKING LOAD
12	GALVANIZED RIGID METAL CONDUIT – 2"
13	GALVANIZED RIGID METAL CONDUIT – 3"
14	PVC CONDUIT, SCH 40 – 2"
15	PVC CONDUIT, SCH 40 – 3"
16	1 5/8" STRUT CHANNEL, STAINLESS STEEL, LENGTH AS REQUIRED
17	STRUT CHANNEL CLAMP 2" CONDUIT, STAINLESS STEEL, LENGTH AS REQUIRED
18	STRUT CHANNEL CLAMP 3" CONDUIT, STAINLESS STEEL, LENGTH AS REQUIRED
19	BOLT, STAINLESS STEEL, 1/2"–13 x REQ'D LENGTH
20	FLAT WASHER, STAINLESS STEEL, 1/2"
21	SPLIT LOCK WASHER, STAINLESS STEEL, 1/2"
22	BOLT, MACH SQ HEAD, 5/8" X REQ'D LENGTH W/NUT
23	WASHER SQ, ID 11/16", WIDTH 2–1/2" X 2–1/2" X 1/4"
24	LOCKNUT, 5/8", MF TYPE
25	FIBERGLASS, SILICONE COATED, 24" STANDOFF BRACKET, VERTICAL LOAD 1900 LBS
26	1/0 AAC, COVERED JUMPER
27	2–HOLE NEMA CU/AL COMPRESSION LUG – 1/0 AL
28	2–HOLE NEMA CU/AL COMPRESSION LUG – 4/0 AL
29	WASHER FLAT 1/2 IN SSTL 1.25" OD X 9/16" ID X 3/16" THICK
30	WASHER BELLEVILLE SSTL, 33/64" ID X 13/16" X 3250 LB RATING
31	NUT, 1/2" – 13 SILICON BRONZE
32	15KV CABLE TERMINATION – 4/0, 110KV BIL, 4 SKIRT, COLD SHRINK
33	CROSS–ARM MOUNTING BRACKETS MB–1, FOR SUPPORTING TERMINATED CABLES
34	SURGE ARRESTER, 10KV, 8.4KV MCOV
35	COMPRESSION CONNECTOR, 3/0 – #6
36	COMPRESSION CONNECTOR, #6 – #6
37	COMPRESSION CONNECTOR, 1/0 – #6
38	#6 CU CONDUCTOR
39	COMPRESSION CONNECTION, 1/0
40	ANCHOR SHACKLE, 5" LONG, 7/8" PIN, 60K LBS
41	YOKE PLATE, 5" LONG, (2) 1.13" DIA HOLES, (1) 1.19" DIA HOLE, 18" HOLE SPACING, 60K LBS
42	HOT LINE CLEVIS EYE, 7" LENGTH, 5/8" PIN, 30K LBS
43	INSULATOR, GLASS BELL, ANSI CLASS 52–6–H, 30K LBS
44	CLEVIS, CLEVIS, 5 1/2" LENGTH, 30K LBS
45	DEAD END, COMPRESSION, FOR 183–T114 ACCR, 15' TERMINAL, 35.35" LENTGH
46	CABLE SUPPORT GRIP – OFFSET EYE, SPLIT MESH, ROD CLOSING – 4/0, 195LB WORKING LOAD
47	GALVANIZED RIGID METAL CONDUIT – 4"
48	STRUT CHANNEL CLAMP 4" CONDUIT, STAINLESS STEEL
49	ANCHOR SHACKLE, 3 3/4" LENGTH, 7/8" PIN, 80K LBS
50	CHAIN LINK, 4" LENGTH, 7/8" THICKNESS, 80K LBS
51	HEAVY DUTY DEADEND THIMBLE CLEVIS, 80K LBS
52	DEADEND GRIP, 37 NO 8 ALUMOWELD
53	DEAD END INSULATOR, 15KV, POLYMER, 7.5K LBS WORKING LOAD
54	HEAVY DUTY THIMBLE CLEVIS, 5 1/2" LENGTH, 40K LBS
55	INSULATED SPACER CABLE DEADEND GRIP
56	144 STRAND SM OPTICAL GROUND WIRE, OPGW, 34,454 LBS RBS
57	144 STRAND SM FIBER OPTIC CABLE
58	NEUTRAL, 1/0 CU BARE
59	2–HOLE NEMA CU/AL COMPRESSION LUG – #6 CU
60	1–HOLE CU/AL COMPRESSION LUG – #6 CU

ITEM #	DESCRIPTION
61	POST INSULATOR, POLYMER, 25/35KV, ANSI C29.18 51–4F
62	INSULATOR TIE, COVERED
63	15KV MV CABLE, 4/0, CONCENTRIC NEUTRAL
64	183 KCMIL ALUMINUM CONDUCTOR COMPOSITE SUPPORTED CORE ACCR
65	1/0 TERMINAL CONNECTOR, 15'
66	37 NO. 8 ALUMOWELD
67	1/0 AAC INSULATED SPACER CABLE, 15 KV
68	COMPRESSION CONNECTOR, 1/0 – 1/0
69	COMPRESSION CONNECTOR, #6 – 1/0
70	#6 CU GROUND WIRE
71	CLAMP, GROUND, VISE TYPE #6 SOLID TO 1/0 STR, W/ 1/2" STUD
72	CLAMP, GROUND ROD, 5/8"
73	ROD, GROUND, 5/8" X 8'
74	CONNECTOR KIT FOR OPGW
75	GALVANIZED RIGID METAL CONDUIT – 2" ELBOW, 36" RADIUS
76	GALVANIZED RIGID METAL CONDUIT – 3" ELBOW, 36" RADIUS
77	GALVANIZED RIGID METAL CONDUIT – 4" ELBOW, 36" RADIUS
78	CONDUIT ADAPTERS – 2", 3", 4"
79	3/4" – 10 UNC THREAD STUD BOLT 1 3/4" LENGTH, SERRATED FLANGE, (2) LOCK WASHERS, HEX NUT
80	PARALLEL GROVE CLAMP (AL TO CU), 37 #8 AW TO #6 CU
81	1–HOLE LUG, COPPER, CODE, LONG BARREL, NO INSPECTION WINDOW, #6

GENERAL NOTES:

A. SEE SHEETS 101E501 THROUGH 101E508 FOR STRUCTURE DETAILS AND SECTION VIEWS.

B. ALL ITEMS MARKED "REQ'D LENGTH" WILL BE DETERMINED BY CONTRACTOR BASED ON FINAL STEEL ATTACHMENT POINT AS PROVIDED BY THE STEEL MANUFACTURER.

C. ALL PARTS AND COMPONENTS TO BE DETERMINED BY THE CONTRACTOR BASED ON SHOWN RATINGS AND SPECIFICATIONS.

REGISTERED PROFESSIONAL ENGINEER OF COLORADO
CERTIFICATE NO. 75014
TOU VANG

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: B. PLANT CADD C. WALCHUK TECH. REVIEW: J. DOHRENDORF DATE: MAY 2026	SUB SHEET NO. 101 E534	TITLE OF SHEET POWER DISTRIBUTION STRUCTURE DETAILS MATERIAL ITEMS NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 50 OF 50

4/17/2026 Vang, Tou C:\Users\TVANG\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\ELEC\101-E534.dwg